The Maturity Insurance Channel of Monetary Transmission*

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Abstract

Long-term interest rates are not sufficient to predict how firms adjust debt maturity to monetary tightening. Using euro-area credit-registry data, we show that firms with greater refinancing needs borrow at longer maturities, but their responses depend on why long-term rates rise. Higher expected future short rates strengthen maturity extension among exposed firms, whereas higher term premia weaken it. Identification relies on using inherited repayment schedules, within-bank-month comparisons, and monetary-policy surprises. A heterogeneous-firm model with financial frictions, calibrated to the loan-level data, reproduces these untargeted patterns and shows that maturity flexibility cushions investment contractions and reduces default following news of higher future rates. Monetary transmission is state dependent: the same expected-rate shock generates larger investment losses when long-duration financing is expensive. Applying the model to the 2022–24 tightening reveals that maturity flexibility enables anticipatory deleveraging, improving financial resilience even as firms shorten debt maturity. Debt maturity is therefore both an inherited exposure and an endogenous margin of monetary transmission.

Keywords: monetary policy transmission, debt maturity, interest rate risk, yield curve, term premium
JEL Codes: E52, E44, G32

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1 Introduction

What determines the strength and timing of monetary transmission through corporate balance sheets? A central insight of the credit channel is that firms' existing financial obligations shape their exposure to monetary tightening. Firms with substantial debt coming due must refinance at prevailing market conditions, while those with longer-maturity debt can postpone refinancing (Bernanke and Gertler, 1995; Gertler and Gilchrist, 1994; Almeida et al., 2012). The maturity structure of outstanding corporate debt thus influences how monetary shocks propagate across firms. Yet this exposure is not fixed: firms can adjust the maturity of new borrowing as financing conditions change, reshaping their future refinancing obligations.

Why should this adjustment matter for monetary transmission? When firms face financial frictions and idiosyncratic uncertainty, refinancing in unfavorable future states can be particularly costly. Longer-maturity debt reduces the frequency with which firms must return to credit markets, providing insurance against future rollover risk. But this protection comes at a price: firms must trade off the benefits of reducing rollover exposure against the cost of long-term financing. Because future refinancing needs affect expected financing costs and default risk, maturity choices can also influence current borrowing and investment decisions. The ability to adjust maturity therefore introduces an additional financing margin through which firms respond to monetary shocks, beyond the exposure determined by their inherited debt structure.

This paper studies how endogenous debt maturity adjustment changes monetary transmission. We examine how firms with different inherited refinancing needs adjust the maturity of new borrowing when aggregate financing conditions change, and quantify how this flexibility affects borrowing, investment, and default. The central question is whether monetary transmission differs when firms can actively reshape their future refinancing obligations, relative to an economy in which maturity cannot respond directly to aggregate financing conditions.

Monetary conditions affect both the value and the price of maturity insurance. An increase in expected future short rates raises anticipated refinancing costs, but is also reflected in current long-term borrowing rates. Firms therefore face a trade-off: extending maturity requires paying the current price of long-term financing but reduces the need to refinance in uncertain future states. With financial frictions, the value of avoiding costly refinancing can be especially high for firms with substantial repayment obligations. By contrast, an increase in term premia raises the cost of long-duration financing without a corresponding increase in expected future short rates, weakening the incentive to extend maturity. Consequently, changes in long-term interest rates need not generate equal maturity responses, it depends on whether they reflect expected future rates or the price of duration. We refer to this mechanism as the *maturity-insurance channel of monetary transmission*.

We study this mechanism using the 2022–23 euro-area monetary tightening cycle, an episode in which both the expected path of policy rates and the cost of long-duration financing changed substantially. Firms entered this period with repayment obligations inherited from earlier financing decisions, generating heterogeneous exposure to tighter credit conditions. The euro area provides

a particularly informative setting because corporate credit is predominantly bank-based and fixed-rate borrowing accounts for a substantial share of outstanding debt. In our data, fixed-rate contracts account for 61% of outstanding loan volume, reaching 75% in Germany and 72% in France. Moreover, approximately 30% of outstanding bank debt is scheduled to mature within the following twelve months, with considerable heterogeneity across firms, industries, and countries. These differences in inherited repayment schedules provide a setting to study how firms adjust their exposure as aggregate financing conditions change.

This paper makes three contributions. First, we provide empirical evidence that firms adjust debt maturity in response to monetary conditions, and that these adjustments depend on inherited rollover exposure and the composition of long-term interest rates. Second, we quantify how endogenous maturity adjustment changes the transmission of monetary shocks to corporate borrowing, investment, and default. Third, we show that the real effects of monetary policy depend on the cost of long-term financing: the same change in expected future interest rates can generate different maturity and investment responses depending on term-premium conditions.

The empirical analysis uses *AnaCredit*, the euro-area credit registry, matched to firm balance sheets from September 2018 through December 2024. These data provide detailed information on the repayment schedules of outstanding bank loans and the maturity of newly originated credit. We measure firms' inherited rollover exposure by the share of outstanding bank debt scheduled to mature within the following twelve months. The empirical design compares firms borrowing from the same bank in the same month but facing different repayment obligations. Bank-by-month fixed effects absorb contemporaneous lender-level conditions, while firm-by-bank fixed effects account for persistent differences within lending relationships. We complement this approach by constructing rollover exposure from repayment schedules determined before the tightening cycle and by exploiting high-frequency monetary policy surprises.

The empirical analysis establishes three results. First, firms with greater inherited rollover exposure obtain longer maturity on new borrowing. Moving from the 25th to the 75th percentile of rollover exposure is associated with approximately 12 percent longer maturity on newly originated loans, roughly 7.5 months at the sample mean. This relationship remains essentially unchanged when rollover exposure is measured using repayment schedules established before monetary tightening. Firms facing greater near-term repayment obligations therefore use new debt maturity to shift refinancing needs further into the future.

Second, and central to the mechanism, maturity responses depend on the source of movements in long-term interest rates. We decompose five-year euro-area yields into expected future short rates and term premia following Joslin et al. (2011). When expected future short rates rise, firms with greater rollover exposure extend the maturity of new borrowing relatively more. When term premia rise, this relative maturity extension weakens. The opposing responses are consistent with the distinction between the value of reducing future refinancing exposure and the current cost of obtaining maturity protection.

We provide complementary evidence using survey expectations and high-frequency monetary-policy surprises following Altavilla et al. (2019). Timing and Forward Guidance surprises, which primarily shift the expected policy-rate path, are associated with relatively longer maturity among firms with greater rollover exposure. Target surprises, which primarily affect current policy rates, instead generate relatively shorter maturity. These patterns are robust to alternative yield horizons and decompositions and to distinguishing monetary-policy shocks from central-bank information shocks. Additional tests show that the maturity response is distinct from coupon-repricing exposure and is not explained by asset-maturity matching, borrower risk, substitution toward committed liquidity, or selection into new borrowing.

Third, maturity adjustment is associated with changes in firms' broader financial positions and real outcomes. Firms with greater inherited rollover exposure shift their overall debt portfolios toward longer maturities, but this adjustment weakens when the cost of long-duration financing rises. Under the same conditions, exposed firms face higher interest expenses and subsequently reduce investment. The investment decline is concentrated among firms that had not previously extended maturity, particularly smaller firms. These findings suggest that the ability to manage refinancing exposure through maturity choices is relevant for the real effects of monetary tightening.

The empirical evidence identifies systematic differences in maturity adjustment across firms and monetary conditions, but does not by itself establish how these choices change the aggregate transmission of monetary shocks. To quantify their role, we develop a heterogeneous-firm model with financial frictions in which firms jointly choose investment, borrowing, and debt maturity. Firms enter each period with outstanding repayment obligations and face idiosyncratic productivity risk, costly refinancing, and endogenous default. Aggregate financing conditions are summarized by the current short rate, news about future short rates, and a term-premium wedge.

The central trade-off in the model is between the insurance provided by longer maturity and its financing cost. By reducing future repayment obligations, longer-term debt limits the need to raise external funds in states in which refinancing may be costly. This benefit depends on firms' financial positions, future productivity realizations, and aggregate financing conditions. At the same time, long-term borrowing is priced endogenously, taking into account repayment risk and the value of outstanding claims. Maturity choices therefore affect not only the timing of refinancing but also firms' current effective financing costs and borrowing capacity.

We discipline the model using loan- and firm-level moments of default, debt issuance, and investment, without targeting the differential maturity responses central to the empirical analysis. Nevertheless, the model reproduces the empirical interaction between rollover exposure and aggregate financing conditions. In particular, firms with greater inherited refinancing needs respond differently to expected-rate news and changes in the term-premium wedge. The model provides a quantitative framework for studying how maturity flexibility affects monetary transmission.

We use the model to establish three quantitative results. First, the ability to adjust debt maturity materially changes the real effects of monetary tightening. We compare the benchmark

economy with a counterfactual in which new-debt maturity cannot respond directly to aggregate financing conditions, while remaining free to evolve with firms' balance sheets. The comparison identifies the effect of restricting the maturity policy: borrowing, investment, and default remain endogenous in both economies, and the inherited maturity distribution is not held fixed.

Following news of higher future interest rates, maturity flexibility preserves 2.4 percentage points of three-year cumulative investment relative to capital and reduces cumulative default by 1.4 percentage points, compared with the restricted economy. Firms with greater inherited refinancing needs place a higher ex ante value on this flexibility. Importantly, the insurance provided by principal maturity is distinct from interest-rate fixation: even when debt coupons reset with the short rate, adjusting principal maturity continues to provide protection against refinancing risk. Principal maturity does not insure against higher coupon payments on floating-rate debt; it insures against having to repay or refinance principal under adverse financing conditions.

Second, the quantitative importance of maturity flexibility depends on the price of long-duration financing. When news of higher future rates coincides with higher duration costs, firms use maturity flexibility to shorten new borrowing and avoid expensive long-term financing. Under this joint shock, flexibility preserves 3.9 percentage points of cumulative investment over three years, compared with 2.4 percentage points following expected-rate news alone. Thus, maturity adjustment need not always take the form of maturity extension: its value also comes from allowing firms to avoid costly long-term funding.

The overall effects of monetary shocks are themselves state dependent. Holding the expected-rate innovation and firms' initial balance sheets fixed, three-year cumulative investment falls by approximately 3 percentage points more when long-duration financing is expensive rather than cheap. Firms also shorten maturity more sharply and persistently when duration costs are high. Consequently, the real effects of a given revision to expected future interest rates depend on the financing environment in which firms make their maturity decisions.

Third, we use the realized 2022–24 euro-area tightening episode to examine how endogenous maturity choices shape the transmission of successive monetary shocks. Expected-rate news induces firms to reduce borrowing before higher short rates materialize, limiting subsequent defaults even as aggregate new-debt maturity shortens. This anticipatory deleveraging largely disappears when maturity cannot respond directly to aggregate financing conditions. Higher duration costs instead discourage long-term borrowing and amplify the investment contraction. Along the full tightening path, maturity flexibility preserves 6.7 percentage points of cumulative investment relative to the restricted economy.

This historical exercise highlights an additional role of maturity adjustment: it affects not only when firms will refinance but also how much they choose to borrow in anticipation of tighter monetary conditions. The interaction between maturity and borrowing choices therefore matters for the timing of monetary transmission, including responses that occur before the anticipated increase in short-term rates is realized.

Taken together, these results show that monetary transmission depends on both inherited and endogenous corporate liability structures. Existing repayment schedules determine which firms are initially exposed to changes in financing conditions, while subsequent maturity choices reshape future repayment obligations and current financing decisions. The option to adjust maturity can attenuate monetary tightening, but the direction and magnitude of the adjustment depend on the cost of obtaining long-duration financing.

These findings have implications for the assessment of monetary transmission. Changes in policy rates and the yield curve alone do not fully characterize how monetary shocks affect corporate financing and investment. Their effects also depend on firms' inherited repayment schedules and on how firms adjust debt maturity in response to changing financial conditions. In particular, the same revision to expected future short rates can have substantially different real effects depending on the term-premium environment. For central banks, this highlights the importance of monitoring corporate debt maturity and refinancing needs alongside the expected policy-rate path and the price of duration. More broadly, debt maturity is not merely a predetermined source of exposure to monetary policy, but an endogenous financing margin that shapes its transmission.

Related literature. Our paper contributes to three strands of literature and is related to a fourth. First, we contribute to research on financial amplification and heterogeneous monetary transmission through corporate balance sheets. A large literature shows that monetary policy affects firms through borrower net worth, external-finance premia, and the supply of bank credit (Gertler and Gilchrist, 1994; Bernanke and Gertler, 1995; Kashyap and Stein, 2000; Jiménez et al., 2012). More recent work emphasizes that transmission depends on firms' pre-existing financial positions. Floating-rate debt exposes interest expenses directly to policy-rate changes (Ippolito et al., 2018), while heterogeneity in financial positions and credit spreads generates differential transmission of monetary tightening (Ottonello and Winberry, 2020; Anderson and Cesa-Bianchi, 2024). Refinancing needs are particularly important when external financing conditions deteriorate: firms with debt coming due reduce investment more sharply during credit contractions (Almeida et al., 2012; Duchin et al., 2010), and rollover risk amplified the decline in investment during the European sovereign debt crisis (Kalemli-Özcan et al., 2022). Closest to our paper, Deng and Fang (2022) and Jungherr et al. (2024) show that debt maturity shapes monetary-policy transmission and that firms trade off rollover risk against the cost of long-term debt. We differ by separating expected-rate and term-premium movements, showing that they have opposite effects on maturity adjustment among rollover-exposed firms, and by quantifying how this endogenous maturity response changes investment and default. Using credit-registry data, we identify these responses within bank-month cells and distinguish the protection provided by principal maturity from that provided by coupon fixation. Our counterfactual further shows that the real effects of expected-rate guidance depend on the term-premium environment.

Second, our paper contributes to research on expectations and the timing of monetary transmission. Monetary-policy announcements convey news about both the current stance and the expected

path of policy rates (Gürkaynak et al., 2005; Altavilla et al., 2019), and expected future rates can affect current decisions before they are realized (McKay et al., 2016). Related work shows that the effects of forward guidance depend on the macroeconomic and financial environment (Keen et al., 2017; Lunsford, 2020; Bauer et al., 2022). We add a corporate-balance-sheet source of state dependence: firms adjust debt maturity in anticipation of future refinancing conditions, and the transmission of expected-rate news depends on the term-premium environment. Because long-term rates reflect both expected future short rates and term premia (Hanson and Stein, 2015), while the relative cost of long-term finance shapes maturity choice (Kozlowski, 2021), the same expected policy-rate path can generate different maturity and real responses depending on the price of long-duration finance.

Third, we contribute to the literature on endogenous corporate debt maturity. Classic theories relate maturity choice to debt overhang, asymmetric information, rollover risk, and the relative cost of short- and long-term finance (Myers, 1977; Flannery, 1986; Diamond, 1991; He and Xiong, 2012; Brunnermeier and Oehmke, 2013). Empirically, firms adjust debt maturity and interest-rate exposure with aggregate financing conditions. Baker et al. (2003) show that firms issue relatively more long-term debt when long-term borrowing is comparatively attractive, Faulkender (2005) relates interest-rate exposure to the slope of the yield curve, and Greenwood et al. (2010) show that issuers respond to changes in the relative supply and pricing of long- and short-term claims. Quantitative work studies maturity jointly with borrowing, investment, and default (Jungherr and Schott, 2021; Poeschl, 2023; Ferrante et al., 2025). Our contribution within this literature is to identify a refinancing-insurance motive behind maturity adjustment. Bräuning et al. (2020) show that term premia affect the maturity composition of aggregate corporate bond issuance, while Fabiani et al. (2024) show that monetary policy changes corporate maturity through movements in the relative cost of long-term financing. We exploit cross-sectional differences in inherited refinancing exposure to distinguish the value of maturity protection from its price. Expected future short rates raise the benefit of extending maturity disproportionately for firms with debt coming due, whereas term premia raise the cost of obtaining that protection. The interaction between aggregate financing conditions and pre-existing refinancing needs therefore helps identify why firms adjust maturity.

Finally, our paper is related to work on firms' intertemporal management of financing risk. Dynamic corporate theories emphasize that firms time external finance, accumulate liquidity, and manage risk in anticipation of future funding needs (Bolton et al., 2013; Eisfeldt and Muir, 2016; Acharya et al., 2020). Empirically, firms refinance early when credit conditions are favorable (Mian and Santos, 2018), while Xu (2018) documents precautionary maturity extension in corporate bond markets. We add to this literature by viewing debt maturity as a liability-side tool for managing future financing risk. By changing the maturity of new borrowing, firms determine when principal must next be refinanced and therefore how much external funding they will need to raise in future states in which financing may be costly. Maturity choice thus allows firms to reshape the timing of their exposure to future monetary and credit conditions.

The remainder of the paper proceeds as follows. Section 2 presents the data and motivating facts.

Section 3 describes the loan-level empirical design and results. Sections 4–5 validate the mechanism and describe the real effects. Section 6 presents the quantitative model, and Sections 7–8 present the model results and counterfactuals. Section 9 concludes.

2 Data

Loan-level data. The main data source is AnaCredit, the euro-area credit registry, which covers the universe of credit transactions above a reporting threshold of EUR 25,000 at origination. In our analysis, we consider bank loans to non-financial corporations from September 2018 to December 2024. AnaCredit includes rich loan-level information such as the loan amount, rate, maturity, collateral, borrower, and lender identifiers for both outstanding balances and newly originated loans. AnaCredit is especially well suited for our analysis because we can observe both the stock of short-term debt—our proxy for rollover exposure—and the terms of newly originated loans [1].

Loan maturity at origination is the main outcome for the loan-level analysis; it is highly dispersed: in the regression sample (Appendix Table 15, winsorized at the 1st and 99th percentiles), the mean is 1,895 days (5.2 years), the median 737 days (2.0 years), the interquartile range runs from 113 to 1,836 days, and the upper tail reaches 7,330 days (20.1 years). The maturity dispersion is present both across borrowers and within lending relationships over time. Appendix Figure 13 plots the evolution of average new-loan maturity between 2018 and 2024, while Appendix Figure 12 documents a clear positive relationship between new-loan maturity and the 5Y–1Y yield-curve slope, after absorbing firm and bank fixed effects.

New borrowing is infrequent and relationship-concentrated. Between 2018 and 2024, 5.8% of indebted non-financial corporations originated a new loan in a given month, and about 32.5% of borrowing firms obtained the new credit from their main bank. The within-bank comparisons in the empirical analysis hold the lender fixed whether or not it is the firm's main bank [2]. Regarding the rate type composition of new lending, fixed-rate contracts are 71 percent of new loans, 51 percent of new credit (including credit lines), but floating-rate loans are on average 2.4 times larger. Appendix Figure 8 reports the distribution of new credit across original-maturity buckets separately for floating- and fixed-rate loans, and shows that fixed-rate credit is disproportionately concentrated at very short and very long maturities. On the outstanding debt stock, fixed-rate contracts account for about 61 percent of term-loan volume, a share stable across 2018–2024, with wide cross-country dispersion: fixed-rate contracts account for 75 percent of term-loan volume in Germany and 72 percent in France against 39 percent in Italy and 14 percent in Finland [3].

[1] Appendix Table 25 extends the new loan sample back to 2014 using Orbis balance-sheet proxy of rollover exposure, since AnaCredit only allows tracking outstanding loans beginning in September 2018.

[2] The main bank is defined as the lender holding the largest share of the firm's outstanding credit in the previous month.

[3] Because our sample excludes credit lines, which are predominantly floating rate, this share is higher than the evidence reported by Core et al. (2025), who find that non-floating debt accounts for about 40% of total credit volume in December 2021.

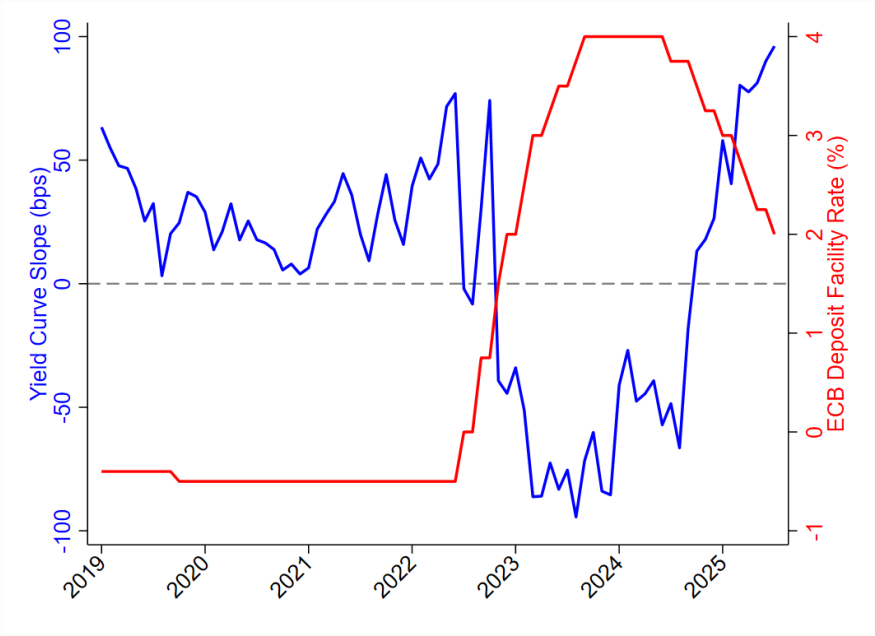


Figure 1: Monetary policy rate and the yield-curve slope

Notes: The figure plots the time series of the ECB Deposit Facility Rate (in red) and the Yield Curve Slope between 10-year AAA government bond yields and 2-year €STR OIS rates (in blue) between 2019 and 2025; the estimation sample ends in December 2024. Source: ECB Statistical Data Warehouse.

Firm- and macro-level data. Annual firm-level balance sheets come from Bureau van Dijk ORBIS, which covers listed and unlisted firms with good coverage of small and medium-sized enterprises. We follow Kalemli-Özcan et al. (2024) in cleaning ORBIS and obtain time-varying controls for total assets, tangibility, liquidity, leverage, and sales growth. Yields and policy rates are from the ECB Statistical Data Warehouse; inflation swaps from Bloomberg.

The final sample runs monthly from September 2018 to December 2024 and comprises 1.67 million firm–bank–month observations covering 811 banks and 109,051 firms, with about 380 banks originating in a typical month. Bank×month cells have a median of 58 borrowing firms. The loan- and firm-level variables are winsorized at 1%. The loan-level analysis is at the firm–bank–month level, while the firm-level analysis is conducted at the firm–year level; summary statistics are reported in the Appendix Table 15.

2.1 Stylized Facts

This section documents three stylized facts that motivate the analysis.

Fact 1: Monetary policy, expected future rates and term premia. The slope of the yield curve varies substantially over monetary-policy cycles. We measure the euro-area yields at a monthly frequency using market-implied zero-coupon rates from AAA euro area government bonds and overnight index swaps (OIS) at maturities ranging from three months to ten years. The yield

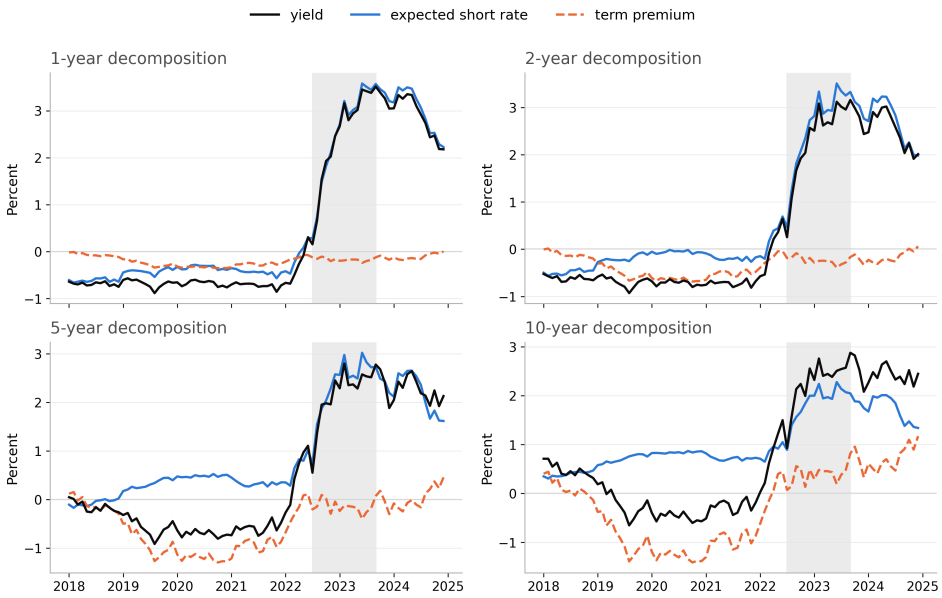


Figure 2: Yield Decomposition into Expected Short Rates and Term Premia

Notes: The figure decomposes euro-area OIS zero-coupon yields into expected future short rates path and term premia. The decomposition uses the affine term-structure model of Joslin et al. (2011). Expected short rates (in blue) capture the anticipated path of future policy rates, while $TP_t(n)$ captures the term-premium component of long-term yields (in orange). The decomposition uses euro-area OIS rates, which provide a measure of the risk-free term structure that is less affected by sovereign credit risk. The shaded region denotes the ECB monetary-policy tightening cycle beginning in 2022. Source: authors' calculation.

curve (YC) slope is defined as the difference between a long-term yield and a short-term risk-free rate, such as the 10-year government bond yield and the 2-year OIS rate, or the 5-year yield and the 1-year OIS rate. Figure 1 plots the spread between the 10-year government bond yield and the 2-year OIS rate together with the ECB deposit facility rate. The slope exhibits substantial time-series variation. Its average level is 0.18 percentage points, with a standard deviation of 0.53 percentage points. The curve flattens sharply during the 2022-23 tightening cycle, reaching a minimum of -0.94 percentage points, and subsequently steepens in late 2024 as financial conditions normalize. To interpret these movements, we decompose yields into expected short rates and term premia using the standard affine term structure model of Joslin et al. (2011); as a robustness check, we also employ the yield decomposition following the methodology of Adrian et al. (2013). See Appendix D for estimation details.

Figure 2 reports the decomposition across yields at different maturities. Two results stand out. First, variation in short-maturity yields is almost entirely driven by changes in expected policy rates. The increase in one- to two-year yields in 2022 reflects a sharp upward revision in the expected path of ECB policy, followed by a gradual decline as expectations of future easing emerge. Second, variation in longer-maturity yields is partially accounted for by movements in term premia. The term premium increases from negative values prior to 2022 to persistently positive levels, accounting for a substantial share of the increase in long-term yields during this period. Thus, these patterns indicate that recent changes in the slope of the yield curve reflect not only revisions in expected

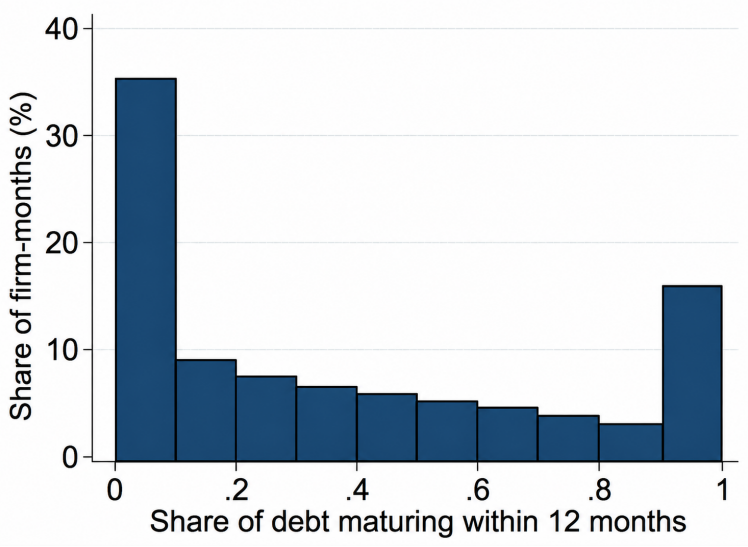


Figure 3: Rollover Exposure across Firms

Notes: This figure reports short-term debt, which is the share of outstanding debt with residual maturity up to 12 months, excluding credit lines at firm-time level. It shows the distribution across firm–months: 35 percent carry no short-term debt and 16 percent have their entire outstanding debt maturing within twelve months. All firms have positive outstanding bank debt; zero denotes no debt scheduled to mature within 12 months. Source: AnaCredit, 2018–2024.

policy, but also a repricing of duration risk.

Fact 2: Corporate rollover exposure is large and highly dispersed. We proxy for rollover exposure with the share of short-term debt, defined as outstanding loans with residual maturity of up to 12 months, excluding credit lines, over total outstanding bank debt. Euro-area non-financial firms rely heavily on short-term debt. On average, approximately 30% of firms' outstanding bank debt matures within twelve months over 2018–2024, with substantial heterogeneity across countries and sectors. Short-term debt is particularly high in Spain, Portugal, Slovenia, and Belgium, as well as in sectors such as mining, trade, wholesale, and manufacturing. In our dataset, Figure 3 shows that 35% of firm–month observations carry no short-term debt at all, while 16% have their entire debt coming due within the year. We exploit this cross-sectional variation in the empirical design: because firms differ in the share of their debt coming due, a given movement in the term structure generates heterogeneous responses across borrowers facing the same credit conditions.

Rollover exposure could in principle proxy for credit quality. Appendix Table 16 compares the difference in means between the bottom and top quartiles of short-term debt. High-exposure firms have lower tangibility and are smaller in terms of fixed assets, but on standard measures of financial health they look, if anything, stronger—they hold lower leverage, substantially higher interest coverage, and indistinguishable sales growth. On observable characteristics, rollover exposure is therefore not a proxy for weaker borrowers; the comparison cannot rule out unobserved differences in credit quality or credit demand, which the fixed-effects design and the legacy-contract measure address only in part; these characteristics are controlled throughout and the legacy-contract mea-

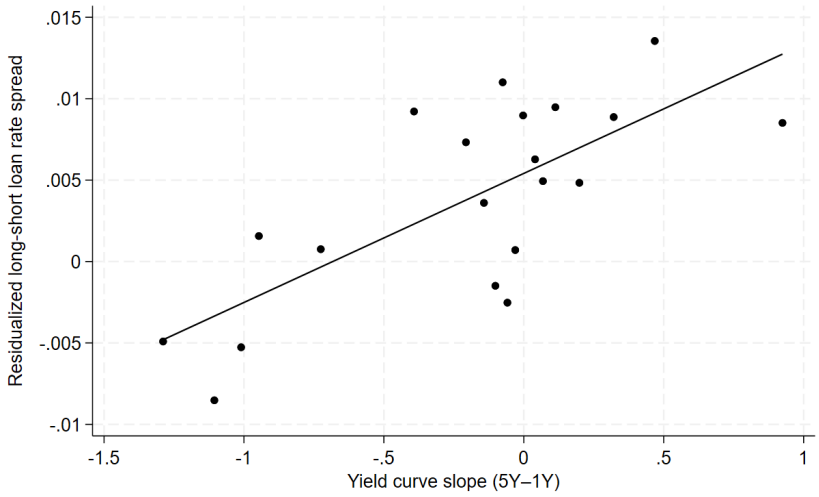


Figure 4: The yield-curve slope and corporate loan spreads

Notes: This figure plots the monthly difference between residualized interest rates on long-term loans (maturity $\geq$ 3 years) and short-term loans (maturity $\leq$ 1 year) against the 5Y–1Y yield-curve slope. Loan rates at origination are residualized using firm and bank fixed effects. For each month, the loan rate spread is calculated as the difference between the average residualized rates on long-term and short-term loan originations. Each dot represents one month, and the solid line shows the linear fitted relationship. Source: AnaCredit, 2018–2024.

sure introduced below addresses the concern that unobserved borrower quality and maturity choice are jointly determined.

Fact 3: The yield-curve slope is strongly associated with loan spreads across maturities. The slope of the yield curve is associated with firms' relative borrowing costs across maturities. Figure 4 shows that a steeper 5Y–1Y yield-curve slope is associated with a larger loan spread between loan rates at origination on long-term loans (with maturity $\geq$ 3 years) and short-term loans (with maturity $\leq$ 1 year). Because the loan rate spread is residualized with respect to firm and bank fixed effects, this relationship reflects time-series variation in relative borrowing costs rather than persistent cross-sectional differences. This is pricing evidence consistent with the term structure affecting the relative prices of long- and short-term borrowing, rather than an estimate of the pass-through of the yield curve into loan spreads, since residualizing on firm and bank fixed effects does not remove time-varying differences in loan composition; it motivates the model's incentive to adjust new maturities[4].

[4]Complementary pricing evidence points the same way. Among newly originated fixed-rate loans, with spreads measured relative to maturity-matched OIS rates, firms with greater rollover exposure pay modestly higher spreads when the long end of the yield curve steepens, consistent with maturity insurance becoming more costly when long-term funding is expensive.

Table 1: Rollover Exposure, the Term Structure, and New Loan Maturity

$\log(\text{Maturity})_{b,f,t}$	(1)	(2)	(3)	(4)	(5)	(6)
ST $\text{Debt}_{f,t-1}$	0.226***	0.233***	0.228***	0.236***	0.252***	0.269***
	(0.0043)	(0.0041)	(0.0049)	(0.0044)	(0.0059)	(0.0064)
YC (5Y–1Y)$_{t-1}\times$ ST $\text{Debt}_{f,t-1}$	-0.0210***	-0.0148**				
	(0.0043)	(0.0053)				
YC (10Y–2Y)$_{t-1}\times$ ST $\text{Debt}_{f,t-1}$				-0.0449***	-0.0812***	
				(0.0050)	(0.0056)	
1Y $\text{Rate}_{t-1}\times$ ST $\text{Debt}_{f,t-1}$			0.129***			
			(0.0055)			
5Y $\text{Rate}_{t-1}\times$ ST $\text{Debt}_{f,t-1}$			-0.050***			
			(0.0068)			
2Y $\text{Rate}_{t-1}\times$ ST $\text{Debt}_{f,t-1}$						0.150***
						(0.0140)
10Y $\text{Rate}_{t-1}\times$ ST $\text{Debt}_{f,t-1}$						-0.095***
						(0.0097)
Observations	1,674,254	1,041,988	1,674,254	1,674,254	1,041,988	1,674,254
Adjusted R^2	0.620	0.620	0.620	0.620	0.620	0.620
Bank × Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm × Bank FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes	Yes	Yes	Yes
Maturity Wall	No	Yes	No	No	Yes	No

Notes: In this table the dependent variable is the logarithm of maturity of newly originated loans. Columns (1)–(3) use the 5Y–1Y yield-curve slope, while columns (4)–(6) use the 10Y–2Y slope as robustness. Columns (1) and (4) report the baseline specification in eq. 1. Columns (2) and (5) replace short-term debt (ST Debt) with the predetermined maturity wall constructed from debt outstanding as of December 2019. Columns (3) and (6) split the corresponding yield-curve slope into its short- and long-rate components. All specifications include lagged firm controls, bank×month fixed effects, and firm×bank fixed effects. Standard errors are clustered at the firm and month level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

3 Empirical Specification and Main Results

The loan-level analysis proceeds in three steps. We first document how the maturity of newly originated loans responds to the slope of the yield curve, allowing the response to vary with firms' lagged rollover exposure. We then exploit high-frequency monetary policy shocks to identify responses to exogenous shifts in the term structure (Altavilla et al., 2019). Finally, we decompose long-term yields into expected future short rates and term premia using standard affine term-structure models (Joslin et al., 2011; Adrian et al., 2013), allowing us to assess which component of the yield curve drives firms' maturity adjustment.

3.1 Rollover-exposed firms lengthen new loan maturity

We begin by estimating at the bank b, firm f, and month t level the following specification:

$$\log(\text{Maturity})_{bft} = \alpha_{bt} + \alpha_{fb} + \beta_1 \text{ST Debt}_{f,t-1} + \beta_2 \left(\text{Yield Curve Slope}_{t-1}^{5Y-1Y} \times \text{ST Debt}_{f,t-1}\right) + \Gamma X_{f,t-1} + \varepsilon_{bft} \tag{1}$$

The dependent variable is the logarithm of the maturity of newly originated loans at the firm–bank–month level, aggregated within each relationship-month using new credit as weights. ST Debt$_{f,t-1}$ is the lagged share of outstanding bank debt with residual maturity of 12 months or less, excluding credit lines. Our baseline measure of the term structure, Yield Curve Slope^{5Y-1Y}, is the difference between the yield on 5-year AAA-rated euro-area government bonds and the 1-year OIS rate. We use the 5-year endpoint because it closely matches the average maturity of newly originated loans in our sample. As a robustness check, we use the corresponding $10Y-2Y$ spread. Bank-by-time fixed effects, α_{bt}, absorb time-varying bank-level credit-supply conditions as well as aggregate shocks (Khwaja and Mian, 2008), while firm-by-bank fixed effects, α_{fb}, absorb persistent relationship characteristics and time-invariant differences in maturity choice across lending relationships. The results are robust to more granular credit-supply controls, including bank–time–industry–location fixed effects (Degryse et al., 2019). The vector $X_{f,t-1}$ includes time-varying firm characteristics such as size, tangibility, the interest coverage ratio, sales growth, and cash over assets.

The first coefficient of interest is β_1, which captures the relationship between rollover exposure and new-loan maturity at the average yield-curve slope.[5] Importantly, rollover exposure does not mechanically imply longer maturity. Greater rollover exposure increases refinancing needs, but firms can meet those needs by issuing new short-term debt without extending maturity. Our dependent variable is the maturity of newly originated loans, conditional on borrowing, rather than the incidence or volume of refinancing. Moreover, firm-by-bank fixed effects absorb persistent differences in maturity across lending relationships, so β_1 is identified from variation within a given firm–bank relationship rather than from exposed firms borrowing from banks that systematically offer longer-maturity credit. The sign of β_1 is therefore a priori ambiguous: firms may lengthen maturity to reduce future refinancing and interest-rate risk, but may instead borrow short if long-duration financing is costly or lenders are unwilling to supply long-term credit. A positive β_1 indicates that greater inherited rollover exposure is associated with longer-maturity new borrowing within the same lending relationship.

The second coefficient of interest is β_2, which measures how the maturity adjustment varies with the term structure. Because the yield-curve slope varies only over time, its direct effect is absorbed by bank-by-month fixed effects[6] . Identification therefore comes from comparing firms borrowing from

[5] We demean the 5Y–1Y yield-curve slope so that β_1 measures this relationship at the average slope rather than at a slope of zero.

[6] Figure 12 complements this evidence by documenting the corresponding aggregate pattern. Average new-loan maturity is strongly positively correlated with the slope of the yield curve. This relationship is descriptive because the term structure co-moves with aggregate macroeconomic and credit conditions. Its role is therefore not identification,

the same bank in the same month, and hence facing the same contemporaneous lender conditions, but entering the period with different inherited rollover exposure. The specification asks whether the same lender allocates relatively longer or shorter maturities to more exposed firms as the term structure changes. Bank-by-month fixed effects also absorb shifts in lenders' aggregate willingness to supply duration. For example, if a bank generally prefers shorter maturities when expected future short rates rise, such a bank-wide response is absorbed by α_{bt} and does not contribute to the identifying variation in β_2. The specification does not rule out banks adjusting their maturity menu differentially across borrowers as a function of rollover exposure. We therefore interpret β_2 as the equilibrium response of the maturity of credit allocated to exposed firms.

A potential concern with the baseline measure of rollover exposure is that short-term debt may itself reflect contemporaneous maturity management. To address this concern, we construct an alternative measure of rollover exposure using only debt contracts outstanding as of December 2019, following the approach of Almeida et al. (2012). For each firm f and month t, we define

$$\text{Maturity Wall}_{f,t}^{2019} = \frac{\text{debt outstanding as of December 2019 scheduled to mature in } [t, t+12]}{\text{total debt outstanding as of December 2019}}$$

This measure fixes the firm's debt portfolio as of December 2019 and uses only the contractual maturity dates of those legacy obligations. It therefore predates both the 2022 monetary-policy tightening cycle and the COVID credit support programs. Although the maturity structure of the December 2019 debt stock reflects earlier financing decisions, its contractual repayment schedule cannot respond to subsequent movements in the yield curve or to contemporaneous credit conditions. Time variation in $\text{Maturity Wall}_{f,t}^{2019}$ is instead generated by the predetermined timing at which these legacy obligations come due. We replace $\text{ST Debt}_{f,t-1}$ in equation (1) with $\text{Maturity Wall}_{f,t}^{2019}$. This specification therefore does not rely on contemporaneous short-term debt being exogenous. Rather, it asks whether firms facing refinancing needs predetermined by their pre-tightening cycle debt contracts adjust the maturity of new borrowing differentially as the term structure changes. The identifying assumption is that, conditional on the fixed effects and controls, the contractual timing of maturities in the December 2019 debt stock is orthogonal to subsequent time-varying shocks that differentially affect maturity choice within a firm–bank relationship.

Table 1 reports the results. Column (1) shows the coefficient on lagged short-term debt is 0.226 and statistically significant. Moving from the 25th to the 75th percentile of rollover exposure—a 50 percentage-point increase in short-term debt in our sample—raises new-loan maturity by approximately 12% ($e^{0.226\times0.5} - 1$), which corresponds to roughly 7.5 months when evaluated at the sample mean maturity of 5.2 years; the conversion to months is an approximation at the mean, not a separately estimated level effect. Thus, firms with more debt coming due do not simply refinance mechanically: conditional on new borrowing, they push the next refinancing date further into the future. Column (4) confirms the finding with the 10Y-2Y yield curve slope. A natural

but to establish that maturity choice moves systematically with the term structure in the aggregate, while equation (1) isolates the differential response associated with inherited rollover exposure.

concern is that contemporaneous rollover exposure may itself reflect recent maturity choices. Firms anticipating future financing conditions could simultaneously alter both the share of debt coming due and the maturity of new borrowing. Columns (2) and (5) therefore replace current rollover exposure with the predetermined maturity wall implied by loans already outstanding as of December 2019. Because the repayment dates of these legacy contracts were fixed before the subsequent tightening cycle, this measure isolates inherited refinancing needs from contemporaneous maturity management. The estimated coefficients remain positive, statistically significant and similar in magnitude to the baseline estimates. Hence, the result survives when exposure is fixed before the tightening: firms facing greater inherited refinancing needs subsequently choose longer maturities on new borrowing. Next, we examine whether this maturity adjustment depends on the term structure of interest rates. In column (1), the interaction between the 5Y–1Y yield-curve slope and rollover exposure is negative and statistically significant: a one percentage point yield curve steepening, when long-term rates are high relative to short-term rates, makes firms with greater rollover exposure extend maturity by 9% less. The same pattern is stronger using the 10Y–2Y slope in column (4), the reduction is about 19%. This is consistent with a trade-off between the benefit of reducing future refinancing needs and the current cost of locking in long-duration funding. Columns (3) and (6) provide a first look at the two sides of this trade-off by entering the short- and long-rate components of each slope separately. In both specifications, higher short rates are associated with greater maturity extension among rollover-exposed firms, whereas higher long rates are associated with less extension. For the 5Y–1Y specification, the interaction coefficients are 0.129 for the 1-year rate and -0.050 for the 5-year rate; the corresponding coefficients for the 2-year and 10-year rates are 0.150 and -0.095. These opposing responses motivate the yield decomposition below, which separates movements in long rates due to expected future short rates from movements due to term premia.

3.2 Future-rate policy news drive maturity extension

Second, a concern with the baseline specification is that movements in the yield curve may reflect macroeconomic and financial conditions that independently affect firms' maturity choices. We therefore exploit monetary-policy surprises measured in narrow windows around ECB announcements using the Euro Area Monetary Policy Event-Study Database of Altavilla et al. (2019). Let X_τ denote the vector of high-frequency changes in risk-free yields around announcement τ. The surprises satisfy

$$X_\tau = \Lambda F_\tau + u_\tau$$

where the factors are extracted by principal component analysis from intraday yield changes and rotated to identify policy news affecting different segments of the term structure[7].

[7]Following Altavilla et al. (2019), the rotation matrix is restricted so that the non-target factors have zero loading on the 1-month OIS surprise. The quantitative-policy factor is further selected, among the admissible orthogonal rotations, to minimize its variance over the pre-crisis period (January 2002–August 2008). The first restriction separates news about future policy from contemporaneous target-rate surprises, while the second identifies a factor

Table 2: High-Frequency Monetary Policy Shocks and New Loan Maturity

$\log(\text{Maturity})_{b,f,t}$	(1)	(2) Pure Monetary Policy	(3) Information
ST Debt$_{f,t-1}$	0.228***	0.229***	0.229***
	(0.00429)	(0.00429)	(0.00428)
Target$_{t-1}\times$ ST Debt$_{f,t-1}$	-0.00778***	-0.0265***	0.000266
	(0.00199)	(0.00531)	(0.00172)
Timing$_{t-1}\times$ ST Debt$_{f,t-1}$	0.0175***	0.0331***	0.00574
	(0.00377)	(0.00640)	(0.00670)
Forward Guidance$_{t-1}\times$ ST Debt$_{f,t-1}$	0.00819***	0.0131***	0.00617***
	(0.00172)	(0.00444)	(0.00205)
Quantitative Policy$_{t-1}\times$ ST Debt$_{f,t-1}$	-0.000678	-0.00801	-0.00278
	(0.00250)	(0.00579)	(0.00301)
Observations	1,674,254	1,674,254	1,674,254
Adjusted R^2	0.620	0.620	0.620
Bank × Month FE	Yes	Yes	Yes
Firm × Bank FE	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes

Notes: The dependent variable is the logarithm of maturity of newly originated loans. ST Debt$_{f,t-1}$ is the firm's lagged short-term debt share, and Shock$_{k,t-1}$ is the cumulated, standardized high-frequency policy-surprise factor from Altavilla et al. (2019). The four factors are Target, Timing, Forward Guidance, and Quantitative Policy. Column (1) uses the high-frequency monetary policy factors directly. Columns (2)–(3) separate the surprises into pure monetary policy and central-bank information components using the high-frequency comovement between short-term OIS rates and equity prices (Jarociński and Karadi, 2020). Shocks are standardized by their monthly standard deviation. All specifications include lagged firm controls, bank×month fixed effects, and firm×bank fixed effects. Standard errors are clustered at the firm and month level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

We obtain four factors with economic interpretation: *Target* captures innovations concentrated at the short end; *Timing* captures revisions to the expected timing of near-term policy changes; *Forward Guidance* captures revisions to the expected policy path at intermediate horizons; and *Quantitative Policy* captures variation associated with asset-purchase policies and longer-horizon rates. Appendix Figure 11 reports the resulting standardized shock series and Appendix Table 19 validates the series. After constructing the shocks, we cumulate announcement-level surprises within calendar months, assigning zero to months without ECB announcements. Then, we estimate:

$$\log(\text{Maturity})_{bft} = \alpha_{bt} + \alpha_{fb} + \beta_1 \,\text{ST Debt}_{f,t-1} + \sum_{k\in\mathcal{K}} \beta_k \left(\text{Shock}_{k,t-1} \times \text{ST Debt}_{f,t-1}\right) + \Gamma X_{f,t-1} + \varepsilon_{bft} \tag{2}$$

that is largely inactive before unconventional monetary policy.

where $\mathcal{K}$ contains the four policy factors. Because the shocks vary only over time, their direct effects are absorbed by bank-by-month fixed effects. Identification of β_k therefore comes from comparing firms borrowing from the same bank in the same month, but entering the period with different inherited rollover exposure, following the same policy surprise. The fixed effects strategy and firm controls are the same as eq. 1. The coefficients β_k measure whether plausibly exogenous policy-induced movements in interest rates generate a differential maturity response among firms with high rollover exposure. The high-frequency identification makes the shocks unlikely to reflect contemporaneous changes in firm-level credit demand or borrower fundamentals. We acknowledge that high-frequency rate surprises may also contain information revealed by the central bank about the economic outlook. Following Jarociński and Karadi (2020), we distinguish pure monetary-policy shocks from central-bank information shocks using the high-frequency co-movement of the 1-month OIS surprise and Euro Stoxx 50 equity prices. Announcements in which interest rates and equity prices move in opposite directions are classified as *pure monetary-policy shocks*; announcements in which they move in the same direction are classified as *central-bank information shocks*. Appendix Figure 11 reports the series, and we complement the above-described analysis.

Table 2 reports the differential response of firms with greater rollover exposure. Column (1) uses the raw factors. Timing shocks generate the largest maturity extension: a one-standard-deviation Timing shock lengthens new-loan maturity by about 0.9% more for firms at the 75th than at the 25th percentile of rollover exposure, or roughly 17 days at the mean maturity. Forward Guidance shocks raise relative maturity by about 0.4%, or 8 days. Target shocks instead reduce it by about 0.4%, or 7 days. This sign difference is consistent with the maturity-insurance mechanism. Target shocks raise near-term refinancing costs precisely when exposed firms must borrow, whereas Timing and Forward Guidance shocks raise expected rates farther ahead, increasing the value of postponing the next refinancing date. The Quantitative Policy interaction is negative but statistically insignificant. Columns (2)–(3) separate pure monetary-policy and central-bank information components following Jarociński and Karadi (2020). The results strengthen for pure policy shocks: the Timing effect rises to about 1.7% (31 days), the Forward Guidance effect to 0.7% (12 days), and the Target effect to -1.3% (25 fewer days), while Quantitative Policy remains insignificant. By contrast, the information components have little differential effect on maturity, apart from a small positive Forward Guidance coefficient. Taken together, the high-frequency evidence reinforces the maturity-insurance interpretation: firms with greater inherited rollover exposure extend maturity primarily when policy news raises expected future rates, whereas unexpected tightening pushes maturity in the opposite direction.

3.3 Expected future rates vs term premia

Third, we decompose long-term yields to distinguish the value and price of maturity insurance. The previous results show that the maturity choices of rollover-exposed firms depend on the term structure of interest rates. However, long-term yields combine two economically distinct compo-

nents with opposing implications for maturity choice. Higher expected future short rates increase anticipated refinancing costs, raising the value of locking in maturity today. By contrast, higher term premia increase the current cost of long-duration funding, raising the price of obtaining maturity insurance. To separate these forces, we use euro-area OIS zero-coupon rates and decompose an n-year yield into an expected-rate component and a term premium as

$$y_t^n = \frac{1}{n}\sum_{j=0}^{n-1}\mathbb{E}_t[r_{t+j}] + \text{Term Premium}_t^n$$

Our baseline decomposition follows Joslin et al. (2011); we use Adrian et al. (2013) as an alternative specification.[8] We use OIS rates because they provide a cleaner measure of the risk-free policy path than sovereign yields. We focus on 5- and 10-year horizons, where the distinction between expected rates and term premia is economically meaningful and term-premium variation is more pronounced. For horizons $n \in \{5, 10\}$, we estimate

$$\begin{aligned}\log(\text{Maturity})_{bft} = \alpha_{bt} + \alpha_{fb} + \beta_1 \text{ST Debt}_{f,t-1} \\ + \beta_{\text{exp}}\left(\text{Expected Rate Path}_{t-1}^n \times \text{ST Debt}_{f,t-1}\right) \\ + \beta_{\text{TP}}\left(\text{Term Premium}_{t-1}^n \times \text{ST Debt}_{f,t-1}\right) + \Gamma X_{f,t-1} + \varepsilon_{bft}\end{aligned} \tag{3}$$

As in the baseline specification, the level effects of the aggregate term-structure components are absorbed by bank-by-month fixed effects, so identification comes from their differential effect across firms with different inherited rollover exposure. The two components have opposite implications for maturity choice. A rise in the expected path of short rates increases the expected cost of refinancing debt that comes due in the future. The value of locking in maturity today therefore rises most for firms with greater refinancing needs, implying $\beta_{\text{exp}} > 0$. By contrast, a rise in the term premium raises the contemporaneous cost of obtaining long-duration financing without a corresponding increase in expected future short rates. Long-term insurance therefore becomes more expensive, implying $\beta_{\text{TP}} < 0$. These opposite sign predictions provide the paper's central mechanism test.

Table 3 separates these two forces. Columns (1)–(2) use the Joslin et al. (2011) decomposition, while Columns (3)–(4) use Adrian et al. (2013). Across both decompositions and at both five- and ten-year horizons, the interaction between rollover exposure and expected rates is positive and statistically significant, whereas the interaction with term premia is negative and significant. The effects are also economically meaningful. A one-standard-deviation increase in the expected-rate component leads firms at the 75th percentile of rollover exposure to choose about 1.5–1.6% longer maturity—roughly 27–30 additional days—relative to firms at the 25th percentile. A comparable

[8]The two approaches recover the same economic objects using different identification strategies. JSZ estimates an affine no-arbitrage term-structure model and separates physical-measure forecasts of future short rates from the risk-premium component of observed long yields. ACM instead uses the predictability of bond excess returns and estimated prices of yield-curve risk to recover the term premium. Appendix Figure 10 and Appendix Table 3 show that the qualitative results are robust across the two decompositions.

Table 3: Yield Decomposition and New Loan Maturity

$\log(\text{Maturity}_{bft})$	(1) 5Y	(2) 10Y	(3) 5Y	(4) 10Y
ST $\text{Debt}_{f,t-1}$	0.179***	0.158***	0.183***	0.157***
	(0.00695)	(0.00828)	(0.00666)	(0.00824)
Expected Rate Path_{t-1} × ST $\text{Debt}_{f,t-1}$	0.0293***	0.0571***	0.0292***	0.0644***
	(0.00313)	(0.00581)	(0.00312)	(0.00639)
Term Premium_{t-1} × ST $\text{Debt}_{f,t-1}$	-0.0382***	-0.0265***	-0.0399***	-0.0297***
	(0.00610)	(0.00440)	(0.00643)	(0.00462)
Observations	1,674,254	1,674,254	1,674,254	1,674,254
Adjusted R^2	0.620	0.620	0.620	0.620
Bank × Month FE	Yes	Yes	Yes	Yes
Firm × Bank FE	Yes	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes	Yes

Notes: The dependent variable is the logarithm of new loan maturity. ST $\text{Debt}_{f,t-1}$ is the share of outstanding bank debt contractually scheduled to mature within twelve months. The Expected Rate Path and Term Premium are obtained from Joslin et al. (2011) in columns (1) and (2) or Adrian et al. (2013) in columns (3) and (4). Columns (1) and (3) use the five-year decomposition, while columns (2) and (4) use the ten-year decomposition. The regressions include bank-by-month and firm-by-bank fixed effects and lagged firm-level controls. Standard errors are clustered at the firm and month level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

increase in the term premium instead reduces relative maturity by about 1.0–1.1%, or 18–19 days. These opposite responses distinguish the value and price of maturity insurance. When expected future short rates rise, firms with greater rollover exposure have a stronger incentive to postpone the date at which they return to the credit market. When term premia rise, long-duration borrowing becomes more expensive today, reducing their incentive to extend maturity. Hence, the maturity response depends not simply on the level of long-term rates but on the source of their movement. The similarity of the estimates across the JSZ and ACM decompositions also indicates that the result is not driven by the choice of affine term-structure model.

We complement the model-based decompositions with direct measures of expected policy rates from the ECB Survey of Professional Forecasters (SPF). For each quarterly survey round q, we construct the average expected policy-rate path over the subsequent year:

$$\text{SPF Expected Path}_q = \frac{E_q i_{3m} + E_q i_{6m} + E_q i_{9m} + E_q i_{12m}}{4}$$

as well as the 12-month-ahead expected rate and the revision in the expected path as $\Delta\text{SPF Expected Path}_q = \text{SPF Expected Path}_q - \text{SPF Expected Path}_{q-1}$. Then, we interact each measure with firms' lagged rollover exposure in the same loan-level specification.

Table 4 shows that higher expected future policy rates are consistently associated with greater maturity extension among rollover-exposed firms. Columns (1)–(2) use the average expected policy-rate

Table 4: ECB Survey of Professional Forecasters (SPF) and New Loan Maturity

$\log(\text{Maturity})_{b,f,t}$	(1)	(2)	(3)
ST $\text{Debt}_{f,t-1}$	0.225***	0.226***	0.224***
	(0.00443)	(0.00438)	(0.00447)
SPF Expected $\text{Path}_{q-1}\times$ ST $\text{Debt}_{f,t-1}$	0.00776***		
	(0.00228)		
SPF 12m $\text{Ahead}_{q-1}\times$ ST $\text{Debt}_{f,t-1}$		0.00682**	
		(0.00298)	
Δ SPF Expected $\text{Path}_{q-1}\times$ ST $\text{Debt}_{f,t-1}$			0.0284***
			(0.00718)
Observations	1,674,254	1,674,254	1,674,254
Adjusted R^2	0.620	0.620	0.620
Bank × Month FE	Yes	Yes	Yes
Firm × Bank FE	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes

Notes: The dependent variable is the logarithm of the maturity of newly originated loans. The ECB Survey of Professional Forecasters (SPF) is used as an alternative measure of expectations. Column (1) interacts firms' rollover exposure with the average expected policy-rate path over the subsequent 12 months. Column (2) uses the SPF 12-month-ahead expected policy rate. Column (3) uses the quarter-to-quarter revision in the expected policy-rate path. All specifications include lagged firm controls, bank×month fixed effects, and firm×bank fixed effects. Standard errors are clustered at the firm and month level. Note that the survey is conducted quarterly. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

path over the subsequent 12 months and the 12-month-ahead expected policy rate, respectively, while Column (3) focuses on revisions in expectations. A one-percentage-point upward revision in the expected path leads firms at the 75th percentile of rollover exposure to choose about 1.4% longer maturity—approximately 26 additional days—relative to firms at the 25th percentile. This specification is especially informative for the mechanism because it links maturity adjustment to news about future policy rates rather than persistent differences in expectations. Since the SPF measures expectations directly, the survey evidence also confirms that the positive expected-rate response is not an artifact of the yield-curve decomposition.

3.4 Robustness

We assess whether the baseline results can be explained by time-varying credit supply, the functional form of the maturity outcome, inflation compensation, or the particular sample period. A first concern is that the interaction between rollover exposure and the yield curve reflects lender behavior rather than borrowers' refinancing needs. The baseline specification already absorbs shocks common to all borrowers of a given bank in a given month through bank×month fixed effects. Column (2) of Appendix Table 18 further replaces these with bank×industry×location×month fixed effects (Degryse et al., 2019). The comparison is therefore among firms borrowing from the same lender, in the same industry and region, in the same month, absorbing credit-supply policies tar-

geted to particular sectors or geographic markets. The interaction remains essentially unchanged, at −0.0219 compared with −0.0210 in the baseline, although the sample declines substantially as more fixed-effect cells become singletons. Column (3) controls directly for total new credit, addressing the possibility that maturity adjustment simply reflects changes in loan size or overall credit availability. The interaction remains negative and similar in magnitude (−0.0175). Column (4) uses the 10Y–2Y slope and yields a larger negative interaction (−0.0812), indicating that the result is not specific to the 5Y–1Y measure of the term structure. Column (5) additionally interacts rollover exposure with the lagged ECB deposit facility rate. The yield-curve interaction remains negative (−0.0180), while the policy-rate interaction is small and statistically insignificant. Column (6) replaces bank×month fixed effects with observable bank controls, while column (7) instead adds firm×year fixed effects, absorbing annual firm-level changes in investment plans, asset composition, and cash-flow expectations. The interaction remains negative and significant in both cases. Taken together, these exercises indicate that the baseline result is not driven by observable lender characteristics, targeted credit supply, or slow-moving changes in firm fundamentals. A second concern is that the baseline log-maturity specification may be influenced by a small number of very long-duration contracts. Appendix Table 22 replaces the logarithm of maturity with indicators for whether a newly originated loan maturity exceeds one, two, three, five, and seven years. Rollover-exposed firms are more likely to borrow above every maturity threshold: the coefficient on short-term debt declines monotonically from 0.076 at the one-year threshold to 0.008 at the seven-year threshold. The interaction with the yield-curve slope is negative and significant at the one-, two-, and three-year thresholds, but small and statistically insignificant at five and seven years. Hence, the baseline interaction is not driven by the extreme upper tail of the maturity distribution; it is concentrated at maturities where most loan contracts are observed. The 2022–2024 period also combined rapid monetary tightening with unusually high inflation, raising the possibility that nominal yield-curve movements capture changes in inflation compensation rather than financing conditions relevant for maturity choice. In Appendix Table 24 we decompose the nominal slope into real-rate and inflation-compensation components following (Gürkaynak et al., 2010). The interaction between rollover exposure and the real slope remains negative and highly significant: −0.079 for the 10Y–2Y slope and −0.071 for the 5Y–1Y slope. The interaction with inflation compensation is also negative and significant. Thus, the negative slope interaction is not solely attributable to inflation compensation; it is present in the real-rate component as well. We next examine whether the result is specific to the 2022–2023 tightening episode. Appendix Table 25 extends the analysis back to 2014 using Orbis balance-sheet data. Because loan-level contractual maturity is not available before 2018, we proxy rollover exposure by current liabilities relative to total debt. This measure is substantially coarser than the AnaCredit measure and includes trade credit and other short-term obligations. Nevertheless, the interaction between the yield-curve slope and rollover exposure remains negative and significant under both the 5Y–1Y and 10Y–2Y definitions. Column (3) further allows both the level effect of rollover exposure and its interaction with the yield curve to differ after 2020. Neither the post-2020 interaction with rollover expo-

sure nor the triple interaction with the yield curve is statistically significant. Thus, although the extended-sample measure is considerably noisier, we find no evidence of a structural break in the maturity response after 2020. Taken together, these results indicate that the mechanism is not specific to the most recent ECB tightening cycle or to the COVID-era sample period. Finally, we assess whether the high-frequency results depend on restricting policy surprises to scheduled ECB Governing Council meetings. The baseline high-frequency analysis uses policy surprises around ECB announcement windows following Altavilla et al. (2019), but this may omit economically relevant policy news communicated between meetings. We therefore repeat the analysis using the extended event-study database of Altavilla et al. (2026), which broadens the event set to include policymakers' intermeeting communication[9]. This extension is particularly useful during periods of frequent policy communication, when restricting measurement to scheduled Governing Council meetings may omit economically relevant policy news. We aggregate the event-level surprises to the monthly frequency and re-estimate equation (2). Appendix A.3 reports the resulting shock series, their loadings across the OIS curve, and the corresponding loan-level maturity estimates. Appendix Table 30 shows that the negative Target response and positive Timing response remain robust in the broader event sample. The Forward Guidance coefficient becomes small and statistically insignificant, while Quantitative Policy shocks are associated with shorter maturity. The key evidence that near-term versus future-rate policy news generates different maturity responses therefore survives the broader event definition.

4 Validating the Mechanism and Alternative Explanations

The baseline results are consistent with firms using debt maturity to manage interest rate and refinancing risk. This section provides additional evidence on that interpretation. We first distinguish rollover risk—the need to refinance principal—from repricing risk—the exposure of outstanding debt to changes in short rates. We then ask whether firms adjust the type of interest rate of new borrowing alongside maturity. Finally, we examine alternative explanations based on asset-maturity matching, borrower risk, liquidity substitution, and selection into new borrowing.

4.1 Rollover Risk, Repricing Risk, and Contract Design

Rollover and repricing risk expose firms to future interest rates through complementary channels. Rollover exposure determines when principal must be refinanced. Repricing exposure instead determines when the interest rate on outstanding debt resets, even if the principal remains outstanding. We measure repricing exposure by the share of outstanding bank debt whose interest rate resets

[9] In this case, *Target* is the 1-month OIS surprise; *Timing* is the component of the 6-month OIS surprise orthogonal to Target; *Forward Guidance* is the component of the 2-year OIS surprise orthogonal to Target and Timing; and *Quantitative Policy* is the residual 10-year surprise orthogonal to the three shorter-horizon components.

within the next 12 months:

$$\text{Reset Share}_{f,t-1} = \frac{\text{Bank debt with an interest-rate reset within 12 months}}{\text{Total bank debt}} \tag{4}$$

Interestingly, the correlation between reset and rollover exposure is only -0.04. Then, we estimate

$$\begin{aligned}\log(\text{Maturity})_{bft} = \alpha_{bt} + \alpha_{fb} + \beta_1 \text{ST Debt}_{f,t-1} + \beta_2 \text{Reset Share}_{f,t-1} \\ + \beta_3 \left(\text{ST Debt}_{f,t-1} \times \text{Reset Share}_{f,t-1}\right) \\ + \beta_4 \left(\text{YC}_{t-1} \times \text{ST Debt}_{f,t-1}\right) + \beta_5 \left(\text{YC}_{t-1} \times \text{Reset Share}_{f,t-1}\right) \\ + \beta_6 \left(\text{YC}_{t-1} \times \text{ST Debt}_{f,t-1} \times \text{Reset Share}_{f,t-1}\right) + \Gamma X_{f,t-1} + \varepsilon_{bft}\end{aligned} \tag{5}$$

Table 5 distinguishes the roles of rollover and repricing exposure. At the average yield curve, rollover exposure continues to predict substantially longer new-loan maturity: the coefficient on Short-Term Debt is 0.229 with the 5Y–1Y slope and 0.236 with the 10Y–2Y slope. Reset Share, by contrast, enters negatively. Importantly, the interaction between Short-Term Debt and Reset Share is small and statistically insignificant in both specifications. Thus, greater repricing exposure does not account for the baseline tendency of firms with more debt coming due to extend maturity.

The key result is the triple interaction with the yield curve. With the 5Y–1Y slope, the marginal effect is only -0.0028 for a firm with no debt repricing within 12 months, but falls to -0.0592 for a firm whose debt fully reprices within the year. Using the 10Y–2Y slope, the corresponding effects are -0.0285 and -0.0740. The triple-interaction coefficients, -0.0564 and -0.0455, are statistically significant at both horizons. Hence, the maturity response of rollover-exposed firms becomes substantially more sensitive to the term structure when their existing liabilities are also exposed to near-term repricing. The magnitudes are economically meaningful. For the 50 percentage-point difference in rollover exposure between the 25th and 75th percentiles, a one-percentage-point steepening of the 5Y–1Y slope reduces relative new-loan maturity by only about 0.1% when Reset Share is zero, but by about 3.0% when Reset Share equals one. For the 10Y–2Y slope, the corresponding effects are about 1.4% and 3.7%. Thus, the price of long-duration funding matters much more for firms that face both near-term refinancing needs and near-term coupon resets.

The distinction between principal maturity and coupon fixation is also visible across newly originated contracts when split by rate type. Appendix Table 21 re-estimates the baseline specification separately by rate type. Rollover exposure predicts longer maturity in both samples, and the yield-curve interaction remains negative. Among fixed-rate originations the borrower has already fixed the coupon, so extending maturity primarily postpones the next principal refinancing date.

Finally, we find evidence that firms adjust the rate structure of new borrowing. Appendix Table 17 uses the floating-rate share of newly originated loans as the dependent variable. The interaction between the yield curve and rollover exposure is negative under both slope measures. For the 5Y–1Y slope, a one percentage-point steepening reduces the floating-rate share of new borrowing by about 0.6 percentage points more for a firm at the 75th than at the 25th percentile of rollover exposure;

Table 5: Rollover Exposure, Repricing Exposure, and New Loan Maturity

$\log(\text{Maturity})_{b,f,t}$	(1) 5Y–1Y	(2) 10Y–2Y
ST $\text{Debt}_{f,t-1}$	0.229***	0.236***
	(0.00549)	(0.00563)
Reset $\text{Share}_{f,t-1}$	-0.0205***	-0.0284***
	(0.00777)	(0.00794)
ST $\text{Debt}_{f,t-1}\times$ Reset $\text{Share}_{f,t-1}$	-0.0128	-0.00215
	(0.0108)	(0.0110)
$\text{YC}_{t-1}\times$ ST $\text{Debt}_{f,t-1}$	-0.00282	-0.0285***
	(0.00599)	(0.00708)
$\text{YC}_{t-1}\times$ Reset $\text{Share}_{f,t-1}$	0.0312***	0.0343***
	(0.00835)	(0.00993)
$\text{YC}_{t-1}\times$ ST $\text{Debt}_{f,t-1}\times$ Reset $\text{Share}_{f,t-1}$	-0.0564***	-0.0455***
	(0.0133)	(0.0152)
Observations	1,659,220	1,659,220
Adjusted R^2	0.617	0.617
Bank × Month FE	Yes	Yes
Firm × Bank FE	Yes	Yes
Firm Controls	Yes	Yes

Notes: The dependent variable is the logarithm of the maturity of newly originated loans. Rollover exposure is proxied by Short-term Debt, which is calculated as the share of outstanding bank debt maturing within 12 months. Repricing exposure is proxied by the Reset Share, which is the share of outstanding bank debt whose interest rate resets within 12 months. Column (1) uses the 5Y–1Y yield-curve slope and column (2) uses the 10Y–2Y slope as robustness. All specifications include lagged firm controls, bank×month fixed effects, and firm×bank fixed effects. Standard errors are clustered at the firm and month level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

the corresponding 10Y–2Y effect is about 0.7 percentage points. Thus, when long-duration funding becomes relatively more expensive, rollover-exposed firms adjust both the maturity and the rate structure of new borrowing. The two margins jointly determine their exposure to future financing conditions, but the maturity insurance channel is not only about rates locking.

4.2 Alternative Explanations

Next, we examine whether the maturity response can instead be explained by asset-maturity matching, borrower risk, substitution toward committed credit lines, or selection into new borrowing.

Asset-maturity matching. One possibility is that firms choose longer debt simply because the assets being financed have longer duration. Table 6 re-estimates the baseline specification across loan-purpose subsamples. The yield-curve interaction remains negative after excluding real-estate loans and after excluding working-capital loans. In the investment-only sample the point estimate is close to zero. More directly, column (7) of Appendix Table 18 includes firm×year fixed

Table 6: New Loan Maturity, Loan Purpose and Maturity Matching

$\log(\text{Maturity})_{b,f,t}$	(1) Full Sample	(2) No Real Estate	(3) No Working Capital	(4) Investment Only
ST $\text{Debt}_{f,t-1}$	0.179***	0.105***	0.153***	0.291***
	(0.0038)	(0.0047)	(0.0046)	(0.0527)
YC (5Y–1Y)$_{t-1}\times$ ST $\text{Debt}_{f,t-1}$	-0.0359***	-0.0183***	-0.0190***	-0.0003
	(0.0042)	(0.0052)	(0.0049)	(0.0613)
Observations	2,030,234	1,131,837	833,305	20,480
Adjusted R^2	0.638	0.724	0.716	0.630
Bank × Month FE	Yes	Yes	Yes	Yes
Firm × Bank FE	Yes	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes	Yes

Notes: The dependent variable is the log maturity of newly originated loans. Column (1) reports the full sample. Note that the number of observations is different from the baseline regression because we include only tangibility, size, and leverage as firm controls to maximize the number of observations. Column (2) excludes loans with a real-estate purpose, column (3) excludes working-capital loans, and column (4) restricts the sample to loans declared to be used for investment purposes. All specifications include lagged firm controls, bank×month fixed effects, and firm×bank fixed effects. Standard errors are clustered at the firm and month level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

effects, which absorb annual changes in investment plans and asset composition. The yield-curve interaction remains -0.0163, close to the baseline estimate. These results provide little support for asset-maturity matching as the source of the exposure gradient.

Borrower risk. A Diamond (1991) mechanism predicts that riskier firms may extend maturity to lock in funding before their expected credit quality deteriorates. If this alternative mechanism explains the baseline results, the sensitivity of maturity to the yield curve should be stronger among firms with higher default risk. To test this prediction, we augment the baseline specification with the probability of default assigned by the lending bank and interact it with both the yield curve and short-term debt exposure. Table 7 controls for the lender-assigned probability of default and allows the entire maturity response to vary with default risk. The yield-curve interaction with rollover exposure remains negative and statistically significant (-0.0248), while the triple interaction with probability of default is statistically insignificant. Borrower risk therefore affects maturity levels but does not account for the differential term-structure response of rollover-exposed firms.

New bank credit. An alternative explanation is that rollover-exposed firms adjust the quantity of new borrowing, rather than its maturity, when financing conditions change. Appendix Table 28 examines this possibility using the volume of newly originated credit as the dependent variable. While firms with greater rollover exposure obtain more new credit on average, the interaction between rollover exposure and the yield curve is small and statistically insignificant for both total new credit and term loans. These results suggest that the differential maturity response is not accompanied by a corresponding adjustment in new credit volumes, conditional on origination. The evidence is therefore consistent with firms adjusting the contractual maturity of new borrowing rather than simply changing its amount.

Table 7: Rollover Exposure and Borrower Riskiness

$\log(\text{Maturity})_{b,f,t}$	(1)	(2)
ST $\text{Debt}_{f,t-1}$	0.192***	0.202***
	(0.00782)	(0.00804)
YC (5Y–1Y)$_{t-1}\times$ ST $\text{Debt}_{f,t-1}$	-0.0170**	-0.0248***
	(0.00785)	(0.00833)
$\text{PD}_{f,t-1}$	0.160***	0.252***
	(0.0311)	(0.0452)
YC (5Y–1Y)$_{t-1}\times$ $\text{PD}_{f,t-1}$		-0.126*
		(0.0748)
ST $\text{Debt}_{f,t-1}\times$ $\text{PD}_{f,t-1}$		-0.251***
		(0.0587)
YC (5Y–1Y)$_{t-1}\times$ ST $\text{Debt}_{f,t-1}\times$ $\text{PD}_{f,t-1}$		0.149
		(0.109)
Observations	559,105	559,105
Adjusted R^2	0.642	0.642
Bank × Month FE	Yes	Yes
Firm × Bank FE	Yes	Yes
Firm Controls	Yes	Yes

Notes: The dependent variable is the logarithm of maturity of newly originated loans. Borrower riskiness is proxied by the firm's lagged probability of default (PD), which is assigned by the lender. Column (1) controls for borrower risk directly, while column (2) allows the relationship between rollover exposure, the yield curve, and loan maturity to vary with borrower risk. All specifications include lagged firm controls, bank×month fixed effects, and firm×bank fixed effects. Standard errors are clustered at the firm and month level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Committed credit lines. Committed credit lines could provide an alternative way to insure against future refinancing needs. If they substitute for maturity insurance, exposed firms should increase their reliance on committed lines when long-duration borrowing becomes expensive. Column (6) of Table 8 provides no evidence of such substitution. The interaction between the yield curve and rollover exposure is negative, indicating that exposed firms do not offset shorter maturity by expanding committed credit lines. Hence, the baseline maturity response is not explained by substitution toward precommitted liquidity.

Pledged collateral. Firms may also respond to refinancing pressure by adjusting the amount of protection pledged to lenders rather than the maturity of new borrowing. Appendix Table 20 examines this alternative margin using the reported value of protection associated with bank credit. Firms with greater rollover exposure pledge significantly more protection on average. However, the interaction between rollover exposure and the yield curve is economically negligible and statistically insignificant. Thus, although collateral and other forms of protection vary systematically among firms, we find no evidence that exposed firms adjust pledged protection in response to changes in the relative cost of long-duration funding.

Selection into new borrowing. Finally, the maturity regressions condition on firms obtaining new credit. Changes in the composition of borrowers could therefore generate the estimated response if rollover-exposed firms selectively exit the new-loan sample when the yield curve changes. Appendix Table 23 estimates firm-month linear probability models for obtaining a new loan. The evidence goes in the opposite direction. A 10 percentage-point increase in rollover exposure raises the monthly probability of new borrowing by about 0.7 percentage points, and the interaction between rollover exposure and the yield curve is positive. Thus, rollover-exposed firms become more, rather than less, likely to appear in the new-loan sample when the curve steepens. These extensive-margin results do not support the specific hypothesis that exposed firms systematically exit the origination sample; a higher probability of borrowing does not by itself rule out changes in the characteristics of the exposed firms that do borrow, which the borrower-risk and loan-purpose tests above address. Overall, the evidence supports an active liability-management interpretation. Rollover and repricing exposure have distinct effects, firms adjust both maturity and coupon structure, and the results are not explained by asset-maturity matching, borrower risk, liquidity substitution, or selection into borrowing.

5 Real Effects

Does maturity adjustment affect firms' balance sheets and real activity? Financing choices are the transmission margin; real outcomes reveal whether that margin matters. In this section, we show that exposed firms rebalance toward longer-term bank debt but do so less when the yield curve steepens; that they then face broader balance-sheet tightening and higher debt-service pressure; and that investment declines are concentrated among firms that had *not* previously obtained maturity insurance, especially small ones.

5.1 Empirical Specification

We estimate at the firm-annual frequency:

$$\Delta Y_{f,t} = \alpha_f + \alpha_{c,s,t} + \beta_1 \text{ST Debt}_{f,t-1} + \beta_2 \left(\text{YC}_{t-1} \times \text{ST Debt}_{f,t-1}\right) + \Gamma X_{f,t-1} + \varepsilon_{f,t} \tag{6}$$

where $\Delta Y_{f,t}$ is the change between $t-1$ and t in a firm-level outcome, $\text{ST Debt}_{f,t-1}$ is the share of bank loans maturing within 12 months, and YC_{t-1} is the lagged 5Y–1Y slope. Firm fixed effects and country–sector–year fixed effects absorb time-invariant heterogeneity and shocks common to firms in the same country–sector–year cell; time-varying firm controls $X_{f,t-1}$ includes total assets, tangibility, interest coverage, sales growth, and cash over assets. The coefficient of interest is β_2: negative for borrowing and investment outcomes means exposed firms contract more when the curve steepens; positive for maturity and long-term-debt outcomes means they shift toward longer liabilities.

Table 8: Rollover Exposure, Firms' Balance Sheet, and the Term Structure

Panel A. Liability Restructuring

	(1) Δ Bank-Debt Maturity	(2) Δ Long Term Bank Loans	(3) Δ Bank Debt	(4) Δ Total Debt
ST Debt$_{f,t-1}$	2.219***	0.792***	0.295***	-0.0483***
	(0.0345)	(0.00583)	(0.00637)	(0.00193)
YC (5Y–1Y)$_{t-1}$× ST Debt$_{f,t-1}$	-0.335***	-0.654***	-0.0339***	-0.0466***
	(0.0554)	(0.00890)	(0.00703)	(0.00578)
Observations	594,168	594,168	594,168	594,168
Adjusted R^2	0.0506	0.0357	0.0416	0.107
Industry × Country × Time FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes	Yes

Panel B. Liquidity and Interest Expenses

	(5) Δ Cash /TA	(6) Δ Credit Lines	(7) Interest Exp. / EBITDA	(8) Debt-Service Ratio
ST Debt$_{f,t-1}$	0.153***	0.00287	-0.00111	-0.00589***
	(0.0405)	(0.00507)	(0.0008)	(0.000495)
YC (5Y–1Y)$_{t-1}$× ST Debt$_{f,t-1}$	-0.444***	-0.0179**	0.0151***	0.0512***
	(0.115)	(0.00739)	(0.0024)	(0.00156)
Observations	594,168	69,785	594,168	594,168
Adjusted R^2	0.0297	0.0368	0.673	0.822
Industry × Country × Time FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes	Yes

Notes: This table reports firm-level estimates of how rollover exposure interacts with the term structure to affect firms' total liability, liquidity, and debt-service burden. Panel A reports changes in bank-debt maturity, the share of long-term bank loans, bank debt, and total debt. Panel B reports cash growth, changes in committed credit lines, interest expenses relative to EBITDA, and the debt-service ratio. Short-term Debt is the lagged share of outstanding bank debt with residual maturity within 12 months. YC (5Y–1Y) is the lagged difference between the 5-year AAA euro-area government bond yield and the 1-year OIS rate. All specifications include lagged firm controls, firm fixed effects, and industry×country×time fixed effects. Standard errors are clustered at the firm level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

5.2 Liability Restructuring and Debt-Service Pressure

Table 8 shows that rollover exposure is associated with substantial balance-sheet adjustment. Panel A documents liability restructuring. Moving from the 25th to the 75th percentile of rollover exposure—a 20 percentage-point increase in this sample—raises average bank-debt maturity by about five months and the share of long-term bank loans by 15.8 percentage points. Bank debt increases by about 5.9%, while total debt declines by about 1.0 percentage point. Thus, exposed firms shift their liability structure toward longer-term bank financing without increasing overall leverage. This adjustment weakens when the yield curve steepens. A one percentage-point increase in the 5Y–1Y slope reduces the maturity extension by about one month and the increase in the long-term bank-loan share by 13.1 percentage points. It also lowers bank debt and total debt, indicating that firms do not fully replace the lost maturity adjustment by expanding other types borrowing. Panel B shows that the same firms experience tighter liquidity and greater debt-service pressure.

A one percentage-point steepening reduces cash growth by about 8.9 percentage points for firms at the 75th relative to the 25th percentile of rollover exposure. At the same time, interest expenses relative to EBITDA rise by about 0.30 percentage points and the debt-service ratio rises by about 1.0 percentage point. Credit-line commitments also decline rather than expand. Taken together, the decline in debt and cash alongside higher debt-service burdens indicates broader balance-sheet tightening, rather than a simple reshuffling across liability categories.

5.3 Investment and Maturity Adjustment

Investment is a demanding outcome, since it responds directly to demand expectations and to information revealed by policy. We therefore run a sharper test: do firms that had already reduced rollover exposure experience smaller investment declines when the curve steepens? We estimate

$$\begin{aligned}\text{Investment}_{f,t} = \alpha_f + \alpha_{c,s,t} + \beta_1 \text{ST Debt}_{f,t-1} + \beta_2 \left(\text{YC}_{t-1} \times \text{ST Debt}_{f,t-1}\right) \\ + \beta_3 \text{Adjust}_{f,t-1} + \beta_4 \left(\text{YC}_{t-1} \times \text{Adjust}_{f,t-1}\right) + \beta_5 \left(\text{ST Debt}_{f,t-1} \times \text{Adjust}_{f,t-1}\right) \\ + \beta_6 \left(\text{YC}_{t-1} \times \text{ST Debt}_{f,t-1} \times \text{Adjust}_{f,t-1}\right) + \Gamma X_{f,t-1} + \varepsilon_{f,t}\end{aligned} \tag{7}$$

$\text{Adjust}_{f,t-1}$ captures predetermined liability adjustment, measured in two ways: an indicator dummy for having increased debt maturity in the previous year, $t-1$, or the lagged change in long-term bank loans. We expect $\beta_2 < 0$ and $\beta_5 > 0$—rollover exposed firms cut investment more when the curve steepens, but less so if they had already lengthened. Table 9 supports both predictions. Because maturity adjustment is lagged, the result is not a mechanical contemporaneous relation—but maturity adjustment is itself a choice, so we read these estimates as evidence on where the real effects of rollover exposure are concentrated, not as the causal effect of maturity extension. The model counterfactual in Section 8 provides the clean answer to this question.

Investment declines concentrate among non-adjusters. In column (2) the slope–exposure interaction is -0.150 and the triple interaction with the maturity-extension indicator is $+0.149$: among firms that had *not* extended maturity, a 1 percentage point steepening lowers investment by about 3.0 percentage points more at the 75th than at the 25th percentile of rollover exposure, while among firms that had extended, this differential is almost fully offset. Column (3) reproduces the pattern using the lagged change in long-term bank loans.

Firm size. Finally, we examine how the channel varies with firm size. We define small firms as those in the bottom quartile of total assets and interact this indicator with rollover exposure and the yield-curve slope, with the outcomes of Tables 1 and 8 (Appendix Table 27). The maturity response is substantially stronger for small firms: when the yield curve steepens, small rollover-exposed firms shorten new-loan maturity more than twice as much as larger firms with the same exposure, and the reductions in bank-debt maturity and in the share of long-term bank loans are 66% and 40% larger. Their interest burden also rises more. The investment response of rollover-exposed firms is

Table 9: Rollover Exposure, Investment and Maturity Adjustment

$\text{Investment}_{f,t}$	(1)	(2)	(3)
$\text{ST Debt}_{f,t-1}$	0.0528***	0.00747	-0.00783
	(0.00454)	(0.00531)	(0.00834)
Yield Curve 5Y–1Y$_{t-1}\times$ $\text{ST Debt}_{f,t-1}$	-0.0404***	-0.150***	-0.0502**
	(0.0124)	(0.0192)	(0.0222)
$\text{Extension Dummy}_{f,t-1}$		-0.195***	
		(0.00220)	
$\text{Short-term Debt}_{f,t-1}\times$ $\text{Extension Dummy}_{f,t-1}$		0.0392***	
		(0.00636)	
YC 5Y–1Y$_{t-1}\times$ $\text{Extension Dummy}_{f,t-1}$		-0.0703***	
		(0.00862)	
YC 5Y–1Y$_{t-1}\times$ $\text{Short-term Debt}_{f,t-1}\times$ $\text{Extension Dummy}_{f,t-1}$		0.149***	
		(0.0261)	
Δ $\text{Long Term Bank Loans}_{f,t-1}$			0.0584***
			(0.00330)
$\text{Short-term Debt}_{f,t-1} \times \Delta$ $\text{LT Bank Loans}_{f,t-1}$			0.0495***
			(0.00909)
YC 5Y–1Y$_{t-1} \times \Delta$ $\text{LT Bank Loans}_{f,t-1}$			0.0274
			(0.0174)
YC 5Y–1Y$_{t-1}\times$ $\text{Short-term Debt}_{f,t-1} \times \Delta$ $\text{LT Bank Loans}_{f,t}$			0.113**
			(0.0503)
Observations	594,168	594,168	594,168
Adjusted R^2	0.134	0.146	0.168
Industry-Country-Time FE	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes

Notes: The dependent variable is the firm investment rate in fixed assets. Short-term debt is the share of outstanding bank debt with residual maturity of 12 months. The yield curve is the 5Y–1Y slope. The extension dummy equals one when the firm increased debt maturity in the previous year. Δ LT Bank Loans is the change in long-term bank loans. All specifications include firm fixed effects, industryxcountryxtime fixed effects and firm controls. Standard errors clustered at the firm level are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

about twice as large among small firms.

Taking stock:

Four findings summarize the empirical analysis. *First*, firms actively manage interest rate and refinancing risk through debt maturity. Moving from the 25th to the 75th percentile of rollover exposure increases new-loan maturity by about 7.5 months, and the effect persists when exposure is measured using repayment schedules fixed before the tightening cycle. *Second*, the response depends on the source of interest-rate movements. Higher expected future short rates induce exposed firms to lengthen maturity, whereas higher term premia induce them to shorten it. High-frequency ECB shocks reinforce this distinction: news about higher future policy rates lengthens

maturity, while unexpected tightening concentrated at the current policy rate compresses it. *Third*, maturity adjustment shapes firms' balance-sheet and real responses. Exposed firms shift toward longer-term bank debt, but this adjustment weakens when long-duration funding becomes relatively costly. Firms that do not extend maturity subsequently experience larger investment declines, with stronger effects among smaller firms. *Fourth*, the evidence is not explained by asset-maturity matching, borrower risk, unusually long contracts, substitution toward committed credit lines, or selection into new borrowing.

6 Quantitative Model

The empirical analysis shows that firms with greater rollover exposure adjust their debt maturity differently as aggregate monetary conditions change. These cross-sectional responses provide evidence of the maturity insurance channel but do not establish its aggregate consequences for monetary transmission. We develop a quantitative model to assess how endogenous maturity adjustment shapes the responses of debt, investment and default to monetary shocks and to quantify the aggregate importance of this channel. The model is annual and builds on quantitative models of firm financing with financial frictions (Cooley and Quadrini, 2001; Gomes, 2001; Hennessy and Whited, 2005; Khan and Thomas, 2013) and endogenous debt maturity (Jungherr and Schott, 2021; Deng and Fang, 2022; Poeschl, 2023; Jungherr et al., 2024). Firms differ in productivity, capital, debt, and inherited repayment obligations. Aggregate financing conditions are characterized by the current short rate, news about future short rates, and the relative cost of long-duration funding.

6.1 Environment and aggregate states

Time is discrete, and the horizon is infinite. Firm i produces using capital (K) according to

$$\pi(z_{it}, K_{it}) = \bar{z} z_{it} K_{it}^{\alpha} - F, \qquad \alpha \in (0,1) \tag{8}$$

where z_{it} is firm-level productivity, $\bar{z}$ scales the level of production, and F is a fixed operating cost. Decreasing returns imply a well-defined optimal firm size, while the fixed cost generates negative operating cash flow at sufficiently low productivity, increasing firms' exposure to adverse productivity realizations.

Firm-level productivity follows an AR(1) process:

$$\log z_{i,t+1} = \rho_z \log z_{it} + \sigma_z \varepsilon^z_{i,t+1}, \qquad \varepsilon^z_{i,t+1} \sim \mathcal{N}(0,1) \tag{9}$$

Capital evolves according to

$$K_{i,t+1} = (1 - \delta_K) K_{it} + I_{it} \tag{10}$$

where I_{it} denotes gross investment and δ_K denotes the depreciation rate. Investment is subject to

the convex adjustment cost:

$$\Phi_K(I_{it}, K_{it}) = \frac{\psi_K}{2}\left(\frac{I_{it}}{K_{it}}\right)^2 K_{it} \tag{11}$$

Adjustment costs make capital accumulation gradual and link current investment to the continuation value of installed capital. Investment is fully reversible: capital trades at a unit price in both directions, with disinvestment limited only by the convex cost and a minimum operating scale $K_{i,t+1} \geq \underline{K}$. This feature gives constrained firms an additional liquidity margin: capital sales can substitute for external refinancing when debt comes due, potentially attenuating the maturity-insurance mechanism.[10]

Aggregate financing conditions are summarized by

$$A_t = (r_t, f_t, \omega_t) \tag{12}$$

where r_t is the current short rate, f_t carries news about future short rates, and ω_t captures the relative cost of supplying long-duration credit. The distinction between current-rate and future-policy-path innovations follows Gürkaynak et al. (2005). Long-term yields also reflect term premia, which can vary for reasons other than revisions to expected future short rates. We represent this additional source of variation through a persistent term-premium wedge ω_t. The three states follow the following processes:

$$r_{t+1} = (1-\rho_r)\bar{r} + \rho_r r_t + f_t + \sigma_r \varepsilon^r_{t+1} \tag{13}$$

$$f_{t+1} = \rho_f f_t + \sigma_f \varepsilon^f_{t+1} \tag{14}$$

$$\omega_{t+1} = (1-\rho_\omega)\bar{\omega} + \rho_\omega \omega_t + \sigma_\omega \varepsilon^\omega_{t+1} \tag{15}$$

where the innovations are mutually independent standard normals.[11]

The short rate determines the risk-free discount rate for one-period debt. A one-unit increase in f_t raises $\mathbb{E}_t[r_{t+1}]$ one for one without changing the current short rate, with longer-horizon effects governed by ρ_r and ρ_f. An increase in ω_t reduces the proceeds from newly issued long-term debt without directly changing the expected short-rate path. We refer to ω_t as a term-premium wedge: it is an exogenous term-premium component of long-term issuance prices rather than a premium generated by the pricing kernel, and it is distinct from the compensation for default risk and recovery that enters the reference price. Its process is disciplined by movements in the empirical term-premium component of long rates, as detailed in Appendix D.3.[12]

[10] In the simulated stationary economy, capital sales finance approximately 12% of principal coming due across the rollover-exposure distribution. Firms default only after liquidating capital to the minimum operating scale.

[11] In practice, expected-rate news and changes in duration-funding conditions need not be independent. We specify independent innovations to isolate the effects of each financing component. Section 8.3 examines their joint effects by combining expected-rate news with a deterioration in the term-premium wedge.

[12] A structural term premium would instead arise from how the stochastic discount factor values long-duration cash flows. Preferred-habitat models show how maturity-specific demand and limited risk-bearing capacity generate variation in the price of duration (Vayanos and Vila, 2021). Related empirical evidence links bond supply to yields and expected returns (Greenwood and Vayanos, 2014).

6.2 Debt structure and rollover exposure

Firm i enters period t with short-term debt B_{it}^S and long-term debt B_{it}^L. Short-term debt matures fully within the period. Long-term debt pays a fixed coupon c_L and amortizes geometrically: a fraction $1-\rho_L$ of principal matures each period and a fraction ρ_L survives. Geometric amortization provides a tractable representation of long-term debt (Leland, 1994; Chatterjee and Eyigungor, 2012). Because all long-term claims share the same coupon and amortization rate, their remaining promised payments are proportional to face value regardless of issuance vintage.

The inherited debt structure can therefore be summarized by total face value (B) and principal due within the period (G):

$$B_{it} = B_{it}^S + B_{it}^L, \qquad G_{it} \equiv B_{it}^S + (1-\rho_L)B_{it}^L \tag{16}$$

The pair (B_{it}, G_{it}) is a sufficient statistic for the inherited debt structure, with feasible states satisfying[13]

$$(1-\rho_L)B_{it} \leq G_{it} \leq B_{it} \tag{17}$$

The lower bound corresponds to a firm financed entirely with long-term debt, whereas the upper bound corresponds to a firm financed entirely with short-term debt.

For firms with positive debt, define rollover exposure (s) as

$$s_{it} \equiv \frac{G_{it}}{B_{it}} \tag{18}$$

Rollover exposure summarizes the concentration of contractual principal repayments in the current period. Holding total debt fixed, a higher s_{it} implies that a larger fraction of outstanding principal must be met from current cash flow, disinvestment, or new borrowing. Thus, when external finance cannot be raised costlessly, the timing of repayment obligations matters independently of the level of debt. This intertemporal liquidity motive is analogous to the broader insurance principle that liquidity constraints increase the value of smoothing concentrated payment obligations across periods (Ericson and Sydnor, 2026). The firm state can be summarized as

$$x_{it} = (z_{it}, K_{it}, B_{it}, G_{it}) \tag{19}$$

Conditional on continuation, the firm chooses next-period capital $K_{i,t+1}$, gross new borrowing

[13] The bounds follow directly from the inversion of (16). Subtracting the two equations gives

$$B_{it} - G_{it} = \rho_L B_{it}^L$$

so that

$$B_{it}^L = \frac{B_{it} - G_{it}}{\rho_L}, \qquad B_{it}^S = \frac{G_{it} - (1-\rho_L)B_{it}}{\rho_L}$$

Since both debt components must be nonnegative, $B_{it}^L \geq 0$ implies $G_{it} \leq B_{it}$, while $B_{it}^S \geq 0$ implies $G_{it} \geq (1-\rho_L)B_{it}$. Hence (B_{it}, G_{it}) is feasible if and only if $(1-\rho_L)B_{it} \leq G_{it} \leq B_{it}$.

$m_{it} \geq 0$, and the long-term share of new borrowing $\ell_{it} \in [0,1]$. Because short-term debt matures fully within the period, next period's short-term debt consists entirely of new short-term borrowing, while next period's long-term debt combines surviving long-term claims with new long-term borrowing:

$$B^S_{i,t+1} = (1-\ell_{it})m_{it}, \qquad B^L_{i,t+1} = \rho_L B^L_{it} + \ell_{it} m_{it}. \tag{20}$$

The benchmark model abstracts from voluntary prepayment of surviving long-term debt [14].

Debt evolves according to

$$B_{i,t+1} = B_{it} - G_{it} + m_{it}, \qquad G_{i,t+1} = (1-\rho_L)B_{i,t+1} + \rho_L(1-\ell_{it})m_{it} \tag{21}$$

Two properties of (21) summarize the roles of borrowing and maturity choice. First, $B_{i,t+1} - B_{it} = m_{it} - G_{it}$, so gross new borrowing in excess of principal coming due equals net debt growth. Second, holding gross issuance fixed, $\frac{\partial G_{i,t+1}}{\partial \ell_{it}}\Big| m_{it} = -\rho_L m_{it} < 0$. This is the sense in which long-term debt provides refinancing insurance: for a given amount of new borrowing, allocating a larger share to long maturity reduces the principal that must be refinanced next period, which is especially valuable when refinancing is costly because of high rates, low productivity, or weak internal funds.

6.3 Firm problem

Timing. Within each period, the aggregate financing state and firm productivity are realized, and the firm produces using its installed capital. Inherited principal and coupon obligations become due. Competitive lenders quote state-contingent price schedules for newly issued short- and long-term debt, reflecting the repayment risk associated with each feasible financing choice. The firm then chooses next-period capital, gross borrowing, and the maturity composition of new debt. If no feasible combination of investment and financing allows the firm to meet its obligations, it defaults. Otherwise, it issues new debt, repays inherited obligations, undertakes investment, distributes any residual cash flow, and transitions to the next period.

Continuation value. The time-varying component of discounting is common to shareholders and competitive lenders:

$$\mathcal{M}_{t,t+1} = e^{-r_t} \tag{22}$$

where r_t is the continuously compounded one-period risk-free rate. Shareholders additionally discount continuation payoffs by the time-invariant factor $\beta_F \equiv 1-\kappa$, whereas lenders discount using $\mathcal{M}_{t,t+1}$ alone. Following Salomão and Varela (2022), we interpret $\beta_F < 1$ as a equity-discounting wedge that captures forces such as the tax advantage of debt or ownership turnover, which shorten

[14] The empirical analysis includes all newly originated loans, including those associated with refinancing existing debt. The benchmark model, however, abstracts from voluntary prepayment of surviving long-term claims. Early refinancing is a distinct margin of maturity management: Mian and Santos (2018) show that firms refinance bank loans before maturity when credit conditions are favorable, allowing them to extend effective maturity and reduce future refinancing risk.

shareholders' effective valuation horizon relative to creditors. This makes current borrowing proceeds relatively valuable to shareholders and is the model's principal motive for positive leverage[15].

Conditional on continuation, the firm solves

$$V^C(x;A) = \max_{(K',m,\ell)\in\mathcal{F}(x;A)} \left\{ d + (1-\kappa)\mathbb{E}_t \left[\mathcal{M}_{t,t+1} V(x';A') \right] \right\} \tag{23}$$

The firm values current distributions and future shareholder payoffs, taking into account how today's investment and financing decisions determine next period's capital, repayment obligations, and exposure to default risk. For firm state x and aggregate state A, define the feasible continuation set as

$$\mathcal{F}(x;A) = \left\{ (K',m,\ell) : K' \geq \underline{K},\ m \geq 0,\ \ell \in [0,1],\ d \geq 0 \right\} \tag{24}$$

Payout and financing constraints. Using the identity $B^L_{it} = (B_{it} - G_{it})/\rho_L$, shareholder payout is

$$d_{it} = \pi(z_{it}, K_{it}) - G_{it} - c_L \frac{B_{it} - G_{it}}{\rho_L} - I_{it} - \Phi_K(I_{it}, K_{it}) - H(m_{it}) + m_{it} \left[(1-\ell_{it}) Q^S_{it} + \ell_{it} Q^L_{it} \right] \tag{25}$$

Operating cash flow finances inherited principal and coupon payments, investment, capital adjustment costs, and issuance costs. The final term is the proceeds from new short- and long-term borrowing. The prices Q^S_{it} and Q^L_{it} depend on the firm's state and financing choices, as specified in Section 6.4. Note that increasing the face value of new borrowing does not generate proportional additional proceeds; in fact higher indebtedness can raise default risk and reduce debt prices.

Gross debt issuance entails the quadratic convex cost

$$H(m_{it}) = \frac{\eta}{2} m^2_{it} \tag{26}$$

which captures increasing marginal fees of raising new bank credit.[16] The cost depends on total gross issuance and is maturity-neutral: conditional on new borrowing m_{it}, reallocating new borrowing between short- and long-term debt does not directly change its costs.

External equity is unavailable, so shareholder payout must satisfy

$$d_{it} \geq 0 \tag{27}$$

This restriction reflects the focus on private, bank-dependent firms and keeps the model centered on debt financing and maturity choice. Firms cannot cover a financing shortfall through an equity

[15] Quantitatively removing the wedge ($\kappa \to 0$) reduces leverage conditional on borrowing from 0.9 to below 0.2. Two-fifths of firms hold no debt, and default disappears.

[16] Berg et al. (2016) document the importance of fees in corporate bank lending, while Altinkilic and Hansen (2000) find evidence of rising marginal underwriting costs in corporate bond and equity issuance. These findings motivate non-interest financing costs and a convex issuance-cost specification; the quadratic functional form remains a modeling assumption.

injection; instead, they must adjust investment, borrowing, or maturity subject to the prices offered by lenders. Together, the no-equity constraint, convex issuance costs, and default-sensitive debt prices generate endogenous debt capacity without an exogenous borrowing limit. These features make the timing of repayment obligations economically consequential, and also make debt maturity relevant for firm value. Longer maturity postpones principal repayment and can reduce exposure to future refinancing conditions and costly debt issuance. At the same time, obtaining longer-term financing may reduce current issuance proceeds when its relative price rises. Firms therefore choose maturity by trading off the current cost of long-duration financing against the continuation-value benefits of postponing repayment.

6.4 Default and debt pricing

Default and recovery. Default is liquidity based. A firm defaults when no feasible continuation policy allows it to meet its current obligations:

$$D(x; A) = \mathbf{1}\{\mathcal{F}(x; A) = \varnothing\} \tag{28}$$

Default is determined by financing feasibility rather than a strategic comparison between continuation and default values. Since current payouts and future equity values are nonnegative, shareholders continue whenever a feasible financing plan exists. When no such plan exists, the firm defaults. Default is therefore a liquidity, or financing-feasibility, event with collateral recovery, not a comprehensive model of corporate bankruptcy. Consistent with limited liability, shareholders receive $V^D = 0$ and are not responsible for unpaid corporate debt. While limited liability protects shareholders ex post, future default represents an ex-ante risk because it extinguishes their claims on future profits. Debt maturity can provide insurance against this risk by reducing near-term refinancing obligations.

Creditors recover a fraction of the firm's depreciated capital. Recovery per unit of face value is

$$\mathcal{R}(x_{i,t+1}) = \min\left\{1,\ \frac{(1-\delta_K)K_{i,t+1}}{B_{i,t+1}}\right\} \tag{29}$$

with $\mathcal{R}(x) = 1$ when $B = 0$. Short- and long-term creditors are pari passu and receive the same recovery per unit of face value.

Default thus extinguishes shareholders' continuation claims and exposes creditors to losses whenever recoverable assets fall short of outstanding debt. These expected losses are reflected in endogenous debt prices and affect firms' ex-ante financing decisions. The benchmark does not impose an additional deadweight bankruptcy cost or liquidation haircut.

Competitive debt pricing. Competitive lenders price new debt taking into account the firm's future default risk and recovery value. Prices depend on the current firm and aggregate states

and on the firm's investment, borrowing, and maturity choices. For compactness, let $D_{i,t+1} \equiv D(x_{i,t+1}; A_{t+1})$ and write $Q^j_{it} \equiv Q^j(x_{it}, K_{i,t+1}, m_{it}, \ell_{it}; A_t)$ for $j \in \{S, L\}$. Next period's price $Q^L_{i,t+1} \equiv Q^L\left(x_{i,t+1}, K^*_{i,t+2}, m^*_{i,t+1}, \ell^*_{i,t+1}; A_{t+1}\right)$ is evaluated at the firm's optimal continuation policy. The price of one unit of newly issued short-term debt is

$$Q^S = \mathbb{E}_t\left[\mathcal{M}_{t,t+1}\left(1 - D_{i,t+1} + D_{i,t+1}\mathcal{R}(x_{i,t+1})\right)\right] \tag{30}$$

Short-term debt matures entirely in the following period. Creditors receive one unit of face value if the firm repays and the recovery value if it defaults. Its price thus reflects the risk-free discount rate and expected credit losses. Long-term debt is priced recursively because a fraction of each claim survives into subsequent periods.

$$Q^L_{i,t} = e^{-\omega_t}\mathbb{E}_t\left[\mathcal{M}_{t,t+1}\Big((1 - D_{i,t+1})\left[c_L + 1 - \rho_L + \rho_L Q^L_{i,t+1}\right] + D_{i,t+1}\mathcal{R}(x_{i,t+1})\Big)\right] \tag{31}$$

Conditional on repayment, creditors receive the coupon c_L, the fraction $1 - \rho_L$ of principal that amortizes, and the value of the surviving fraction ρ_L of the claim; upon default, they receive the recovery value. A higher ω_t lowers the proceeds firms receive from long-term debt relative to the value of the underlying lender claim. The wedge is borne by the issuer: lenders value the claim with the common kernel and pay $e^{-\omega_t}$ per unit of that value, so the yield they require on long-term claims rises with ω_t while the firm's cash proceeds fall. Because long-term claims survive beyond the next period, their value depends on future interest rates, default risk, and the firm's subsequent investment and financing decisions. This recursive structure also gives rise to debt-overhang and dilution effects. Higher investment can reduce future default risk and increase the value of outstanding long-term debt. Part of the benefit of investment then accrues to existing creditors rather than shareholders, potentially generating the classic debt-overhang problem of Myers (1977). Conversely, additional future borrowing can increase default risk and reduce the value of surviving pari passu claims, diluting existing long-term creditors. Lenders anticipate these effects when pricing new debt.

6.5 The maturity trade-off

The maturity choice trades off current financing proceeds against future rollover exposure. A larger long-term share ℓ lowers next-period principal due by the second property of (21), but long-term credit can be more expensive than short-term credit today. This breaks Modigliani–Miller irrelevance and firms are not indifferent between short- and long-term debt even though issuance costs are maturity-neutral: maturity changes both the current price of funding and the timing of future repayment obligations. Let $\mu \geq 0$ denote the multiplier on the no-equity constraint (27). For an issuing firm with an interior maturity choice, the first-order condition with respect to long-debt

ℓ is

$$Q^S - Q^L - (1-\ell)\frac{\partial Q^S}{\partial \ell} - \ell\frac{\partial Q^L}{\partial \ell} = -\frac{(1-\kappa)\rho_L}{1+\mu}\mathbb{E}_t\left[\mathcal{M}_{t,t+1}V_G(z', K', B', G'; A')\right] \tag{32}$$

The left-hand side is the current cost of extending maturity: the price gap $Q^S - Q^L$, together with endogenous price-feedback terms. By lowering future principal due, a longer maturity can reduce default risk, which competitive lenders internalize in both short- and long-term debt prices; whether this feedback offsets or adds to the direct cost of duration depends on the signs of the maturity-share price derivatives, which the model determines endogenously. The endogenous pricing of debt also makes the cost of obtaining maturity protection sensitive to firms' borrowing decisions. Appendix E.8 shows that long-term debt prices are substantially more sensitive to gross issuance than short-term debt prices, particularly for firms with high rollover exposure. While these issuance sensitivities differ from the maturity-share derivatives in equation (32), they highlight the endogenous financing costs faced by firms seeking to insure against refinancing risk. The right-hand side is the intertemporal benefit. Raising ℓ does not eliminate liabilities; it shifts principal repayments away from the next period. When external financing is costly, concentrating a larger obligation in a single period lowers firm value ($V_G < 0$), so longer maturity protects against refinancing and interest-rate risk—worth more when future refinancing conditions are unfavorable and when the firm has little capacity to absorb a concentrated repayment. Maturity becomes irrelevant if short- and long-term funding have the same price and future refinancing is frictionless. In that case, shifting principal across dates has no effect on firm value. The remaining first-order conditions govern investment and the amount of new borrowing. For an interior capital choice,

$$1 + \Phi_I(I, K) = m\left[(1-\ell)\frac{\partial Q^S}{\partial K'} + \ell\frac{\partial Q^L}{\partial K'}\right] + \frac{1-\kappa}{1+\mu}\mathbb{E}_t\left[\mathcal{M}_{t,t+1}V_K\right] \tag{33}$$

The left-hand side is the marginal resource cost of investment. The right-hand side captures its two benefits: additional capital raises continuation value and can improve current borrowing terms by increasing recovery values and reducing default risk. For an interior gross-borrowing choice,

$$(1-\ell)Q^S + \ell Q^L - H'(m) + m\left[(1-\ell)\frac{\partial Q^S}{\partial m} + \ell\frac{\partial Q^L}{\partial m}\right] = -\frac{1-\kappa}{1+\mu}\mathbb{E}_t\left[\mathcal{M}_{t,t+1}\left(V_B + (1-\rho_L\ell)V_G\right)\right] \tag{34}$$

The left-hand side is the marginal amount of funds raised today, net of issuance costs and the endogenous effect of additional borrowing on debt prices. The right-hand side is the continuation cost of the additional liability: one unit of new borrowing raises next-period total debt by one and next-period principal due by $1 - \rho_L\ell$. Appendix B provides the full derivations.

6.6 Recursive equilibrium and stationarity

Given the exogenous productivity process in (9) and the processes for aggregate financing conditions in (13)–(15), these objects satisfy the following equilibrium conditions. Firms solve (23) subject to

the feasible set $\mathcal{F}(x; A)$ in (24); default follows (28), with $V(x; A) = 0$ upon default and $V(x; A) = V^C(x; A)$ otherwise; debt-price schedules satisfy (30), (31); and expectations are consistent with the exogenous state transitions and firms' optimal policies. Let $\mathcal{X}$ denote the firm-state space, with $x = (z, K, B, G) \in \mathcal{X}$, and let $\mathcal{A}$ denote the state space for aggregate financing conditions. Let μ_t denote the cross-sectional distribution of firm states conditional on the realized history of aggregate financing conditions. For any measurable set $E \subseteq \mathcal{X}$, let $\Pi(E \mid x, A)$ denote the probability that a continuing firm currently in state x and facing aggregate state A has next-period state in E. Let $\nu(E)$ denote the probability that a new entrant's initial state lies in E. Entrants begin with capital K_0, zero debt, and productivity drawn from its stationary distribution. The cross-sectional distribution then evolves according to:

$$\mu_{t+1}(E) = \int_{\mathcal{X}} \big[\big(1 - D(x; A_t)\big)\Pi(E \mid x, A_t) + D(x; A_t)\nu(E)\big]\, \mu_t(dx). \tag{35}$$

Thus, $\mu_{t+1}(E)$ is the mass of firms whose next-period state lies in E: continuing firms evolve according to their optimal policies and the productivity process, while defaulting firms are replaced one for one by entrants drawn from ν. Let $P(E' \mid A)$ denote the probability that next-period aggregate financing conditions lie in the measurable set $E' \subseteq \mathcal{A}$, conditional on current aggregate state A. Conditional on (x, A), idiosyncratic productivity innovations and entrants' initial states are independent of the next aggregate innovation. In addition, next-period inherited balance sheets are determined by current firm choices. The transition in firm states therefore factors from the transition in aggregate financing conditions. An invariant joint distribution Γ^* over firm and aggregate states, if it exists, satisfies

$$\Gamma^*(E \times E') = \int_{\mathcal{X}\times\mathcal{A}} \big[\big(1 - D(x; A)\big)\Pi(E \mid x, A) + D(x; A)\nu(E)\big]\, P(E' \mid A)\, \Gamma^*(dx, dA). \tag{36}$$

Unconditional model moments are computed under Γ^*.

6.7 Calibration

We calibrate the model annually to the matched AnaCredit–Orbis sample of Section 2; Table 10 summarizes the parameterization. Three parameters—the fixed operating cost F, the issuance cost η, and the capital adjustment cost ψ_K—are estimated jointly by simulated method of moments; the remaining parameters are measured outside the model. No maturity, rollover-exposure, or leverage moment enters the SMM objective, so those outcomes and the heterogeneous responses to financing shocks are untargeted tests of the mechanism.

Parameters determined outside the model. Profit curvature is $\alpha = 0.60$ and depreciation rate $\delta_K = 0.10$; productivity scale is normalized by $\bar{z} = [e^{\bar{r}} - (1 - \delta_K)]/\alpha$, so the frictionless capital stock of a median-productivity firm is one. The persistence and dispersion of idiosyncratic productivity are estimated from the Orbis panel $\rho_z = 0.85$, $\sigma_z = 0.15$ and the estimation procedure

Table 10: Calibration

Parameter	Description	Value	Source or target
Panel A. Externally assigned and normalizations			
α	Profit curvature	0.60	Cooper and Haltiwanger (2006)
δ_K	Depreciation rate	0.10	Literature
$\bar{z}$	Productivity scale	$[e^{\bar{r}} - (1 - \delta_K)]/\alpha$	Normalization, $K^{ss} = 1$
Panel B. Directly measured or estimated outside the model			
K_0	Entrant capital	0.30	Orbis: median fixed assets new firms
c_L	Long-debt coupon	0.0083	Par at the mean risk-free rate, $e^{\bar{r}} - 1$
κ	Equity-horizon wedge	0.05	Orbis: Annual exit rate
ρ_z	Productivity persistence	0.85	Firm-level panel estimates
σ_z	Productivity innovation s.d.	0.150	Firm-level panel estimates
$\bar{r}$	Mean short rate	0.0082	One-year OIS rate
ρ_r	Short-rate persistence	0.950	One-year OIS rate
σ_r	Short-rate innovation s.d.	0.0058	One-year OIS rate
ρ_f	Expected-rate news persistence	0.363	JSZ expectations component, normalized
σ_f	Expected-rate news innovation s.d.	0.0088	JSZ expectations component, normalized
$\bar{\omega}$	Term-premium wedge, level	0.0315	JSZ term premium, 2022–2024
ρ_ω	Term-premium wedge persistence	0.908	JSZ Term premium via Eq. (76)
σ_ω	Term-premium innovation s.d.	0.0160	JSZ Term premium via Eq. (76)
Panel C. Estimated by simulated method of moments			
F	Fixed operating cost	0.024	Annual SME default rate, 0.020
η	Debt issuance cost	0.32	Gross issuance rate, 0.267
ψ_K	Capital adjustment cost	0.02	Investment rate, 0.297

Notes: Aggregate processes are annual. The short-rate process is estimated on the one-year OIS rate alone; f_t is the normalized expected-rate-path component of Joslin et al. (2011), orthogonalized to the current rate; ω_t is recovered from the observed term premium through the measurement equation (76). Panel C values are selected on the star identification design (Section 7.1); the moments reported in Section 7 are computed at those values.

is described in Appendix D. Entrant capital is set to $K_0 = 0.30$ as in the Orbis panel data: excluding firms first observed in 2019 to avoid left censoring, median fixed assets among newly observed firms are about 30% of the median among firms observed for at least three prior years. Entrants start with no debt and draw productivity from its stationary distribution. Under geometric amortization, expected maturity is $1/(1 - \rho_L)$, so $\rho_L = 1 - 1/5.19 = 0.807$, where 5.19 years is the credit-weighted mean original maturity of newly issued term loans in AnaCredit. The model thus matches the effective maturity composition of new borrowing rather than the distribution of contractual maturities: a five-year bullet or installment loan and a geometrically amortizing claim with a five-year expected maturity differ in the timing of principal repayments, and the model's long-term share is mapped to log maturity through this expected-maturity equivalence (Appendix Table 31). The coupon is normalized to par at the mean risk-free rate, $c_L = e^{\bar{r}} - 1 = 0.00826$—a normalization, not a measurement: it places no credit spread in the coupon, so all compensation for default risk and duration enters through the issuance price. Away from the mean rate the

claim trades away from par, and that valuation channel is exactly what maturity choice trades off against refinancing insurance. The equity-horizon wedge κ captures in reduced form the shorter effective horizon of the equity claim: ownership changes hands without extinguishing the firm or its liabilities, and current owners do not fully capture post-turnover value. It is a *valuation* wedge, not an exit probability. Its magnitude is disciplined by the average 5% annual exit hazard in Orbis, used as an observable proxy for the equity horizon rather than as a matched moment; leverage remains an untargeted implication. The short rate follows (13), estimated on the one-year OIS rate. The state f_t is the Joslin et al. (2011) expectations component orthogonalized to the contemporaneous short rate; because the pass-through coefficient and the scale of f_t are only jointly identified, we rescale the raw factor by its estimated pass-through, normalizing its loading in the short-rate transition to one. The implied $\sigma_f = 0.0088$, so a one-standard-deviation innovation raises the expected next-period short rate by about 0.9 percentage points. The wedge ω_t is a price of duration on newly issued corporate loans. Its level is preset at $\bar{\omega} = 0.0315$, which corresponds to a term premium of about 0.6 percent per year on the model's claim. This is the average five-year term premium over 2004–2025 in the Joslin et al. (2011) decomposition, 0.6 percent, and above its average over the 2018–2024 estimation window, -0.5 percent, which is dominated by the years of asset purchases and negative policy rates; the historical exercises in Section 8.3 feed the measured wedge of those years into the model directly. The dynamics come from the same decomposition: a persistence of $\rho_\omega = 0.908$ and a stationary standard deviation of 0.038, implying $\sigma_\omega = 0.016$. The duration-consistent measurement equation (76) maps observed term-premium movements into movements of the log price wedge on the model's long claim, so the mapping is independent of the coupon c_L.

Internally estimated parameters. The remaining parameters $\Theta = (F, \eta, \psi_K)$ are estimated jointly by SMM. With $m(x)$ the empirical moments and $\widehat{m}(\widetilde{x} \mid \Theta)$ their simulated counterparts, the estimator solves

$$\widehat{\Theta}_{\text{SMM}} = \arg\min_{\Theta} g(\Theta)' g(\Theta), \qquad g_j(\Theta) = \frac{\widehat{m}_j(\widetilde{x} \mid \Theta) - m_j(x)}{m_j(x)} \tag{37}$$

so the objective weights proportional deviations equally. Each target maps primarily to one parameter. The fixed operating cost F governs proximity to the liquidity-default boundary and is disciplined by the 2.0 percent annual default rate for euro-area SME exposures in EBA 2019 reporting. The issuance cost η is quadratic in gross issuance, $H(m_{it}) = \frac{\eta}{2} m_{it}^2$, it governs borrowing intensity. Its target is the aggregate gross issuance rate (gross borrowing during the year relative to the outstanding debt stock) from AnaCredit term-loan originations,

$$\frac{\sum_i m_{it}}{\sum_i B_{it}} = 0.267$$

A quadratic capital adjustment cost governs adjustment intensity rather than the arrival of

expansion opportunities, so ψ_K is disciplined by investment conditional on a financed expansion: among firm-years with a net increase in bank debt,

$$\mathbb{E}\left[\frac{I_{it}}{K_{i,t-1}}\,\middle|\, B_{it} > B_{i,t-1}\right] = 0.297$$

This moment is monotone in ψ_K and leaves the other two targets nearly unchanged, so the moment–parameter mapping is close to diagonal.[17] The criterion is minimized in two stages under common random numbers. A scrambled-Sobol scan of the parameter box screens for competing basins; its leading candidates are re-evaluated at full accuracy, since the scan itself uses small panels and coarse tolerances, and none improves on the neighborhood of the initial calibration. A Newton step from one-at-a-time probe designs then locates the minimum, verified by re-solving at the selected vector; the probe table doubles as the identification exhibit for the moment–parameter Jacobian. At $\widehat{\Theta}$ the criterion is 0.0013. Moments are averaged across 50 simulated histories of 10,000 firms after 300-period burn-in.

7 Quantitative Results

This section evaluates the model in three steps: untargeted cross-sectional moments, policy functions linking rollover exposure to the two components of long rates, and the paper's empirical regressions re-estimated on simulated panels.

7.1 Parameter estimates and targeted moments

Panel A of Table 11 reports the targeted moments. The estimated internal parameters are $F = 0.024$, $\eta = 0.32$, and $\psi_K = 0.02$. All three targets are matched closely: the default rate is 0.019 against 0.020, the gross issuance rate 0.272 against 0.267, and investment conditional on net debt expansion 0.281 against 0.297, with the small residual concentrated in the last moment.

7.2 Untargeted model fit

Panels B–D of Table 11 report untargeted moments. A measurement note is useful when comparing rollover exposure in the model and the data. The model state is

$$s_{it} = \frac{G_{it}}{B_{it}} = \frac{B^S_{it} + (1-\rho_L)B^L_{it}}{B_{it}},$$

the fraction of outstanding principal contractually due within the period. Because long-term claims amortize geometrically, s_{it} includes the scheduled repayment $(1-\rho_L)B^L_{it}$ on long-term debt

[17] The unconditional mean absolute capital growth rate, the natural alternative, bundles this intensive margin with an extensive margin—the frequency of net debt expansions—that a convex issuance cost cannot move; asking ψ_K to fit it drives the parameter to a boundary. Both statistics are left outside the estimation.

Table 11: Model fit: targeted and untargeted moments

Moment	Data	Model
Panel A: Targeted		
Annual default rate	0.020	0.019
Gross issuance / debt	0.267	0.272
$\mathbb{E}[I/K \mid \Delta B > 0]$	0.297	0.281
Panel B: Untargeted		
Mean long-term issuance share $\mathbb{E}[\ell]$	0.70	0.69
Dispersion sd(ℓ)	0.43	0.37
All-short issuers $\Pr(\ell = 0)$	0.22	0.17
Mean, $\mathbb{E}[s]$	0.27	0.26
Dispersion sd(s)	0.32	0.10
Leverage $\mathbb{E}[B/K \mid B > 0]$	0.85	0.90
Size–exposure corr($\log K, s$)	−0.15	−0.19
Debt expansion $\Pr(\Delta B > 0)$	0.47	0.23

Notes: Data moments from the matched AnaCredit–Orbis panel; construction in Section 2. Two conventions for refinancing needs are used and must not be crossed. Rollover exposure $s = G/B$ is the share of principal due within the year and includes the amortizing fraction of long-term debt, so $s \geq 1 - \rho_L$ for any indebted firm; its data counterpart is the raw maturity wall in the regression estimation sample ($N = 2.26$ million firm-years). The debt moments are measured among firms with lagged bank debt above its 5th percentile; conditional growth rates are winsorized at the 1st and 99th percentiles; the same sample restrictions are applied to the simulated panel. The capital-adjustment moments are winsorized fixed-asset growth rates. The targeted investment moment conditions on firm-years with a net increase in bank debt, with investment winsorized within the expander sample. Panel A moments enter the calibration objective; Panels B–D do not. The model column reports moments of the stationary distribution at the estimated parameters $\widehat{\Theta} = (0.024, 0.32, 0.02)$, averaged across 50 independently simulated histories of 10,000 firms after a 300-period burn-in; the criterion at $\widehat{\Theta}$ is 0.0013.

and is bounded below by $1 - \rho_L = 0.193$ for any indebted firm. The empirical rollover measure likewise captures debt whose principal comes due within the following twelve months. We therefore compare it directly with s_{it} in Table 11.[18] Maturity composition is reproduced closely. The mean long-term share of new issuance is 0.69 against 0.70 in the data, its standard deviation 0.37 (0.43), and the frequency of all-short issuers 0.17 (0.22). The model thus matches the level and the spread of maturity choice without any maturity moment in the SMM objective. Regarding inherited rollover exposure, the mean is 0.26 against 0.27, so the model reproduces how much principal comes due on average, but its dispersion is 0.10 against 0.32. With most issuance long-term and long-term debt amortizing smoothly, the model meets the average repayment burden through

[18]The mapping is necessarily imperfect because AnaCredit reports contractual residual maturity rather than the full amortization schedule. The empirical measure includes the outstanding balance of long-term loans that have entered their final twelve months, but does not capture scheduled amortization on loans whose final maturity lies further in the future. Geometrically amortizing long-term claims in the model, by contrast, generate scheduled principal repayment every period but have no deterministic terminal maturity date. Thus, the two measures are not mechanically identical, but both are designed to capture the principal repayment burden falling due within the year.

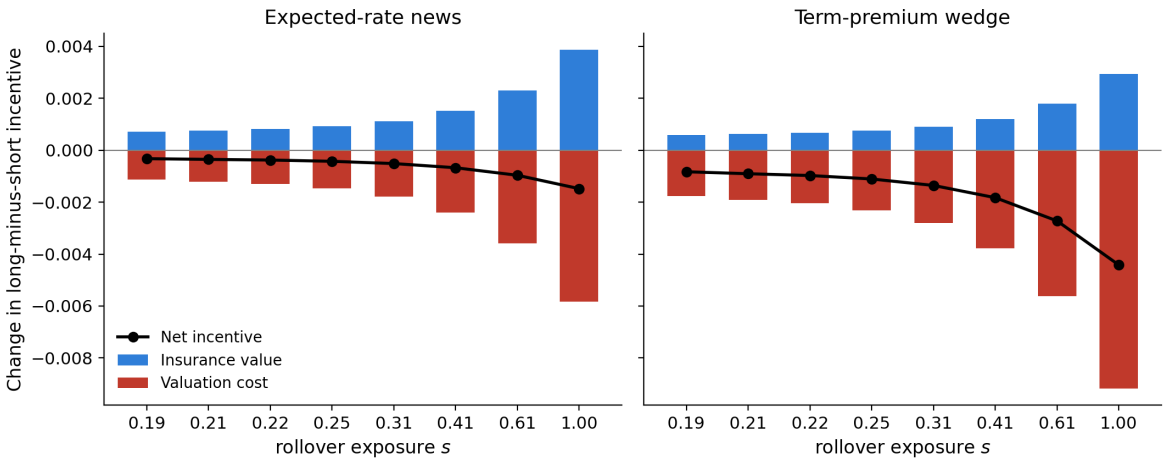


Figure 5: Maturity-insurance mechanism. The figure decomposes the change in the marginal incentive to issue long rather than short debt following one-standard-deviation shocks to expected-rate news (left) and to the term-premium wedge (right), relative to the mean aggregate state, by rollover exposure s. Bars report the change in the continuation-value benefit of postponing refinancing (insurance value) and in the valuation cost of long-duration funding; black dots report the change in the net incentive, which also includes the quantitatively smaller credit-price feedback. Higher expected-rate news and a higher term-premium wedge both reduce the net incentive to issue long debt. Expected-rate news, however, also increases the continuation value of postponing refinancing, substantially cushioning its valuation effect. This insurance offset grows with rollover exposure in proportion to the valuation cost, whereas the term-premium wedge generates a considerably larger deterioration in the net incentive at high exposure. The decomposition is evaluated at the rollover margin ($K' = K$, $B' = B$) for the median idiosyncratic state.

amortization rather than through short-term contracts, and it lacks the data's thick upper tail of predominantly short-term borrowers. This is a mechanical consequence of geometric amortization—there are no discrete repayment dates to concentrate—and cannot be resolved by the calibration. The negative size–rollover exposure relation is matched closely ($\text{corr}(\log K, s) = -0.19$ against -0.15). In addition, the frequency of net debt expansion is 0.23 against 0.47: the extensive margin of borrowing is understated, since a convex issuance cost prices intensity rather than participation. Leverage conditional on borrowing is close—0.90 against 0.85—so the model generates the observed intensity of debt usage without a tax advantage or a leverage target. Because the model understates both the dispersion of rollover exposure and the frequency of net debt expansion, its quantitative statements about heterogeneous refinancing needs and financing responses are conditional on the distribution it represents.

7.3 The Value and Price of Maturity Insurance

The trade-off in equation (32) highlights two opposing forces shaping maturity choice. An increase in expected future short rates raises the current cost of long-duration borrowing but also increases the value of postponing principal refinancing. By contrast, a higher term-premium wedge raises the cost of long-duration finance without directly increasing expected future refinancing rates. Expected-rate news therefore affects both the value and the price of maturity insurance, whereas

the term-premium wedge operates primarily through its price.

Figure 5 evaluates the change in the marginal incentive to substitute long- for short-term issuance following one-standard-deviation shocks, at a common rollover point and across levels of rollover exposure. Both shocks reduce this net incentive, but their decompositions differ. Following expected-rate news, the increase in the marginal continuation benefit of postponing principal refinancing offsets about two-thirds of the rise in the marginal valuation cost of long-duration funding. Following a term-premium shock, the continuation benefit also rises, but offsets less than one-third of the valuation-cost increase. At this shock size and normalization, the term-premium wedge therefore produces the larger deterioration in the net marginal incentive to issue long debt. Both components scale with the principal coming due, so the insurance gain is larger in absolute terms for exposed firms while the offset share is nearly constant across exposure, 0.67 to 0.68 under news and 0.28 to 0.30 under the wedge. These offsets are ratios of marginal changes at the evaluation point, not fractions of firms protected or changes in maturity shares.

These are local incentive comparisons, not optimal maturity responses, and the exposure profile does not isolate rollover exposure from the other balance-sheet states that move with it: leverage, capital, productivity, and the distance to the switching threshold between short- and long-term issuance. Endogenous debt prices and the location of those thresholds determine the actual policy response. We therefore read the decomposition jointly with the conditional maturity policies in Appendix E.10 and the simulated fixed-effects regressions in Section 7.4.[19]

7.4 Model counterparts to the empirical regressions

The main test runs the empirical specification on simulated data, in the same units. Appendix Table 31 states the mapping from each empirical object to its model counterpart. We estimate

$$\log(\text{Maturity}_{it}) = \alpha_i + \alpha_t + \beta_s\, s_{i,t-1} + \beta_E \left(\text{ESR}^{5Y}_{t-1} \times s_{i,t-1}\right) + \beta_{TP} \left(\text{TP}^{5Y}_{t-1} \times s_{i,t-1}\right) + \gamma' X_{i,t-1} + \varepsilon_{it} \tag{38}$$

with firm and time fixed effects, lagged factors, and the same lagged firm controls as in the data (size and leverage); time effects absorb the levels of the aggregate states. Exposure enters as the model state $s_{i,t-1} = G_{i,t-1}/B_{i,t-1}$, the same object compared with the data in Table 11. Since each simulation contains one realized aggregate history, we report means across 50 independent histories with Monte Carlo standard errors. The model reproduces the responses to both components. Table 12 compares the standardized interactions. For expected-rate news the log-maturity interaction is 0.424 against 0.430 in the data and the long-term-share interaction 0.245 against 0.201; for the term premium, -0.191 against -0.375 and -0.127 against -0.175. The model reproduces qualitatively the sign of the expected-rate and term premium interactions with rollover exposure and it is quantitatively close to the size of the coefficients in the data. These interactions are relative

[19]The decomposition is evaluated at the rollover point ($K' = K$, $B' = B$) using the model's debt-price and continuation-value tables. It characterizes marginal incentives rather than an optimality condition evaluated at each firm's chosen maturity. The qualitative comparison is unchanged for one-half- and two-standard-deviation shocks.

Table 12: Model Counterparts to the Empirical Estimates

	log(Maturity)		Long-Term Debt Share	
	Data	Model	Data	Model
Rollover Exposure	1.171	0.366	0.526	0.309
	(0.014)	(0.102)	(0.006)	(0.067)
Expected Rate$^{5Y}_{t-1}\times$ Rollover Exposure	0.430	0.424	0.201	0.245
	(0.009)	(0.054)	(0.004)	(0.037)
TP$^{5Y}_{t-1}\times$ Rollover Exposure	-0.375	-0.191	-0.175	-0.127
	(0.009)	(0.072)	(0.004)	(0.048)

Notes: Data: firm-year specification of Table 26, standard errors in parentheses. Model: equation (38) on simulated panels at $\widehat{\Theta}$, averaged across fifty histories, with Monte Carlo standard errors in parentheses.Entries are per unit of exposure. Interaction coefficients are per one standard deviation of the lagged factor, standardized separately in the data and in the model.

statements: time fixed effects absorb the aggregate maturity response in the data, and the model regressions measure the same conditional interaction over simulated equilibrium histories. In the model, expected-rate news reduces the net incentive to issue long debt at every level of rollover exposure, but the accompanying increase in the insurance value of long maturity provides a stronger offset for firms with greater rollover exposure (Figure 5). An aggregate decline in the long-term share, as in the counterfactuals below, and a positive cross-sectional interaction with rollover exposure therefore coexist. The marginal-incentive decomposition explains why exposure changes the relative attractiveness of long debt; it is not itself an estimate of the regression interaction, which is an equilibrium object that also reflects the distribution of firms across states and their dynamic adjustment (Appendix E.10). Appendix Figure 20 shows the maturity responses themselves by rollover exposure, evaluated for the same firms at the same balance sheets at average financing conditions: exposed firms shorten less than firms at the amortization floor following expected-rate news and more following the term-premium wedge, the signs of the two interactions. This discipline is stricter than in the closest quantitative maturity models, which target maturity composition directly—the mean long-term share, the share of debt due within a year, or the response coefficients themselves.

8 Counterfactuals

We use the model to quantify how endogenous debt maturity shapes the transmission of monetary shocks. Our main counterfactual removes the direct response of maturity choice to aggregate financing conditions while allowing maturity to adjust as firms' balance sheets evolve. We then examine how the value of this flexibility varies across firms and distinguish the insurance provided by contractual maturity from that provided by coupon fixation. Finally, we use the model to study how the value of maturity flexibility changes when higher expected refinancing costs coincide with a higher price of long-duration finance, and how the same expected-rate shock transmits

across different term-premium environments. To quantify these effects, we compute generalized impulse response functions (GIRFs), following Koop et al. (1996). For each restricted economy, we re-solve firms' policies and lenders' price schedules. We simulate 1,000 economies of 10,000 firms, initialized from the benchmark stationary cross-section, and compare outcomes under a one-standard-deviation financing shock with those under a no-shock path. After the initial innovation, aggregate states follow (13)–(15) without further aggregate shocks. Under expected-rate news, the short rate is unchanged on impact and rises from the following year. Idiosyncratic productivity draws are held fixed across scenarios. For each economy, the GIRF is the difference between the shock and no-shock paths. We report averages across simulations and Monte Carlo standard errors. Because aggregate financing conditions are exogenous, these comparisons capture firm responses conditional on the specified financing shock.

8.1 Removing the Aggregate Maturity Response

We quantify the contribution of endogenous maturity adjustment to monetary transmission by restricting the long-term issuance share (ℓ) to its benchmark policy evaluated at the no-shock aggregate state:

$$\ell^{CF}(x_{it}, A_t) = \ell^{B}(x_{it}, \bar{A}) \tag{39}$$

The restriction removes the direct response of maturity choice to aggregate financing conditions without fixing maturity across firms or over time: firms can still change maturity as their balance sheets evolve. We re-solve investment, borrowing, and lenders' debt-price schedules under the restricted rule. Both economies begin from the benchmark stationary distribution, so firms enter each shock with identical inherited financial positions. We measure the contribution of maturity adjustment as the difference between the benchmark and counterfactual impulse responses, reported in Table 13.[20] Maturity adjustment affects both real activity and firm survival. It substantially attenuates the investment contraction: within three years, cumulative investment relative to capital is 4.4, 2.4, and 1.6 percentage points higher in the benchmark following current-rate, expected-rate news, and term-premium shocks, respectively. These effects are economically large relative to the corresponding benchmark contractions of 4.5, 8.4, and 1.1 percentage points. Their timing differs across shocks. Following a current-rate shock, the investment benefit emerges immediately. Following expected-rate news, investment initially falls slightly more in the benchmark, but the difference reverses as the anticipated increase in refinancing costs materializes. Following a term-premium shock the benefit is small and fades by year five: a higher price of duration by itself does not raise future refinancing costs, so there is little for maturity adjustment to insure against.

The investment benefit does not require an aggregate extension of debt maturity. In the benchmark economy, firms predominantly shorten new borrowing following contractionary shocks. On impact, the long-term share of new issuance falls by 3.4, 8.8, and 4.8 percentage points more than

[20] This counterfactual isolates the contribution of maturity adjustment conditional on inherited benchmark balance sheets; it does not compare economies that have permanently adapted to different maturity regimes.

Table 13: Value of Maturity Flexibility

Horizon (years)	1	3	5
Investment (cumulative, pp)			
Current rate	+2.0 (0.5)	+4.4 (0.3)	+4.7 (0.4)
Expected-rate news	−0.9 (0.3)	+2.4 (0.3)	+1.9 (0.4)
Term-premium wedge	+1.0 (0.3)	+1.6 (0.2)	+0.6 (0.3)
Debt issuance (cumulative, pp)			
Current rate	+0.8 (0.5)	+5.1 (0.6)	+8.1 (0.8)
Expected-rate news	−2.5 (0.4)	+0.5 (0.5)	+3.1 (0.6)
Term-premium wedge	−0.6 (0.3)	+2.2 (0.4)	+3.3 (0.6)
Long-term issuance share (response at horizon, pp)			
Current rate	−3.4 (0.3)	−1.1 (0.4)	−1.2 (0.3)
Expected-rate news	−8.8 (0.6)	−8.5 (0.3)	−7.4 (0.3)
Term-premium wedge	−4.8 (0.3)	−2.0 (0.4)	+1.9 (0.3)
Credit spread (average through horizon, pp)			
Current rate	−4.5 (0.6)	−0.4 (0.2)	+0.9 (0.1)
Expected-rate news	−0.5 (0.4)	−3.4 (0.3)	−2.2 (0.3)
Term-premium wedge	−1.7 (0.2)	−0.7 (0.2)	−0.5 (0.1)
Default (cumulative, pp)			
Current rate	—	−0.5 (0.2)	+0.9 (0.3)
Expected-rate news	—	−1.4 (0.2)	−2.7 (0.3)
Term-premium wedge	—	−0.2 (0.2)	+0.7 (0.3)

Notes: Entries are flexible-minus-restricted differences in shock responses to one-standard-deviation innovations in the current rate, expected-rate news, or the term-premium wedge. Positive values indicate a larger response in the flexible economy. Investment, debt issuance, and default are cumulative through the reported horizon; the long-term issuance share is the response at that horizon; credit spreads are averaged through the horizon. Horizon 1 is the shock year. Default is omitted at horizon 1 because it does not respond on impact under the model's timing. The credit spread is the issuance-weighted yield spread of newly issued debt over its default-free counterpart. Because the restricted maturity policy does not respond directly to aggregate financing conditions, the horizon-1 entries for the long-term issuance share coincide with the flexible economy's own responses. After the initial innovation, aggregate states follow their estimated laws of motion without further aggregate shocks. Results average 1,000 simulated economies of 10,000 firms, initialized from the flexible economy's stationary cross-section. Idiosyncratic productivity draws are held fixed across scenarios. Monte Carlo standard errors of the mean are in parentheses.

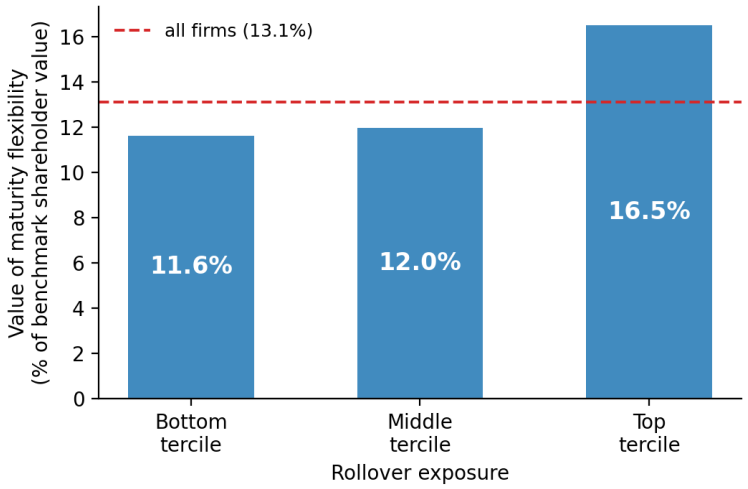


Figure 6: Value of Maturity Flexibility by Rollover Exposure. The figure reports the shareholder value preserved by maturity flexibility: benchmark minus counterfactual value, as a percentage of benchmark shareholder value, where the counterfactual permanently restricts maturity policy to its aggregate-state-invariant rule. Bars are terciles of rollover exposure among indebted firms. Values are evaluated at identical initial firm states drawn from the benchmark stationary distribution. The dashed line reports the average across indebted firms. The calculation is an ex ante value comparison over all future states, not the effect of a single financing shock.

in the restricted economy following current-rate, expected-rate news, and term-premium shocks, respectively. This adjustment allows firms to substitute away from costly long-duration funding while preserving financing capacity. Within three years, cumulative gross issuance is 5.1 percentage points higher in the benchmark following a current-rate shock and 2.2 points higher following a term-premium shock. The average credit-spread response over the first three years is also lower in the benchmark across all three shocks. Following expected-rate news, the financing benefit emerges later: gross issuance initially declines more in the benchmark, but the difference reverses as the anticipated tightening materializes. Maturity flexibility also provides substantial protection against default when the shock raises expected future refinancing costs. Following expected-rate news, cumulative default is 1.4 percentage points lower in the benchmark after three years and 2.7 points lower after five years. Thus, the ability to adjust maturity before higher expected short rates fully materialize reduces subsequent financial distress as firms encounter more costly refinancing conditions. The default effects are smaller and less persistent for the other shocks: following a current-rate shock, cumulative default is 0.5 percentage points lower after three years and 0.9 points higher by year five, and following a term-premium shock it is 0.7 points higher after five years. Maturity flexibility therefore does not uniformly reduce cumulative default. Following a current-rate shock, the reversal reflects the restricted economy's sharper initial default wave and deleveraging, followed by lower subsequent default rates, rather than rising defaults in the flexible economy. By contrast, expected-rate news allows firms to adjust their financing decisions ahead of higher refinancing costs, providing more persistent protection against financial distress.

Figure 6 reports the ex ante shareholder value of retaining maturity flexibility. Unlike the impulse responses, this calculation integrates over future financing shocks and firm dynamics rather than evaluating a single tightening episode. The value preserved averages 13.5% among indebted firms and increases from 11.8% in the bottom tercile of inherited rollover exposure to 16.9% in the top tercile. Firms facing greater refinancing needs therefore place a higher ex ante value on the ability to adjust repayment schedules, even though the realized investment and default effects depend on the shock.[21]

8.2 Maturity and Interest-Rate Fixation

Long-term fixed-rate debt protects firms along two distinct margins. A longer maturity postpones principal refinancing, while a fixed coupon prevents changes in short rates from resetting interest payments on the outstanding debt stock. To separate these margins, we solve an alternative economy in which long-term debt retains the benchmark amortization schedule but its coupon resets with the short rate. Maturity flexibility continues to provide substantial survival protection in this economy: restricting the aggregate maturity response raises five-year cumulative default by 1.2, 4.2, and 1.9 percentage points following current-rate, expected-rate news, and term-premium shocks, respectively (Figure 7). The effect is largest following expected-rate news, showing that firms can insure against rollover risk by changing when principal must be refinanced even when they cannot lock in the interest rate on outstanding debt. Comparing fixed- and floating-coupon economies further shows that coupon fixation provides more default protection when maturity adjustment is restricted. Maturity and rate fixation therefore provide distinct but partially substitutable forms of insurance. Appendix E.4 reports the full investment, financing, maturity, and default responses.

8.3 Implications for Monetary Transmission

The preceding counterfactuals establish that endogenous maturity adjustment materially affects the aggregate investment and default responses to monetary shocks. We now draw two broader implications for monetary transmission. First, the benefits of maturity flexibility depend on the interaction between the value of refinancing insurance and the price of obtaining it. We quantify this interaction by combining monetary shocks with changes in the cost of long-duration finance, retaining the benchmark-versus-restricted comparison of Section 8.1. Second, the real effects of expected-rate news depend on the prevailing price of long-duration finance. Holding the expected-rate innovation and the initial distribution of firm balance sheets fixed, we examine how the same news transmits across different costs of long-duration financing. Finally, we feed the model the realized EA 2022–24 path of its three aggregate states to assess how expected-rate news and duration costs shaped the transmission of the ECB tightening cycle.

[21] The valuation compares a permanent restriction imposed on firms with benchmark balance sheets. The valuation excludes other sources of financial flexibility, including cash buffers, equity issuance, and credit lines, and should not be interpreted as a comprehensive welfare measure.

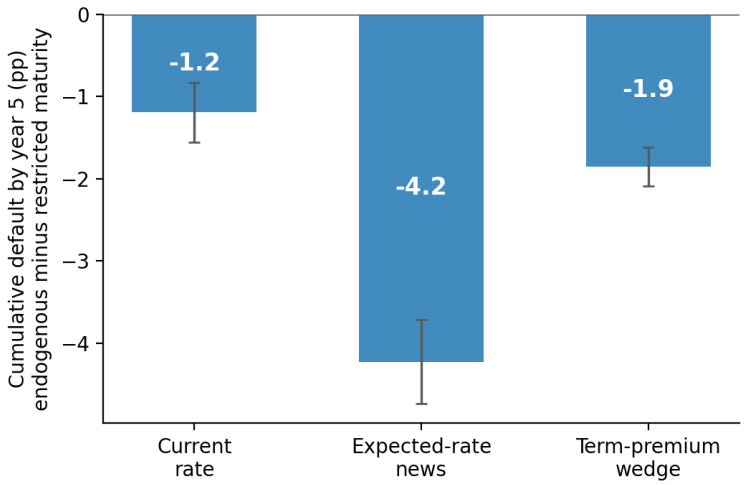


Figure 7: Maturity Flexibility under Floating-Rate Contracts. Bars report the five-year difference in cumulative default responses between the floating-coupon economies with endogenous and restricted maturity, in percentage points. Negative values indicate lower default with maturity flexibility. Long-term debt retains the benchmark amortization schedule, but its coupon resets with the short rate. Whiskers denote 95% Monte Carlo confidence intervals across 1,000 simulated economies.

Maturity flexibility under joint tightening. The transmission of monetary policy depends not only on expected future refinancing costs but also on the price of long-duration finance. Both increased during the 2022–23 tightening cycle. We therefore extend the counterfactual of Section 8.1 by combining either a one-standard-deviation expected-rate news shock or a current-rate shock with a one-standard-deviation increase in the term-premium wedge. Table 14 reports the value of maturity flexibility under these joint shocks. When expected refinancing costs and the price of duration rise together, maturity flexibility becomes more valuable for investment. Under expected-rate news alone, flexibility preserves 2.4 percentage points of cumulative investment after three years; with higher duration costs, this benefit increases to 3.9 points. The long-term share of new issuance falls 16.4 percentage points more in the flexible than in the restricted economy on impact, compared with 8.8 points under news alone (Table 13). Firms thus use maturity flexibility to avoid costly long-duration borrowing while preserving investment. Five-year cumulative default remains 2.8 percentage points lower in the flexible economy, compared with 2.7 points under news alone.

The interaction between duration costs and maturity flexibility is substantially stronger when monetary tightening raises expected future refinancing costs. Adding the same term-premium wedge to a current-rate shock increases the three-year investment benefit of maturity flexibility by only 0.1 percentage points, compared with 1.5 points under expected-rate news. The mirror experiment with lower duration costs confirms this distinction (Appendix Table 32). Flexibility does not uniformly reduce cumulative default: under the joint current-rate shock, it preserves 4.5 points of investment after three years, but five-year cumulative default is 1.5 points higher than in the restricted economy. This reversal reflects the restricted economy's sharper deleveraging and front-loaded defaults, followed by lower default rates among surviving firms, rather than a

Table 14: Value of Maturity Flexibility under Joint Long-Rate Tightening

Horizon (years)	1	3	5
Investment (cumulative, pp)			
Expected-rate news + higher duration costs	−0.5 (0.3)	+3.9 (0.6)	+2.2 (0.6)
Current rate + higher duration costs	+1.9 (0.6)	+4.5 (0.5)	+4.7 (0.4)
Debt issuance (cumulative, pp)			
Expected-rate news + higher duration costs	−2.6 (0.4)	+4.3 (0.7)	+7.0 (0.9)
Current rate + higher duration costs	−0.3 (0.7)	+7.6 (0.8)	+13.3 (0.8)
Long-term issuance share (response at horizon, pp)			
Expected-rate news + higher duration costs	−16.4 (0.8)	−12.8 (0.4)	−7.6 (0.3)
Current rate + higher duration costs	−9.6 (0.7)	−4.6 (0.5)	+1.1 (0.3)
Credit spread (average through horizon, pp)			
Expected-rate news + higher duration costs	−1.1 (0.8)	−5.3 (0.4)	−2.7 (0.2)
Current rate + higher duration costs	−6.5 (0.6)	−2.6 (0.3)	−0.5 (0.2)
Default (cumulative, pp)			
Expected-rate news + higher duration costs	—	−1.2 (0.2)	−2.8 (0.4)
Current rate + higher duration costs	—	−0.5 (0.2)	+1.5 (0.4)

Notes: Entries are flexible-minus-restricted differences in shock responses, as in Table 13, to a one-standard-deviation expected-rate news shock or current-rate shock combined with a one-standard-deviation increase in the term-premium wedge in the shock year; each factor subsequently follows its estimated law of motion. Responses to each monetary shock in isolation are reported in Table 13. Restricted firm policies and debt-price schedules are the same as in that table, as are units, horizons, and the simulation design. Investment, debt issuance, and default are cumulative through the reported horizon; the long-term issuance share is the response at that horizon; credit spreads are averaged through the horizon. Horizon-1 default is omitted because default does not respond on impact under the model's timing. Monte Carlo standard errors of the mean are in parentheses.

subsequent wave of defaults in the flexible economy.

State-dependent costs of forward guidance. We next examine whether the same news about future policy rates has different real effects depending on the prevailing price of long-duration finance. We subject the benchmark economy to an identical one-standard-deviation expected-rate news shock, starting from a low and a high term-premium wedge state, $\omega_0 \in \{0.005, 0.065\}$, corresponding to annualized yield premia of approximately 0.1 and 1.2 percent. We hold the initial distribution of firms' balance sheets fixed across states and allow the term-premium wedge to revert toward its mean according to its estimated law of motion. Responses are measured relative to the corresponding no-news path.[22]

[22]The exercise compares different initial aggregate financing conditions while holding the expected-rate innovation and firms' inherited balance sheets fixed. It does not compare economies that have permanently adapted to different

The state dependence is particularly pronounced in firms' maturity choices (Appendix Figure 17). When duration is cheap, the long-term share of new issuance falls by 7.5 percentage points on impact and remains 2.1 points below its no-news path after three years. When duration is expensive, the corresponding declines are 13.1 and 10.8 points. Thus, costly duration induces both sharper and more persistent maturity shortening. The real effects are also larger: three-year cumulative investment falls by 5.7 percentage points relative to capital in the low-cost state and by 8.9 points in the high-cost state, a 3.2-point difference. Cumulative default rises by 0.4 and 0.9 percentage points in the low- and high-cost states, respectively. Under a current-rate shock the investment ordering is the same, but the maturity response is small in both states (Appendix E.3). These results show that the real effects of forward guidance depend on the price of maturity insurance. When duration is cheap, firms can respond to higher expected future rates while retaining relatively affordable long-term financing. When duration is costly, the same news induces sharper and more persistent maturity shortening and a larger investment contraction. This exercise establishes state dependence in equilibrium transmission, rather than isolating the contribution of maturity adjustment from other financing channels. The price of long-duration finance is therefore an important state variable for assessing the real effects of forward guidance via corporate balance sheets.[23]

The 2022–24 tightening through the model. We conclude by examining how the composition of monetary tightening shaped firms' financing decisions during the 2022–24 euro-area episode. We feed the realized annual paths of the short rate, expected-rate news, and the term-premium wedge into the model, constructed from the one-year OIS rate and the Joslin et al. (2011) decomposition, starting from balance sheets conditioned on the 2021 aggregate state. We compare the realized short-rate path alone with paths that also incorporate expected-rate news and changes in duration costs (Appendix Figure 18 and Table 34). These are conditional experiments that isolate the role of each component within the model, not an additive decomposition and not an identified estimate of the effect of news. Appendix E.5 reports the full responses and discusses their magnitudes.

The exercise reveals a role for anticipatory balance-sheet adjustment. Under the realized short-rate path alone, firms do not adjust borrowing in 2022, before most of the tightening materializes, and the 2023 short-rate increase then raises cumulative default by 4.3 percentage points. Adding expected-rate news brings financing decisions forward: firms enter 2023 with 4.5 percent less outstanding debt, and cumulative default falls by approximately 3.3 percentage points relative to the rate-only path, with little change in cumulative investment. Although firms shorten new borrowing, the principal due for refinancing in 2023 declines by only about 1 percent. The improvement in financial resilience therefore operates through *anticipatory deleveraging* rather than through postponing principal repayments. This adjustment depends on maturity flexibility. When maturity choices cannot respond to aggregate financing conditions, expected-rate news does not induce the

duration-cost environments.

[23]The exercise does not determine the optimal degree of forward guidance, since the model does not quantify the broader stabilization benefits of policy communication.

same reduction in borrowing ahead of tightening and lowers cumulative default by only 0.4 percentage points. Endogenous maturity adjustment thus affects not only when firms refinance but also how much they borrow in anticipation of future financing conditions.

Higher duration costs have a different effect: adding them to the realized rate path deepens the cumulative investment contraction by 2.9 points and nearly doubles the average shortening of new borrowing, from 14 to 27 points, with little effect on default. Along the full realized path, maturity flexibility preserves 6.7 percentage points of cumulative investment and avoids 7.3 points of cumulative default relative to the restricted economy. The episode therefore illustrates how expected-rate news and duration costs jointly shape financing decisions, with maturity flexibility affecting both the timing of refinancing and the evolution of firms' balance sheets.[24] The magnitudes are conditional model predictions rather than estimates of the realized aggregate effects of the ECB tightening: the episode involves unusually large shocks, and the model abstracts from general-equilibrium adjustment. The exercise is informative about the timing and the mechanisms of firms' financing responses.

8.4 Additional Counterfactuals and Diagnostics

Appendix E reports additional exercises that clarify the interpretation of the main counterfactual. First, we verify that its investment effects do not arise mechanically from changes in the composition of surviving firms: the investment-preservation result remains when outcomes are computed for the initial cohort and when the capital of defaulting firms is counted as destroyed. We then isolate the role of inherited refinancing exposure. Holding the remaining firm state fixed, firms entering a financing shock with more principal coming due reduce investment more on impact. Yet reallocating a given aggregate maturity wall across firms has only modest aggregate effects when firms remain free to reoptimize maturity after the shock. Inherited exposure therefore shapes firm-level vulnerability, but its cross-sectional distribution is not persistent: subsequent maturity choices endogenously rebuild refinancing exposure. This distinction also explains why the raw cross-sectional relation between rollover exposure and default need not reveal the structural effect of exposure. Rollover exposure is itself an equilibrium outcome; holding productivity, capital, and debt fixed, default risk rises sharply with the share of principal coming due, even though the corresponding relationship is not monotone in the equilibrium cross-section. Finally, we document the endogenous price of obtaining maturity insurance. Long-term debt prices are substantially more sensitive to additional issuance than short-term debt prices, and this sensitivity rises with refinancing needs. Thus, the firms for whom long maturity provides the greatest protection also face a steeper price schedule for obtaining that protection.

[24]The annual model compresses the sequence of expectations repricing and policy-rate increases within 2022, so that news arrives one period before the rate increase. The 6.7-point investment benefit of maturity flexibility is measured relative to the restricted economy, whereas the responses in Appendix Figure 18 are measured relative to the benchmark economy's no-shock path.

Taking stock.

The quantitative analysis delivers three main findings. *First*, endogenous maturity adjustment materially shapes monetary transmission. Conditional on firms' inherited balance sheets, restricting the response of maturity choices to aggregate financing conditions deepens the three-year cumulative investment contraction by 1.6–4.4 percentage points of capital across the three financing shocks, including 2.4 points following news of higher future rates. Firms with greater inherited refinancing needs place a higher ex ante value on this flexibility. Maturity therefore matters not only as an inherited source of exposure but also as a margin through which firms adjust their financing decisions as monetary conditions change.

Second, maturity insurance is distinct from interest-rate fixation. Longer debt postpones principal refinancing even when coupons reset with the short rate. In the floating-coupon economy, restricting maturity adjustment raises five-year cumulative default by 1.2–4.2 percentage points across the three financing shocks. Principal maturity and coupon repricing thus provide distinct, partially substitutable forms of protection against monetary tightening.

Third, the value of maturity flexibility depends on the price of long-duration finance. When news of higher future rates coincides with higher duration costs, firms shorten new borrowing more aggressively, yet maturity flexibility preserves 3.9 rather than 2.4 percentage points of three-year cumulative investment relative to the restricted economy. Firms benefit not only from the ability to postpone refinancing but also from the option to avoid expensive maturity protection. The same interaction makes the real effects of expected-rate guidance state dependent: holding the expected-rate innovation fixed, three-year cumulative investment falls by 5.7 percentage points when duration is cheap, compared with 8.9 points when it is expensive.

The 2022–24 tightening illustrates the relevance of these mechanisms. Expected-rate news induces firms to reduce borrowing ahead of the realized rate increases, limiting subsequent defaults even as they shorten maturity. Higher duration costs instead amplify the investment contraction by making long-term financing more expensive. Along the full tightening path, maturity flexibility preserves 6.7 percentage points of cumulative investment relative to the restricted economy. These results show that maturity adjustment shapes not only when firms refinance but also how much they borrow and how they absorb monetary shocks.

The policy implication is that the expected path of short rates alone is insufficient to assess monetary transmission. The price of long-duration finance determines the cost of refinancing protection, while corporate repayment schedules and endogenous maturity choices shape firms' exposure to future financing conditions. Assessing the strength and timing of monetary transmission therefore requires considering these dimensions jointly.

9 Conclusion

This paper identifies a maturity-insurance channel of monetary transmission. Firms enter monetary-policy cycles with repayment schedules inherited from past borrowing, but their exposure to future financing conditions is not fixed. By choosing the maturity of new debt, firms determine when they will next refinance and reshape their exposure to future monetary conditions. Using euro-area credit-registry data matched to firm balance sheets, we show that firms with greater refinancing needs obtain longer maturity on new borrowing, but their response depends on why long-term interest rates rise. Higher expected future short rates increase the value of postponing refinancing and strengthen maturity extension among exposed firms, whereas higher term premia raise the cost of long-duration financing and weaken that response.

A quantitative model shows that endogenous maturity adjustment materially shapes monetary transmission. Restricting maturity choices from responding to aggregate financing conditions amplifies the investment contraction and financial distress following anticipated tightening. Maturity insurance operates through the timing of principal refinancing and remains valuable even when borrowing rates can reset. Moreover, its value depends on the price of long-duration finance. When expected refinancing costs and duration costs rise together, firms use maturity flexibility to avoid expensive long-term borrowing while preserving investment. Maturity flexibility thus provides not only the ability to postpone refinancing but also the option to economize on that protection when it becomes costly. The 2022–24 euro-area tightening illustrates the relevance of this mechanism. Feeding the realized paths of short rates, expected future rates, and duration costs into the model shows that their composition shaped the transmission of tightening. Expected-rate news induced firms to reduce borrowing ahead of the realized rate increases, improving balance-sheet resilience and limiting subsequent defaults, even as firms shortened new debt maturity. Higher duration costs instead amplified maturity shortening and the investment contraction. The episode demonstrates that maturity flexibility changes not only when firms refinance but also how much they borrow as monetary conditions evolve.

These findings have implications for assessing monetary transmission. The real effects of expected-rate guidance are state dependent: the same increase in expected future short rates generates a larger investment contraction when long-duration financing is expensive. The expected policy-rate path alone is therefore insufficient to characterize corporate transmission. The price of long-duration finance and firms' inherited repayment schedules jointly shape their incentives and ability to adjust liabilities. More broadly, debt maturity is not merely an inherited source of exposure to monetary policy, but an endogenous financing margin that determines how firms absorb and propagate monetary shocks.

References

Acharya, V. V., Byoun, S., and Xu, Z. (2020). The sensitivity of cash savings to the cost of capital. NBER Working paper No. w27517.

Adrian, T., Crump, R. K., and Moench, E. (2013). Pricing the term structure with linear regressions. *Journal of Financial Economics*, 110(1):110–138.

Almeida, H., Campello, M., Laranjeira, B., and Weisbenner, S. (2012). Corporate debt maturity and the real effects of the 2007 credit crisis. *Critical Finance Review*, 1(1):3–58.

Altavilla, C., Brugnolini, L., Gürkaynak, R. S., Motto, R., and Ragusa, G. (2019). Measuring euro area monetary policy. *Journal of Monetary Economics*, 108:162–179.

Altavilla, C., Gürkaynak, R. S., Kind, T., and Laeven, L. (2026). Monetary transmission with frequent policy events. *Journal of Monetary Economics*, 163:103986.

Altinkilic, O. and Hansen, R. S. (2000). Are there economies of scale in underwriting fees? evidence of rising external financing costs. *The Review of Financial Studies*, 13(1):191–218.

Anderson, G. and Cesa-Bianchi, A. (2024). Crossing the credit channel: Credit spreads and firm heterogeneity. *American Economic Journal: Macroeconomics*, 16(3):417–446.

Baker, M., Greenwood, R., and Wurgler, J. (2003). The maturity of debt issues and predictable variation in bond returns. *Journal of Financial Economics*, 70(2):261–291.

Bauer, M. D., Lakdawala, A., and Mueller, P. (2022). Market-based monetary policy uncertainty. *The Economic Journal*, 132(644):1290–1308.

Berg, T., Saunders, A., and Steffen, S. (2016). The total cost of corporate borrowing in the loan market: Don't ignore the fees. *The Journal of Finance*, 71(3):1357–1392.

Bernanke, B. S. and Gertler, M. (1995). Inside the black box: The credit channel of monetary policy transmission. *Journal of Economic Perspectives*, 9(4):27–48.

Bolton, P., Chen, H., and Wang, N. (2013). Market timing, investment, and risk management. *Journal of Financial Economics*, 109(1):40–62.

Bräuning, F., Fillat, J. L., and Wang, J. C. (2020). Corporate debt maturity and monetary policy. Current policy perspectives, Federal Reserve Bank of Boston.

Brunnermeier, M. K. and Oehmke, M. (2013). The maturity rat race. *Journal of Finance*, 68(2):483–521.

Chatterjee, S. and Eyigungor, B. (2012). Maturity, indebtedness, and default risk. *American Economic Review*, 102(6):2674–2699.

Cooley, T. F. and Quadrini, V. (2001). Financial markets and firm dynamics. *American Economic Review*, 91(5):1286–1310.

Cooper, R. W. and Haltiwanger, J. C. (2006). On the nature of capital adjustment costs. *Review of Economic Studies*, 73(3):611–633.

Core, F., De Marco, F., Eisert, T., and Schepens, G. (2025). Inflation and floating-rate loans: Evidence from the euro area. Working paper, European Central Bank. Available at SSRN.

Degryse, H., De Jonghe, O., Jakovljević, S., Mulier, K., and Schepens, G. (2019). Identifying credit supply shocks with bank-firm data: Methods and applications. *Journal of Financial Intermediation*, 40:100813.

Deng, M. and Fang, M. (2022). Debt maturity heterogeneity and investment responses to monetary policy. *European Economic Review*, 144:104095.

Diamond, D. W. (1991). Debt maturity structure and liquidity risk. *Quarterly Journal of Economics*, 106(3):709–737.

Duchin, R., Ozbas, O., and Sensoy, B. A. (2010). Costly external finance, corporate investment, and the subprime mortgage credit crisis. *Journal of Financial Economics*, 97(3):418–435.

Eisfeldt, A. L. and Muir, T. (2016). Aggregate external financing and savings waves. *Journal of Monetary Economics*, 84:116–133.

Ericson, K. M. and Sydnor, J. (2026). Liquidity constraints and the value of insurance. *American Economic Journal: Microeconomics*, 18(3):426–453.

Fabiani, A., Heineken, J., and Falasconi, L. (2024). Monetary policy and the maturity structure of corporate debt. Unpublished manuscript.

Faulkender, M. (2005). Hedging or market timing? selecting the interest rate exposure of corporate debt. *The Journal of Finance*, 60(2):931–962.

Ferrante, F., Prestipino, A., and Schott, I. (2025). Corporate debt maturity and business cycle fluctuations. Technical Report 1409, Board of Governors of the Federal Reserve System.

Flannery, M. J. (1986). Asymmetric information and risky debt maturity choice. *Journal of Finance*, 41(1):19–37.

Gertler, M. and Gilchrist, S. (1994). Monetary policy, business cycles, and the behavior of small manufacturing firms. *Quarterly Journal of Economics*, 109(2):309–340.

Gomes, J. F. (2001). Financing investment. *American Economic Review*, 91(5):1263–1285.

Greenwood, R., Hanson, S. G., and Stein, J. C. (2010). A gap-filling theory of corporate debt maturity choice. *Journal of Finance*, 65(3):993–1028.

Greenwood, R. and Vayanos, D. (2014). Bond supply and excess bond returns. *The Review of Financial Studies*, 27(3):663–713.

Gürkaynak, R. S., Sack, B., and Swanson, E. T. (2005). Do actions speak louder than words? the response of asset prices to monetary policy actions and statements. *International Journal of Central Banking*, 1(1):55–93.

Gürkaynak, R. S., Sack, B., and Wright, J. H. (2010). The TIPS yield curve and inflation compensation. *American Economic Journal: Macroeconomics*, 2(1):70–92.

Hanson, S. G. and Stein, J. C. (2015). Monetary policy and long-term real rates. *Journal of Financial Economics*, 115(3):429–448.

He, Z. and Xiong, W. (2012). Rollover risk and credit risk. *Journal of Finance*, 67(2):391–430.

Hennessy, C. A. and Whited, T. M. (2005). Debt dynamics. *Journal of Finance*, 60(3):1129–1165.

Ippolito, F., Ozdagli, A. K., and Pérez-Orive, A. (2018). The transmission of monetary policy through bank lending: The floating rate channel. *Journal of Monetary Economics*, 95:49–71.

Jarociński, M. and Karadi, P. (2020). Deconstructing monetary policy surprises. *American Economic Journal: Macroeconomics*, 12(2):1–43.

Jarociński, M. and Karadi, P. (2020). Deconstructing Monetary Policy Surprises—The Role of Information Shocks. *American Economic Journal: Macroeconomics*, 12(2):1–43.

Jiménez, G., Ongena, S., Peydró, J.-L., and Saurina, J. (2012). Credit supply and monetary policy: Identifying the bank balance-sheet channel. *American Economic Review*, 102(5):2301–2326.

Joslin, S., Singleton, K. J., and Zhu, H. (2011). A new perspective on gaussian dynamic term structure models. *Review of Financial Studies*, 24(3):926–970.

Jungherr, J., Meier, M., Reinelt, T., and Schott, I. (2024). Corporate debt maturity matters for monetary policy. Working paper.

Jungherr, J. and Schott, I. (2021). Optimal debt maturity and firm investment. *Review of Economic Dynamics*, 42:110–132.

Kalemli-Özcan, Ş., Laeven, L., and Moreno, D. (2022). Debt overhang, rollover risk, and corporate investment: Evidence from the european crisis. *Journal of the European Economic Association*, 20(6):2353–2395.

Kalemli-Özcan, Ş., Sørensen, B. E., Villegas-Sanchez, C., Volosovych, V., and Yeşiltaş, S. (2024). How to construct nationally representative firm-level data from the orbis global database: New facts on smes and aggregate implications for industry concentration. *American Economic Journal: Macroeconomics*, 16(2):353–374.

Kashyap, A. K. and Stein, J. C. (2000). What do a million observations on banks say about the transmission of monetary policy? *American Economic Review*, 90(3):407–428.

Keen, B. D., Richter, A. W., and Throckmorton, N. A. (2017). Forward guidance and the state of the economy. *Economic Inquiry*, 55(4):1593–1624.

Khan, A. and Thomas, J. K. (2013). Credit shocks and aggregate fluctuations in an economy with production heterogeneity. *Journal of Political Economy*, 121:1055–1107.

Khwaja, A. I. and Mian, A. (2008). Tracing the impact of bank liquidity shocks: Evidence from an emerging market. *American Economic Review*, 98(4):1413–1442.

Koop, G., Pesaran, M. H., and Potter, S. M. (1996). Impulse response analysis in nonlinear multivariate models. *Journal of Econometrics*, 74:119–147.

Kozlowski, J. (2021). Long-term finance and investment with frictional asset markets. *American Economic Journal: Macroeconomics*, 13(4):411–448.

Leland, H. E. (1994). Corporate debt value, bond covenants, and optimal capital structure. *Journal of Finance*, 49(4):1213–1252.

Lunsford, K. G. (2020). Policy language and information effects in the early days of federal reserve forward guidance. *American Economic Review*, 110(9):2899–2934.

McKay, A., Nakamura, E., and Steinsson, J. (2016). The power of forward guidance revisited. *American Economic Review*, 106(10):3133–3158.

Mian, A. and Santos, J. A. C. (2018). Liquidity risk and maturity management over the credit cycle. *Journal of Financial Economics*, 127(2):264–284.

Myers, S. C. (1977). Determinants of corporate borrowing. *Journal of Financial Economics*, 5(2):147–175.

Ottonello, P. and Winberry, T. (2020). Financial heterogeneity and the investment channel of monetary policy. *Econometrica*, 88(6):2473–2502.

Poeschl, J. (2023). Corporate debt maturity and investment over the business cycle. *European Economic Review*, 152:104348.

Salomão, J. and Varela, L. (2022). Exchange rate exposure and firm dynamics. *The Review of Economic Studies*, 89(1):481–514.

Vayanos, D. and Vila, J.-L. (2021). A preferred-habitat model of the term structure of interest rates. *Econometrica*, 89(1):77–112.

Xu, Q. (2018). Kicking maturity down the road: Early refinancing and maturity management in the corporate bond market. *The Review of Financial Studies*, 31(8):3061–3097.

A Empirics Appendix

A.1 Tables

Table 15: Summary statistics

Variable	Obs.	Mean	Std. Dev.	Min	p25	Median	p75	Max
Loan-level variables (monthly)								
Loan maturity, days	14,792,489	1,895	1,525	29	113.42	737	1,836	7,330
Annualized interest rate	9,317,565	0.033	0.023	0	0.015	0.030	0.049	0.137
Log(Total credit)	11,319,955	11.12	1.92	3.26	10.05	11.00	12.21	16.32
Probability of default	4,765,303	0.050	0.146	0	0.004	0.011	0.031	1
Short-Term Debt (monthly)	11,319,955	0.28	0.34	0	0	0.12	0.50	1
Maturity Wall (monthly)	11,319,955	0.34	0.37	0.00	0.18	0.66	0.80	1
Firm-level variables (yearly)								
Short-Term Debt (annual)	10,457,475	0.18	0.31	0.00	0.00	0.00	0.20	1.00
Interest expense/assets	6,364,645	0.01	0.01	0.00	0.00	0.01	0.01	0.08
Debt-service ratio	6,364,462	0.44	0.28	0.01	0.21	0.40	0.62	1.24
Interest coverage ratio	6,087,148	40.74	110.66	-55.96	2.44	8.54	28.38	640.43
Δ Investment rate	7,374,655	0.19	1.03	-0.93	-0.11	-0.02	0.10	7.70
Δ Bank Maturity, years	3,478,903	-0.10	4.25	-24.85	-1.97	0.00	1.94	24.83
Maturity extension dummy	10,457,475	0.67	0.47	0.00	0.00	1.00	1.00	1.00
Δ long-term bank loans	3,653,379	0.01	0.57	-1.00	0.00	0.00	0.06	1.00
Δ bank loans	1,585,898	2.59	8.61	-1.00	-0.68	-0.04	1.59	50.30
Δ net total debt	6,957,930	0.00	1.43	-7.08	-0.24	-0.01	0.18	7.45
Log total assets	10,452,781	13.72	1.55	10.33	12.66	13.60	14.65	18.26
Tangibility	10,247,641	0.30	0.30	0.00	0.05	0.20	0.50	0.99
Cash/assets	9,853,643	0.15	0.18	0.00	0.02	0.09	0.23	0.78
Sales growth	4,721,460	0.23	1.24	-0.99	-0.21	0.00	0.25	9.26
Leverage	10,452,781	0.70	0.36	0.05	0.47	0.69	0.88	2.48
Time-level variables (monthly)								
Yield Curve 5Y–1Y	76	-0.16	0.55	-1.51	-0.42	-0.05	0.17	1.02
Yield Curve 10Y–2Y	76	0.18	0.53	-0.94	-0.18	0.25	0.55	1.10
Expected Rate Path 5Y, JSZ	76	1.133	1.005	-0.033	0.336	0.495	2.217	3.020
Expected Rate Path 5Y, ACM	76	1.047	0.965	-0.023	0.269	0.413	2.136	2.954
Expected Rate Path 10Y, JSZ	76	1.159	0.567	0.422	0.721	0.842	1.770	2.278
Expected Rate Path 10Y, ACM	76	1.041	0.525	0.392	0.641	0.738	1.618	2.141
Term Premium 5Y, JSZ	76	-0.531	0.506	-1.296	-1.039	-0.433	-0.134	0.455
Term Premium 5Y, ACM	76	-0.423	0.532	-1.182	-0.938	-0.354	-0.035	0.713
Term Premium 10Y, JSZ	76	-0.276	0.801	-1.411	-1.073	-0.249	0.459	1.172
Term Premium 10Y, ACM	76	-0.178	0.847	-1.335	-1.005	-0.223	0.617	1.305
Target shock	76	0.13	1.09	-1.88	-0.15	0.00	0.00	8.00
Timing shock	76	0.02	0.34	-1.14	-0.03	0.00	0.15	1.32
Forward guidance shock	76	0.05	0.87	-2.61	-0.26	0.00	0.10	2.96
Quantitative policy shock	76	-0.06	0.73	-3.13	-0.23	0.00	0.14	1.88

Notes: This table reports the summary statistics of the variables used in the empirical analysis. Loan-level and firm-level variables are winsorized at the 1st and 99th percentiles.

Table 16: Firm Characteristics and Rollover Exposure

	Low exposure $\leq p25$	High exposure $\geq p75$	Mean Difference High − Low
Fixed assets$_{f,t-1}$	12.325	12.266	-0.059^{***}
Tangibility$_{f,t-1}$	0.347	0.233	-0.114^{***}
Sales growth$_{f,t-1}$	20.569	18.106	-2.463
Leverage$_{f,t-1}$	0.715	0.673	-0.041^{***}
Interest coverage$_{f,t-1}$	50.808	61.709	10.901^{***}

Notes: This table compares firms in the bottom and top quartiles ($\leq$p25 and $\geq$p75) of the within-year distribution of rollover exposure proxied by short-term bank debt, calculated from AnaCredit. Firm characteristics are measured in the previous year; the final column reports the difference in group means, with standard errors clustered at the firm level. Sample period 2018–2023. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 17: Rollover Exposure and Floating-Rate New Loans

Floating-rate Share of New Loans	(1) 5Y–1Y	(2) 10Y–2Y
ST Debt$_{f,t-1}$	0.0246***	0.0283***
	(0.0011)	(0.0012)
YC$_{t-1}\times$ ST Debt$_{f,t-1}$	-0.0121***	-0.0137***
	(0.0015)	(0.0017)
Observations	1,674,254	1,674,254
Adjusted R^2	0.696	0.696
Bank × Month FE	Yes	Yes
Firm × Bank FE	Yes	Yes
Firm Controls	Yes	Yes

Notes: The dependent variable is the share of new loans with a floating rate over total new loans. Column (1) uses the 5Y–1Y yield-curve slope and column (2) uses the 10Y–2Y slope as robustness. All specifications include lagged firm controls, bank×month fixed effects, and firm×bank fixed effects. Standard errors are clustered at the firm and month level.
*, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 18: Robustness of the baseline result

$\log(\text{Maturity}_{bft})$	(1) Excl. COVID period	(2) Richer supply FE	(3) New credit	(4) Alt. slope	(5) Policy rate	(6) Bank controls	(7) Firm × year FE
ST $\text{debt}_{f,t-1}$	0.131***	0.242***	0.230***	0.252***	0.226***	−0.427***	0.438***
	(0.00489)	(0.00795)	(0.00436)	(0.0059)	(0.00432)	(0.00342)	(0.00733)
Yield Curve Slope_{t-1}× ST $\text{debt}_{f,t-1}$	−0.0224***	−0.0219***	−0.0175***	−0.0812***	−0.0180**	−0.150***	−0.0163**
	(0.00463)	(0.00811)	(0.00427)	(0.00503)	(0.00704)	(0.00611)	(0.00745)
DFR_{t-1}× ST $\text{debt}_{f,t-1}$				0.0050	0.00152		
				(0.0086)	(0.00283)		
Firm × bank FE	Yes	Yes	Yes	Yes	Yes	Yes	No
Firm × year FE	No	No	No	No	No	No	Yes
Bank × month FE	Yes	No	Yes	Yes	Yes	No	Yes
Bank × ind. × loc. × month FE	No	Yes	No	No	No	No	No
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bank controls	No	No	No	No	No	Yes	No
New credit control	No	No	Yes	No	No	No	No
Yield-curve slope	5Y–1Y	5Y–1Y	5Y–1Y	10Y–2Y	5Y–1Y	5Y–1Y	5Y–1Y
Observations	1,139,504	992,254	1,674,254	1,674,254	1,674,254	1,765,750	1,624,347
Adjusted R^2	0.613	0.583	0.620	0.620	0.620	0.056	0.607

Notes: Column (1) estimates the baseline specification excluding the COVID period, defined as 2019Q4–2021Q1. Column (2) replaces bank × month fixed effects with bank × industry × location × month fixed effects, so that identification comes from firms borrowing from the same lender, in the same industry and region, in the same month. Column (3) adds total new credit as a control. Column (4) uses the 10Y–2Y slope instead of the 5Y–1Y slope and adds the lagged deposit facility (DFR) rate interacted with rollover exposure. Column (5) adds the lagged deposit facility rate interacted with rollover exposure. Column (6) replaces bank × month fixed effects with observable bank controls. Column (7) is the most saturated specification: firm × year fixed effects absorb annual investment plans, asset composition, and cash-flow expectations at the firm level, so identification comes from within-year monthly variation in the term structure among firms borrowing from the same bank in the same month. Standard errors are clustered at the firm and month level. *, **, *** denote significance at the 10%, 5%, and 1% levels.

Table 19: High-Frequency Monetary Policy Factors across the Yield Curve

	1M	3M	6M	1Y	2Y	5Y	10Y
Target	3.083***	2.638***	2.241***	1.560***	0.860	0.282	-0.047
	(0.071)	(0.228)	(0.216)	(0.383)	(0.562)	(0.527)	(0.284)
Timing	0.823***	1.225***	1.413***	1.704***	1.520***	1.095***	0.490**
	(0.109)	(0.107)	(0.097)	(0.176)	(0.205)	(0.197)	(0.221)
Forward Guidance	0.028	0.092	1.036***	2.108***	3.011***	3.349***	2.656***
	(0.083)	(0.287)	(0.283)	(0.406)	(0.410)	(0.375)	(0.274)
Quantitative Policy	-0.054	-0.429***	-1.213***	-2.025***	-2.258***	-1.206***	0.024
	(0.081)	(0.100)	(0.139)	(0.296)	(0.305)	(0.313)	(0.249)
Observations	128	128	128	128	128	127	128
R^2	0.965	0.831	0.801	0.659	0.589	0.579	0.618

Notes: The table validates the high-frequency monetary policy factors of Altavilla et al. (2019) by relating them to changes in OIS yields at different maturities. Each entry reports the response of the corresponding OIS yield to a one-standard-deviation monetary policy factor. Target primarily loads on the short end of the yield curve, Timing affects short- and intermediate-maturity rates, Forward Guidance loads most strongly on intermediate maturities, and Quantitative Policy primarily affects intermediate-maturity yields. The sample covers ECB monetary policy events from July 2011 to October 2025. Standard errors are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 20: Rollover Exposure and Protection Value

$\log(\text{Protection Value})_{bft}$	(1)
$\text{ST Debt}_{f,t-1}$	0.306***
	(0.0105)
$\text{Yield Curve}_{t-1} \times \text{ST Debt}_{f,t-1}$	-0.00113
	(0.0133)
Observations	675,691
Adjusted R^2	0.706
Bank × Month FE	Yes
Firm × Bank FE	Yes
Firm Controls	Yes

Notes: The dependent variable is the logarithm of the protection value reported in AnaCredit, which measures the monetary value of collateral or guarantees securing bank credit, based on their reported notional amount or most recent valuation. Short-Term Debt proxy rollover exposure as the share of the firm's outstanding bank debt maturing within the following 12 months. Yield Curve is the lagged 5Y–1Y yield-curve slope. The specification includes bank×month fixed effects, firm fixed effects, and lagged firm controls. The main effect of the yield curve is absorbed by bank×month fixed effects. Standard errors are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 21: Rollover Exposure and New Loan Maturity by contract rate type

$\log(\text{Maturity}_{bft})$	(1) Floating rate	(2) Fixed rate
ST $\text{Debt}_{f,t-1}$	0.0883***	0.154***
	(0.00740)	(0.00439)
Yield Curve $\text{Slope}_{t-1}\times$ ST $\text{debt}_{f,t-1}$	−0.0165**	−0.0108**
	(0.00668)	(0.00444)
Firm FE	Yes	Yes
Bank × month FE	Yes	Yes
Firm controls	Yes	Yes
Observations	359,175	1,179,346
Adjusted R^2	0.753	0.616

Notes: The sample is split by the interest-rate type of the newly originated loan. Column (1) uses new loans with a floating rate at origination , column (2) new loans with fixed-rate at origination. Firm FE are used to maximize the number of observations. Firm controls are lagged log total assets, cash over assets, interest coverage, tangibility and sales growth. The slope is the lagged 5Y–1Y spread. Standard errors clustered at the firm and month level. *, **, *** denote significance at the 10%, 5% and 1% levels.

Table 22: Rollover Exposure and New Loan Maturity across the maturity distribution

$\mathbf{1}\{\text{Maturity}_{bft} \geq h\}$	≥1Y	≥2Y	≥3Y	≥5Y	≥7Y
ST $\text{Debt}_{f,t-1}$	0.0763***	0.0761***	0.0664***	0.0323***	0.00799***
	(0.00159)	(0.00143)	(0.00135)	(0.00104)	(0.000512)
$\text{Slope}_{t-1}\times$ ST $\text{debt}_{f,t-1}$	−0.0108***	−0.00685***	−0.00427**	0.00206	−0.00114
	(0.00242)	(0.00204)	(0.00192)	(0.00143)	(0.000821)
Bank × month FE	Yes	Yes	Yes	Yes	Yes
Firm × bank FE	Yes	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes	Yes
Observations	1,414,098	1,414,098	1,414,098	1,414,098	1,414,098
Adjusted R^2	0.564	0.543	0.537	0.470	0.569

Notes: Each dependent variable is an indicator equal to one if the new loan has maturity at least the indicated threshold; the yield-curve slope is 5Y–1Y. Standard errors are clustered at the firm and month level. *, **, *** denote significance at the 10%, 5% and 1% levels.

Table 23: Extensive margin: selection into new borrowing

Dependent variable: $\mathbf{1}\{\text{New loan}_{f,t}\}$	(1) 10Y–2Y	(2) 5Y–1Y
ST Debt$_{f,t-1}$	0.0665***	0.0693***
	(0.000291)	(0.000291)
YC$_{t-1}$ × ST Debt$_{f,t-1}$	0.0122***	0.00678***
	(0.000431)	(0.000364)
Firm FE	Yes	Yes
Country × industry × month FE	Yes	Yes
Firm controls	Yes	Yes
Observations	53,200,020	53,200,020
Adjusted R^2	0.301	0.301

Notes: Firm-month linear probability models. The dependent variable equals one if firm f obtains at least one new loan in month t. The yield curve slope is 10Y-2Y in column (1) and 5Y-1Y in column (2). Firm controls include size, cash over assets, interest coverage, and tangibility. Standard errors, in parentheses, are clustered at the firm and month level. *, **, *** denote significance at the 10%, 5%, and 1% level.

Table 24: Real-rate and inflation-compensation components

Dependent variable: log(Loan maturity)$_{bft}$	(1) 10Y–2Y	(2) 5Y–1Y
ST Debt$_{f,t-1}$	0.258***	0.223***
	(0.00491)	(0.00450)
YC$^{real}_{t-1}$ × ST Debt$_{f,t-1}$	−0.0792***	−0.0707***
	(0.00648)	(0.00652)
YC$^{inflation}_{t-1}$ × ST Debt$_{f,t-1}$	−0.114***	−0.0975***
	(0.00857)	(0.00832)
Bank × month FE	Yes	Yes
Firm × bank FE	Yes	Yes
Firm controls	Yes	Yes
Observations	1,674,254	1,674,254
Adjusted R^2	0.620	0.620

Notes: The nominal yield-curve slope is decomposed into real-rate and inflation-compensation components following Gürkaynak et al. (2010), using inflation swaps. Firm controls are included. Standard errors, in parentheses, are clustered at the firm and month level. *, **, *** denote significance at the 10%, 5%, and 1% level.

Table 25: Extended sample using balance-sheet exposure

Dependent variable: $\log(\text{Loan maturity})_{bft}$	(1) 5Y–1Y	(2) 10Y–2Y	(3) 5Y–1Y
ST Debt$_{f,t-1}$	−0.200***	−0.166***	−0.198***
	(0.00879)	(0.00936)	(0.0109)
YC$_{t-1}$ × ST Debt$_{f,t-1}$	−0.0750***	−0.115***	−0.0683*
	(0.0138)	(0.0118)	(0.0363)
Post2020$_t$ × ST Debt$_{f,t-1}$			−0.00234
			(0.00962)
Post2020$_t$ × YC$_{t-1}$ × ST Debt$_{f,t-1}$			−0.00985
			(0.0393)
Bank × month FE	Yes	Yes	Yes
Firm × bank FE	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes
Trade-credit controls	Yes	Yes	Yes
Observations	1,523,207	1,523,207	1,523,207
Adjusted R^2	0.636	0.636	0.636

Notes: The dependent variable is new loan maturity. Rollover exposure is proxied by ST Debt measured as current liabilities over total debt, which is coarser than the loan-level measure and includes trade credit and other short-term obligations. Columns (1)–(2) report the extended-sample estimates using the 5Y–1Y and 10Y–2Y yield-curve slopes. Column (3) tests for a structural break after 2020 by allowing both the level effect of refinancing exposure and its interaction with the yield curve to differ in the post-2020 period. Firm controls include size, cash over assets, sales growth, tangibility, and receivables over assets. Standard errors, in parentheses, are clustered at the firm and month level. *, **, *** denote significance at the 10%, 5%, and 1% level.

Table 26: Yield Decomposition and New Loan Maturity at year level

	(1) log(Maturity)	(2) Maturity, years	(3) LT debt share
ST Debt$_{f,t-1}$	1.171***	3.975***	0.526***
	(0.0135)	(0.751)	(0.00585)
ESR^{5Y}_{t-1} (1 s.d.) × ST Debt$_{f,t-1}$	0.430***	1.921***	0.201***
	(0.0085)	(0.489)	(0.0036)
TP^{5Y}_{t-1} (1 s.d.) × ST Debt$_{f,t-1}$	−0.375***	−1.925***	−0.175***
	(0.0085)	(0.492)	(0.0036)
Firm controls	Yes	Yes	Yes
Observations	1,232,782	1,243,865	1,261,899
Adjusted R^2	0.588	0.262	0.459

Notes: The dependent variables are the volume-weighted maturity of newly originated bank loans and the long-term share of bank debt in columns 1-3 respectively. ST Debt is the lagged share of outstanding bank debt with residual maturity of twelve months or less. ESR^{5Y} and TP^{5Y} are the lagged five-year expected-rate and term-premium components of the Joslin et al. (2011) decomposition. Interaction coefficients are standardized: each reports the effect of a one-standard-deviation change in the lagged factor. Controls include lagged log assets and lagged outstanding loan volume. Standard errors in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels.

Table 27: Rollover Exposure, the Term Structure, and Firm Size

Panel A. Maturity and Liability Restructuring

	(1) log(Maturity) Loan level	(2) Δ Bank-Debt Maturity	(3) Δ Long Term Bank Loans	(4) Δ Total Debt
ST Debt$_{f,t-1}$	0.200***	2.390***	0.577***	-0.0343***
	(0.00450)	(0.0280)	(0.00484)	(0.00230)
YC (5Y–1Y)$_{t-1}$× ST Debt$_{f,t-1}$	-0.0309***	-1.701***	-0.535***	-0.0382***
	(0.00496)	(0.0742)	(0.0125)	(0.00632)
YC (5Y–1Y)$_{t-1}$× ST Debt$_{f,t-1}$× Low TA$_f$	-0.0387***	-1.128***	-0.216***	-0.0472***
	(0.0149)	(0.164)	(0.0308)	(0.0156)
Observations	1,674,254	594,168	594,168	594,168
Adjusted R^2	0.587	0.084	0.044	0.098
Bank × Time FE	Yes	No	No	No
Firm × Bank FE	Yes	No	No	No
Industry × Country × Time FE	No	Yes	Yes	Yes
Firm FE	No	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes	Yes

Panel B. Liquidity, Interest Expenses, and Investment

	Δ Cash /TA	Interest Exp. / EBITDA	Debt-Service Ratio	Investment Rate
ST Debt$_{f,t-1}$	0.140***	-0.00520***	-0.00846***	0.103***
	(0.0523)	(0.000943)	(0.000576)	(0.00961)
YC (5Y–1Y)$_{t-1}$× ST Debt$_{f,t-1}$	-0.442***	0.00920***	0.0554***	-0.124***
	(0.134)	(0.00261)	(0.00158)	(0.0242)
YC (5Y–1Y)$_{t-1}$× ST Debt$_{f,t-1}$× Low TA$_f$	-0.0615	0.0195***	0.0155***	-0.137**
	(0.280)	(0.00650)	(0.00421)	(0.0674)
Observations	594,168	594,168	594,168	594,168
Adjusted R^2	0.029	0.490	0.821	0.137
Industry × Country × Time FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes	Yes

Notes: This table reports how the interaction between rollover exposure and the term structure varies with firm size. Low TA is an indicator for firms in the bottom quartile of total assets. Column (1) is the loan-level maturity regression of Table 1; columns (2)–(8) are firm-level regressions with the outcomes of Table 8 and the investment rate, the latter in the full firm-year sample (7) and in the sample with observed bank-debt maturity changes (8). All specifications include the lower-order interactions of Low TA with rollover exposure and with the slope, lagged firm controls, and the fixed effects indicated. Short-Term Debt is the lagged share of outstanding bank debt with residual maturity within 12 months. YC (5Y–1Y) is the lagged difference between the 5-year AAA euro-area government bond yield and the 1-year OIS rate. Standard errors are clustered at the firm level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 28: Rollover Exposure, Yield Curve, and New Credit

Log New Credit$_{f,b,t}$	(1) Total Credit	(2) Term Loans
ST Debt$_{f,t-1}$	0.381***	0.363***
	(0.00661)	(0.00618)
Yield Curve$_{t-1}^{5Y-1Y}$ × ST Debt$_{f,t-1}$	0.00026	-0.00668
	(0.00827)	(0.00772)
Observations	1,822,621	1,776,785
Adjusted R^2	0.697	0.719

Notes: The dependent variable is the logarithm of newly originated credit at the firm–bank–month level. Column (1) considers total new credit, while column (2) restricts the outcome to new term loans. ST Debt is the lagged share of outstanding bank debt maturing within twelve months. The Yield Curve (5Y–1Y) is the difference between the five-year and one-year yields. Both specifications include firm and bank-by-month fixed effects and firm-level controls. Standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

A.2 Figures

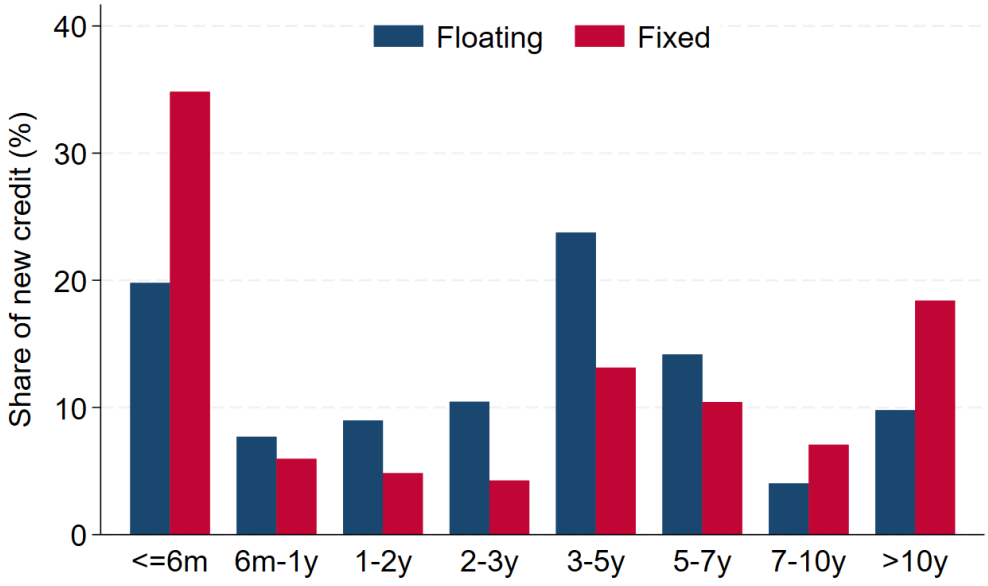
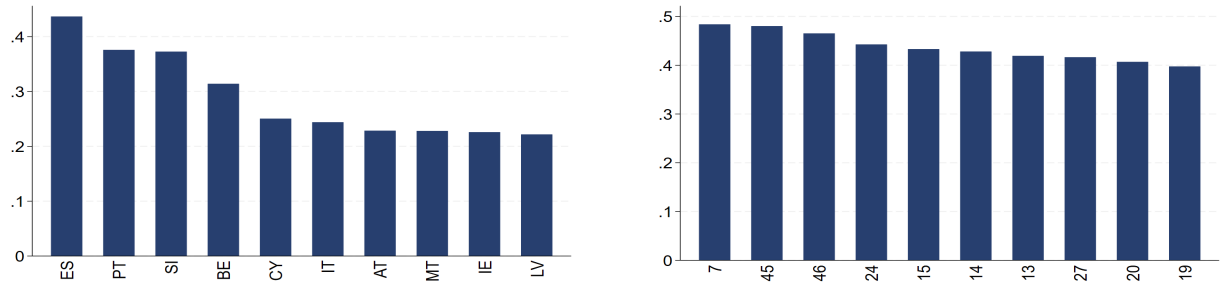


Figure 8: New Loan Maturity by rate type

Notes: Distribution of new credit across original-maturity buckets, separately for floating-rate and fixed-rate originations, September 2018 to December 2024. Maturity is the new-credit-weighted average original maturity of originations within a firm–bank–month; shares are computed over new credit and sum to 100 within each rate type. Fixed-rate credit is bimodal, concentrated at maturities of six months or less (30 percent) and beyond ten years (24 percent), while floating-rate credit is concentrated between one and seven years (57 percent against 32 percent for fixed-rate credit). Duration and coupon fixation are therefore not the same contractual margin.



(b) By country **(c) By sector (NACE2)**

Figure 9: Rollover Exposure by Country and Sector

Notes: Median short-term debt, the share of outstanding debt with residual maturity up to 12 months excluding credit lines, by country (left) and by sector, NACE2 (right), in 2018. Source: AnaCredit.

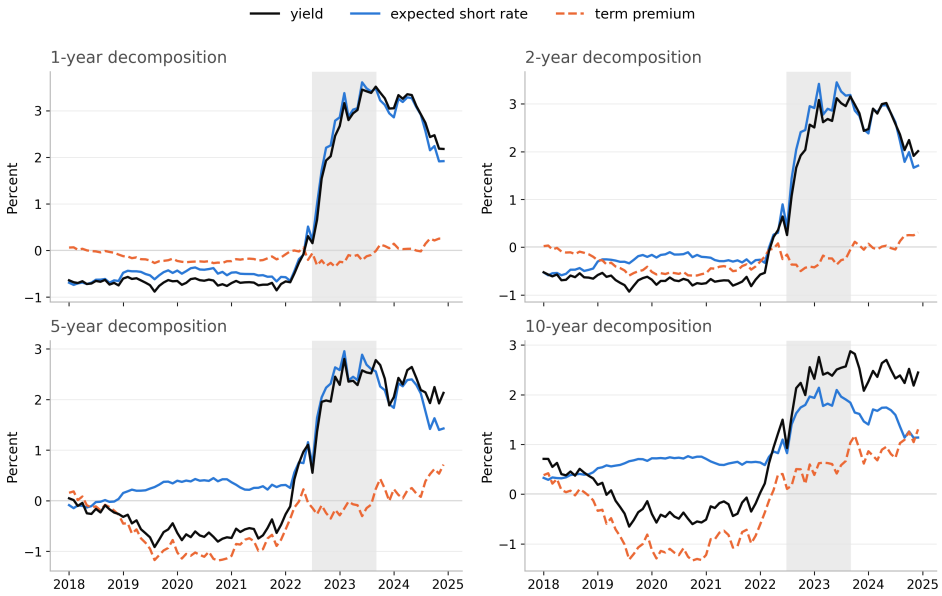
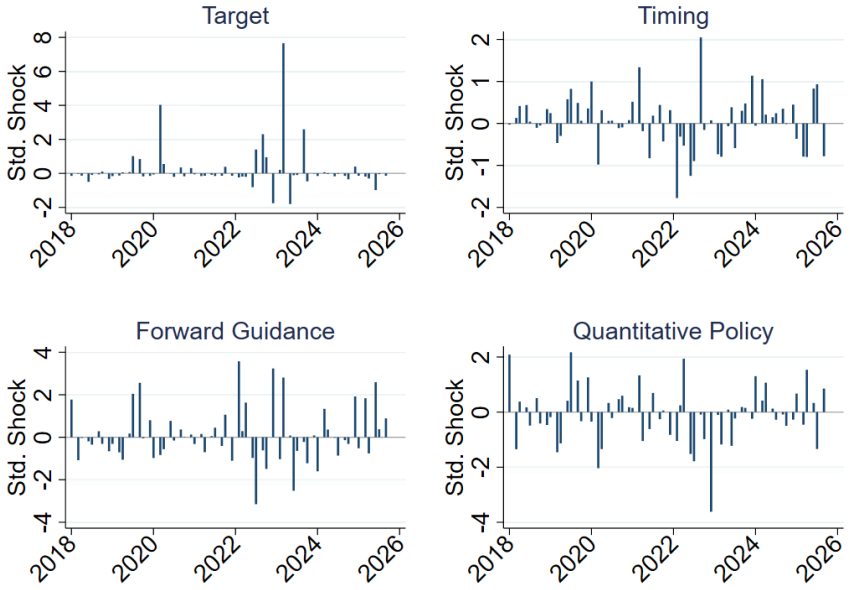


Figure 10: Yield Decomposition: Adrian, Crump, and Moench (2013)

Notes: Same decomposition as Figure 2, using the methodology of Adrian et al. (2013) in place of the affine model of Joslin et al. (2011). Euro-area OIS zero-coupon yields, monthly, September 2018 to December 2024; the shaded region denotes the ECB tightening cycle beginning in 2022.

Panel A. High-frequency monetary-policy surprises



Panel B. Pure monetary policy and information shocks

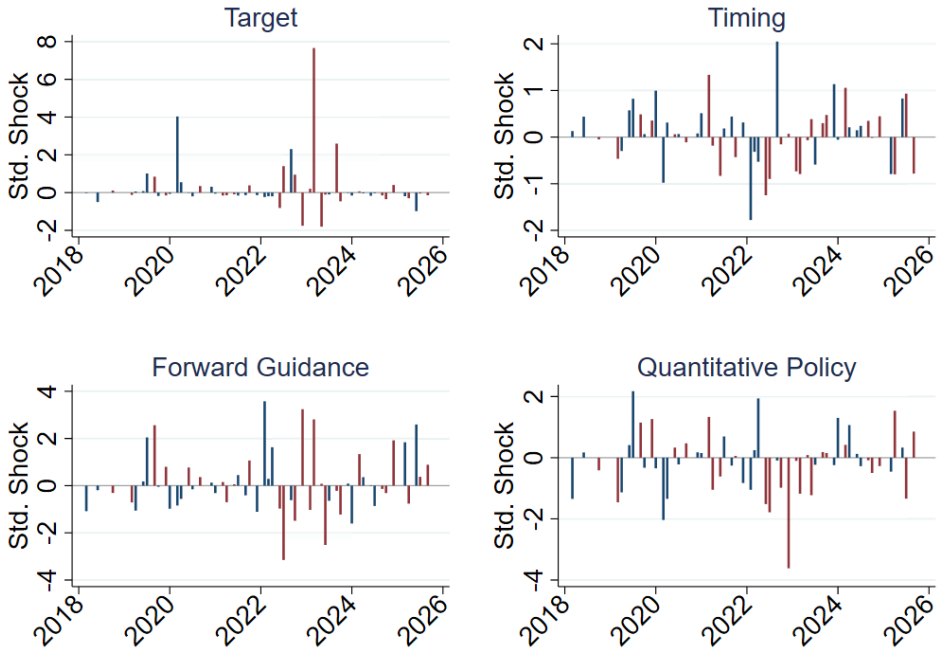


Figure 11: High-Frequency Monetary Policy Surprises

Notes: The figure reports monthly high-frequency monetary-policy surprises around narrow ECB announcements, following Altavilla et al. (2019). Panel A reports the Target, Timing, Forward Guidance, and Quantitative Policy factors. The *Target* factor captures unexpected changes in the current policy rate; *Timing* and *Forward Guidance* capture revisions to the expected policy path at short and medium horizons; and *Quantitative Policy* captures variation associated with asset- purchase policies and longer-horizon yields. Panel B separates monetary-policy events into *pure monetary policy shocks* (in blue) and *central-bank information shocks* (in red) (Jarociński and Karadi, 2020). Events are classified as pure monetary policy shocks when the 1-month OIS surprise and Euro Stoxx 50 equity prices move in opposite directions, and as central-bank information shocks when they move in the same direction. Positive monetary-policy surprises correspond to unexpected policy tightening.

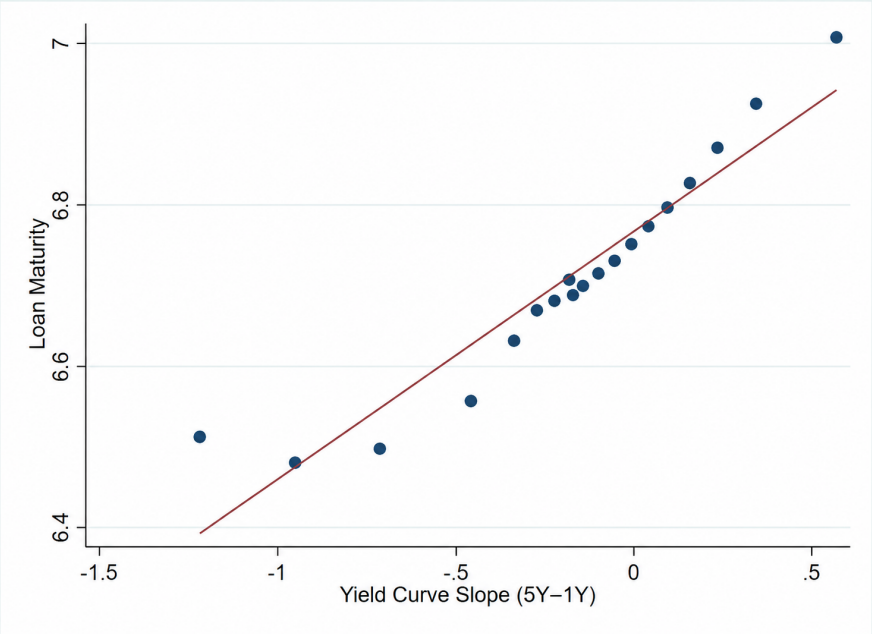


Figure 12: Yield-curve slope 5Y-1Y and new-loan maturity

Notes: Binscatter of average new loan maturity against the 5Y–1Y euro-area yield-curve slope. Loan maturity is residualized on firm and bank fixed effects. AnaCredit, 2018–2024.

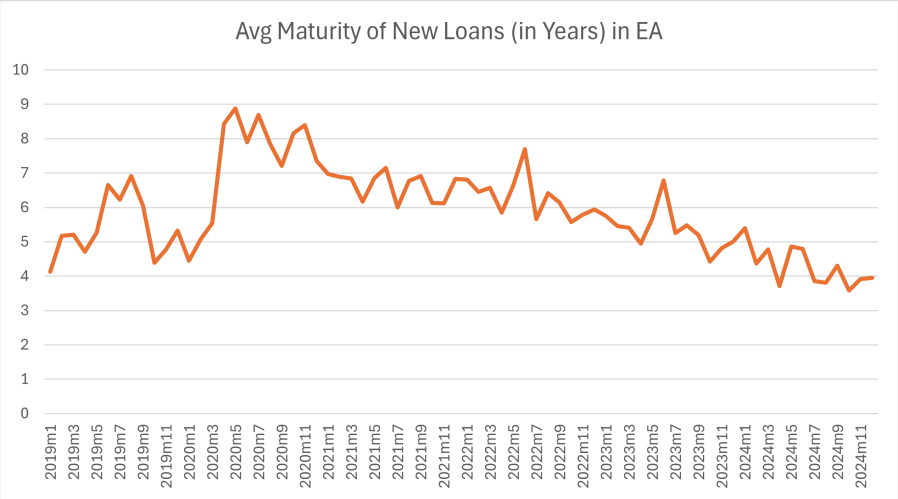


Figure 13: New Loan Maturity in the Euro Area

Notes: The figure reports the average maturity (in years) of newly originated bank loans to non-financial corporations in the euro area. The sample covers September 2018 through December 2024 and is based on AnaCredit.

A.3 Robustness to Frequent Policy Events

This appendix repeats the high-frequency analysis using the extended event-study database of Altavilla et al. (2026). Relative to the baseline EA-MPD, the extended database incorporates a broader set of monetary-policy events, including policymakers' communication between scheduled Governing Council meetings. We construct four mutually orthogonal components of monetary-

policy news. *Target* is the 1-month OIS surprise; *Timing* is the component of the 6-month OIS surprise orthogonal to Target; *Forward Guidance* is the component of the 2-year OIS surprise orthogonal to Target and Timing; and *Quantitative Policy* is the residual 10-year surprise orthogonal to the three shorter-horizon components. Announcement-level surprises are aggregated to the monthly frequency before being merged with the loan-level data.

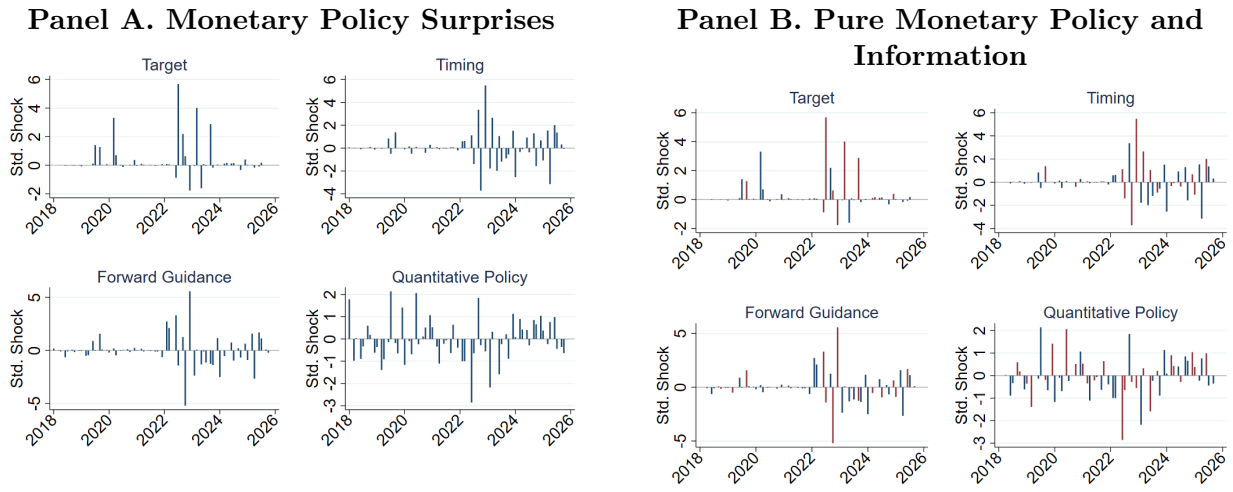


Figure 14: High-Frequency Monetary Policy Surprises: Extended Event Sample

Notes: Panel A reports the monthly Target, Timing, Forward Guidance, and Quantitative Policy surprises constructed from the extended event-study database of Altavilla et al. (2026). Panel B separates these surprises into pure monetary-policy and central-bank information components using the high-frequency co-movement of the 1-month OIS surprise and Euro Stoxx 50 equity prices (Jarociński and Karadi, 2020).

Table 29: Validation of High-Frequency Policy Factors

	1M	3M	6M	1Y	2Y	5Y	10Y
Target	3.190***	2.953***	2.640***	2.242***	1.792***	0.847***	0.147***
	(0.000)	(0.043)	(0.000)	(0.117)	(0.000)	(0.077)	(0.000)
Timing	-0.000	1.378***	1.993***	1.286***	0.000*	-0.647***	-0.490***
	(0.000)	(0.111)	(0.000)	(0.250)	(0.000)	(0.191)	(0.000)
Forward Guidance	-0.000	-0.642***	-0.000	2.274***	4.345***	4.632***	3.202***
	(0.000)	(0.127)	(0.000)	(0.226)	(0.000)	(0.193)	(0.000)
Quantitative Policy	0.000	0.078*	0.000	-0.120***	0.000	1.191***	2.074***
	(0.000)	(0.046)	(0.000)	(0.043)	(0.000)	(0.099)	(0.000)
Observations	128	128	128	128	128	128	128
R^2	1.000	0.984	1.000	0.983	1.000	0.959	1.000

Notes: The table reports the response of OIS yields at different maturities to a one-standard-deviation innovation in each high-frequency policy factor. The sample covers July 2011 through October 2025 and contains 128 ECB policy events. Target loads primarily on the short end of the curve. Timing is concentrated at short and intermediate maturities. Forward Guidance loads most strongly at intermediate and long maturities, while Quantitative Policy primarily raises longer-maturity yields. Standard errors are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 30: High-Frequency Policy Shocks and Loan Maturity: Extended Event Sample

$\log(\text{Maturity})_{b,f,t}$	(1) All Shocks	(2) Pure MP	(3) Information
$\text{Target}_{t-1} \times$ ST $\text{Debt}_{f,t-1}$	-0.00942***	-0.0363***	0.00249
	(0.00248)	(0.00651)	(0.00205)
$\text{Timing}_{t-1} \times$ ST $\text{Debt}_{f,t-1}$	0.00725***	0.0184***	0.00661***
	(0.00153)	(0.00373)	(0.00188)
Forward $\text{Guidance}_{t-1} \times$ ST $\text{Debt}_{f,t-1}$	-0.000200	-0.000974	0.00241
	(0.00159)	(0.00223)	(0.00209)
Quantitative $\text{Policy}_{t-1} \times$ ST $\text{Debt}_{f,t-1}$	-0.00756***	-0.0129***	-0.00199
	(0.00238)	(0.00366)	(0.00356)
Observations	1,674,254	1,674,254	1,674,254
Adjusted R^2	0.620	0.620	0.620
Bank × Month FE	Yes	Yes	Yes
Firm × Bank FE	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes

Notes: The dependent variable is the logarithm of maturity of newly originated loans. Short-term Debt is the lagged share of firm bank debt maturing within 12 months. Monetary-policy surprises are standardized by their monthly standard deviation. Column (1) uses the full set of high-frequency surprises. Columns (2)–(3) separate them into pure monetary-policy and central-bank information components using the high-frequency co-movement between the 1-month OIS surprise and Euro Stoxx 50 equity prices. All specifications include lagged firm controls, bank×month fixed effects, and firm×bank fixed effects. Standard errors are clustered at the firm and month level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

B Firm Problem and Optimality Conditions

This appendix states the firm's continuation problem and derives the first-order conditions for capital, total issuance, and debt maturity. Firm and time subscripts are suppressed. The arguments of debt-price schedules are also suppressed when there is no ambiguity.

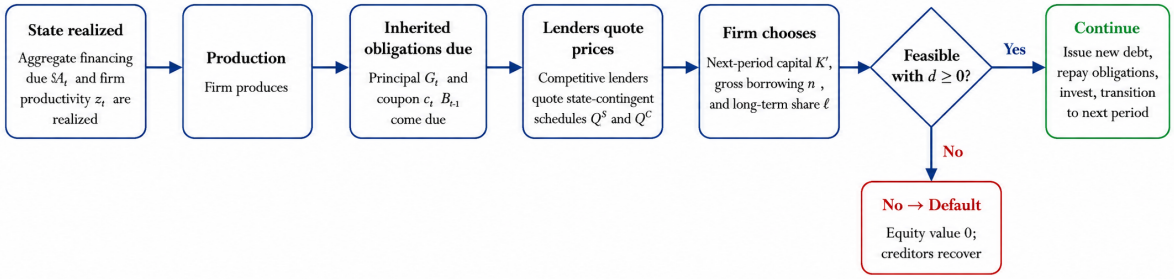


Figure 15: Model Timing

Notes: The figure summarizes the within-period timing of decisions and shocks in the model.

B.1 Firm problem

Given the firm state

$$x = (z, K, B, G) \tag{40}$$

and aggregate financing conditions

$$A = (r, f, \omega) \tag{41}$$

a continuing firm chooses next-period capital, gross new borrowing, and the long-term share of new borrowing:

$$K' \geq 0, \qquad m \geq 0, \qquad \ell \in [0, 1] \tag{42}$$

New short- and long-term debt are

$$M^S = (1 - \ell)m, \qquad M^L = \ell m \tag{43}$$

Thus, $M^S \geq 0$ and $M^L \geq 0$, and the benchmark does not allow early repayment of surviving long-term debt. When $m = 0$, the value assigned to ℓ is immaterial.

Investment is

$$I = K' - (1 - \delta_K)K \tag{44}$$

Total debt evolves according to

$$B' = B - G + m \tag{45}$$

Next-period principal due is

$$G' = (1 - \rho_L)B' + \rho_L(1 - \ell)m \tag{46}$$

Equivalently,

$$G' = (1 - \rho_L)(B - G) + (1 - \rho_L\ell)m \tag{47}$$

The relevant derivatives of the state transition are

$$\frac{\partial B'}{\partial m} = 1, \qquad \frac{\partial G'}{\partial m} = 1 - \rho_L\ell \tag{48}$$

and

$$\frac{\partial B'}{\partial \ell} = 0, \qquad \frac{\partial G'}{\partial \ell} = -\rho_L m \tag{49}$$

Dividends are

$$d = \pi(z, K) - G - c_L \frac{B - G}{\rho_L} - I - \Phi_K(I, K) - H(m) + m\left[(1-\ell)Q^S + \ell Q^L\right] \tag{50}$$

The coupon term follows from

$$B^L = \frac{B - G}{\rho_L} \tag{51}$$

The firm cannot issue external equity:

$$d \geq 0 \tag{52}$$

Conditional on continuation, the value function is

$$V(x; A) = \max_{K', m, \ell} \left\{ d + \beta \mathbb{E}_t \left[\mathcal{M}_{t,t+1} V(z', K', B', G'; A') \right] \right\} \tag{53}$$

subject to (42), (45), (46), and (52). The benchmark sets

$$\beta = 1 - \kappa, \qquad \mathcal{M}_{t,t+1} = e^{-r_t} \tag{54}$$

where κ is the reduced-form equity-horizon wedge of Section 6: shareholders and lenders use the same stochastic discount factor, and shareholders additionally discount continuation payoffs by $1-\kappa$. The wedge is a valuation object only—it generates no exit transition in the firm distribution and does not enter creditor pricing.

Let $\mu \geq 0$ denote the multiplier on the no-equity constraint. For an interior solution, the Lagrangian is

$$\mathcal{L} = (1 + \mu) d + \beta \mathbb{E}_t \left[\mathcal{M}_{t,t+1} V(z', K', B', G'; A') \right] \tag{55}$$

with complementary slackness

$$\mu d = 0 \tag{56}$$

The conditions below apply at differentiable interior choices with

$$K' > 0, \qquad m > 0, \qquad \ell \in (0, 1) \tag{57}$$

At a boundary, the corresponding equality is replaced by the appropriate Kuhn–Tucker inequality.

For compactness, define

$$\Phi_I(I, K) \equiv \frac{\partial \Phi_K(I, K)}{\partial I} = \psi_K \frac{I}{K} \tag{58}$$

and

$$H'(m) = \eta m \tag{59}$$

B.2 Capital choice

An increase in K' raises investment one-for-one and may change debt prices through its effect on next-period default risk. The first-order condition for capital is

$$0 = (1+\mu)\left[-1 - \Phi_I(I,K) \; + m\left((1-\ell)\frac{\partial Q^S}{\partial K'} + \ell\frac{\partial Q^L}{\partial K'}\right)\right] + \beta\mathbb{E}_t\left[\mathcal{M}_{t,t+1}V_K(z',K',B',G';A')\right] \tag{60}$$

Equivalently,

$$1 + \Phi_I(I,K) = m\left[(1-\ell)\frac{\partial Q^S}{\partial K'} + \ell\frac{\partial Q^L}{\partial K'}\right] + \frac{\beta}{1+\mu}\mathbb{E}_t\left[\mathcal{M}_{t,t+1}V_K(z',K',B',G';A')\right] \tag{61}$$

The left-hand side is the current marginal resource cost of investment. The right-hand side contains the effect of capital on current borrowing terms and the discounted continuation value of additional installed capital.

B.3 Total issuance

An additional unit of gross borrowing raises total debt by one and next-period principal due by $1-\rho_L\ell$. The first-order condition for gross issuance is

$$\begin{aligned} 0 = (1+\mu)\left[(1-\ell)Q^S + \ell Q^L - H'(m) \; + m\left((1-\ell)\frac{\partial Q^S}{\partial m} + \ell\frac{\partial Q^L}{\partial m}\right)\right] \\ + \beta\mathbb{E}_t\left[\mathcal{M}_{t,t+1}\left\{V_B(z',K',B',G';A') + (1-\rho_L\ell)V_G(z',K',B',G';A')\right\}\right] \end{aligned} \tag{62}$$

The firm equates the current proceeds from additional borrowing, net of issuance costs and endogenous price effects, to the continuation cost of higher total debt and higher future principal obligations.

If debt prices are locally insensitive to the size of the candidate issuance, the condition simplifies to

$$(1-\ell)Q^S + \ell Q^L - H'(m) = -\frac{\beta}{1+\mu}\mathbb{E}_t\left[\mathcal{M}_{t,t+1}\left\{V_B + (1-\rho_L\ell)V_G\right\}\right] \tag{63}$$

B.4 Maturity choice

Holding gross issuance fixed, a higher ℓ replaces short-term borrowing with long-term borrowing. It leaves total debt unchanged but lowers next-period principal due by $\rho_L m$. The first-order condition for maturity is

$$0 = (1+\mu)m\left[Q^L - Q^S \; + (1-\ell)\frac{\partial Q^S}{\partial \ell} + \ell\frac{\partial Q^L}{\partial \ell}\right] - \beta\rho_L m\mathbb{E}_t\left[\mathcal{M}_{t,t+1}V_G(z',K',B',G';A')\right] \tag{64}$$

For $m > 0$, this can be written as

$$Q^S - Q^L - (1-\ell)\frac{\partial Q^S}{\partial \ell} - \ell\frac{\partial Q^L}{\partial \ell} = -\frac{\beta\rho_L}{1+\mu}\mathbb{E}_t\left[\mathcal{M}_{t,t+1}V_G(z', K', B', G'; A')\right] \tag{65}$$

The left-hand side is the current marginal funding cost of replacing short-term debt with long-term debt. The right-hand side is the continuation value of the associated reduction in next-period principal due.

If debt prices are locally insensitive to the maturity composition of the candidate issuance, the condition simplifies to

$$Q^S - Q^L = -\frac{\beta\rho_L}{1+\mu}\mathbb{E}_t\left[\mathcal{M}_{t,t+1}V_G(z', K', B', G'; A')\right] \tag{66}$$

When inherited principal obligations reduce firm value,

$$V_G < 0 \tag{67}$$

the right-hand side is positive. The firm then accepts a lower current price for long-term debt in exchange for lower future refinancing needs.

A higher expected-rate state f may increase this insurance value by raising expected future refinancing costs. It can also affect the current valuation of long-duration claims through expected future discount rates. By contrast, the long-duration funding component directly affects the borrower price

$$\left.\frac{\partial Q^L}{\partial \omega}\right|_{Q^L} = -Q^L < 0 \tag{68}$$

The maturity response to either aggregate financing shock is therefore an equilibrium outcome rather than an imposed sign restriction.

C Numerical Algorithm

The model is solved on a discretized state space $x = (z, K, B, s)$, $A = (r, f, \omega)$, where $s = G/B$ replaces G as the debt-structure coordinate (equation (16)). Choices of (K', B', ℓ) are restricted to grid points, so continuation values are interpolated only in the exogenous states.

Grids. Productivity uses a five-node Rouwenhorst discretization. Capital and debt are log-spaced, with 11 nodes on $[0.20, 2.00]$ for capital and 21 nodes for debt including an exact zero, giving uniform relative steps of 26 and 21 percent; log spacing is used because the estimated adjustment and issuance costs are identified by relative growth moments, which a uniform grid would resolve well at the top of the grid and poorly at the bottom. Rollover exposure s has eight log-spaced nodes on $[1-\rho_L, 1]$ and the maturity share ℓ nine uniform nodes on $[0, 1]$. The aggregate states use uniform grids with seven nodes on $[0.001, 0.045]$ for r, five on $[-0.018, 0.018]$ for f, and seven on $[0.005, 0.095]$ for ω.

Expectations. Expectations over next-period aggregate states use Gauss–Hermite quadrature with three nodes per innovation in (13)–(15), 27 joint nodes per state, with next-period states clipped to the grid and continuation objects interpolated. The same rule is used for every result in the paper.

Backward induction. Because default is determined by feasibility and debt prices depend on next-period default, prices and default are coupled within a period. We solve a long finite-horizon economy backward from a terminal date at which $V = 0$, default probabilities are zero, and long-debt values equal their risk-free counterparts. At each earlier date we construct Q^S, $\widetilde{Q}^L$, and Q^L from the already determined date-t+1 value function, default rule, and claim values using the lenders' break-even conditions (30) and (31); solve the date-t firm problem state by state, with default when no choice satisfies (24); and carry $(V, D, \widetilde{Q}^L)$ back one period as a damped average with the previous iterate, re-optimizing policies against the damped continuation objects. Iteration stops when values, prices, the default rule, and every cross-sectional policy aggregate are unchanged, after which the long-debt price recursion is re-solved with policies and the default rule frozen to drive the residual to tolerance. The terminal condition dies out geometrically at rate $(1 - \kappa)e^{-r}$ per step, and lengthening the horizon leaves policies and moments unchanged.

Simulation. The stationary distribution is computed by Monte Carlo panel simulation, with exposure tracked continuously between grid nodes and policies evaluated at the nearest node. Aggregate states are drawn from their estimated processes, clipped to the grid, and mapped to the nearest node. Defaulting firms are replaced by entrants with capital K_0, zero debt, and productivity drawn from its invariant distribution. Counterfactual economies are re-solved in full, prices included, and simulated under common shock paths from the benchmark stationary distribution.

D Measurement of Aggregate Processes

D.1 Yield Decomposition

To distinguish changes in long-term yields due to expected future short rates from changes in term premia, we decompose euro-area zero-coupon yields using the affine term-structure model of Joslin et al. (2011). We use the Adrian et al. (2013) decomposition as an alternative measure. The two approaches differ in how they identify risk compensation, but both express an n-period yield as

$$y_t^{(n)} = ER_t^{(n)} + TP_t^{(n)} \tag{69}$$

where

$$ER_t^{(n)} = \frac{1}{n} \sum_{j=0}^{n-1} \mathbb{E}_t^P[r_{t+j}] \tag{70}$$

is the average expected short rate over the life of the bond and $TP_t^{(n)}$ is the corresponding term premium.

JSZ decomposition. The Joslin et al. (2011) approach summarizes the zero-coupon yield curve with a small set of pricing factors, X_t. Factor dynamics under the physical measure follow

$$X_{t+1} = \mu^P + \Phi^P X_t + \Sigma \varepsilon_{t+1} \tag{71}$$

and the short rate is affine in the factors,

$$r_t = \delta_0 + \delta_1' X_t \tag{72}$$

No-arbitrage restrictions map these factor dynamics into zero-coupon bond prices. The expected-rate component is obtained from forecasts of future short rates under the physical measure, and the term premium is the portion of the observed long yield not accounted for by those expected short rates.

ACM decomposition. As a robustness check, we use the methodology of Adrian et al. (2013). ACM also summarizes the yield curve using a small number of factors but identifies time-varying prices of risk from the predictability of excess bond returns. For an n-period bond, excess returns are given by

$$rx_{t+1}^{(n)} = p_{t+1}^{(n-1)} - p_t^{(n)} - r_t \tag{73}$$

and are related to innovations in the yield factors and to their lagged values. These estimates identify the compensation investors require for bearing yield-curve risk. The expected-rate component is then the portion of the yield implied by expected future short rates, with the residual attributed to the term premium.

Despite their different identification strategies, the JSZ and ACM decompositions produce very similar expected-rate and term-premium series in our sample. Table 3 reports the corresponding maturity estimates under both decompositions.

Table 31: Mapping the Model to the Data

Empirical Object	**Model Object**	**Mapping**
Rollover exposure Debt due within 1y / total bank debt	Principal due, $s = G/B$	Both count principal due within the year;
Expected-rate component 5Y JSZ	Rate states (r_t, f_t)	$\mathrm{ESR}_t^{5Y} = \frac{1}{5}\sum_{j=1}^{5} \mathbb{E}_t[r_{t+j}]$
Term premium 5Y JSZ	Term-premium wedge ω_t	$TP_t = y(e^{-\omega_t} Q_t^E) - y(Q_t^E)$; ω_t is inverted period by period to match the observed term premium
Maturity outcomes Log original maturity; LT share of new loans	LT share of new issuance, ℓ	$\mathrm{Mat} = (1-\ell)\cdot 1 + \ell \cdot \frac{1}{1-\rho_L}$; model outcomes are log(Mat) and ℓ

Notes: For the long-term claim, $y(Q) = \log\left(\rho_L + \frac{c_L + 1 - \rho_L}{Q}\right)$. Both maturity outcomes are transformations of the same underlying maturity choice and are reported to match the two empirical specifications rather than as independent model moments. Differences in log-maturity units up to an additive constant are absorbed by fixed effects.

D.2 Model financing states

The model's aggregate block is estimated at annual frequency from monthly series. The current-rate process is an AR(1) estimated on the one-year OIS rate. The news state is the five-year

expected-rate-path component orthogonalized to the current rate; monthly persistence is converted to annual as $\rho_a = \rho_m^{12}$, with the innovation variance point-sampled accordingly.

D.3 The term-premium wedge measurement equation

The empirical long-duration component is observed in yield units, whereas ω_t is a price wedge. We map the yield decomposition into the model using the duration of the model's long-term claim. Let Q_t^E denote the price of its risk-free counterpart when only the expected path of short rates varies:

$$Q_t^E = e^{-r_t}\mathbb{E}_t\left[c_L + 1 - \rho_L + \rho_L Q_{t+1}^E\right] \tag{74}$$

Q_t^E is the expectations-only counterpart of the claim value $\widetilde{Q}^L$ in (31); the corresponding issuance price after the wedge is $Q_t^D = e^{-\omega_t}Q_t^E$, as in Section 6.4. For a geometrically amortizing claim, the continuously compounded yield associated with price Q is

$$y(Q) = \log\left(\rho_L + \frac{c_L + 1 - \rho_L}{Q}\right) \tag{75}$$

The measurement equation sets the observed long-duration yield component equal to the yield differential generated by the model wedge:

$$TP_t = y\big(e^{-\omega_t}Q_t^E\big) - y(Q_t^E) \tag{76}$$

We apply (76) to deviations of the term premium from its sample mean, which identifies movements in ω_t and hence its law of motion (15); the level $\bar{\omega}$ is preset, as discussed in Section 7. Because (74) is linear in $c_L + 1 - \rho_L$, Q_t^E is proportional to this cash-flow term, which cancels from the yield differential. Hence the recovered ω_t depends on the duration parameter ρ_L and the expected short-rate path, but not on the coupon normalization c_L.

D.4 Firm-level productivity

Firm-level productivity is estimated using the matched AnaCredit–Orbis panel. Given the production function

$$Y_{fct} = \bar{z} z_{fct} K_{fct}^{\alpha}$$

we measure productivity using Orbis sales for output and fixed assets for capital. With $\alpha = 0.60$, log productivity is recovered from

$$\log(\text{Sales}_{fct}) = \alpha \log(K_{fct}) + \delta_c + \delta_s + \log z_{fct}$$

where δ_c and δ_s denote country and sector fixed effects.

We then estimate

$$\log z_{fct} = \rho_z \log z_{fc,t-1} + \varepsilon_{fct}^z$$

Following the literature, we instrument lagged productivity with its second lag to correct for transitory measurement error. The estimated coefficient gives the persistence ρ_z, while σ_z is the standard deviation of the residuals ε_{fct}^z.

E Additional Model Counterfactuals and Diagnostics

This appendix supports the counterfactual exercises in Section 8. It reports the full dynamics of the main counterfactual and the complete results for the separation of maturity from interest-rate fixation. It then examines the scope of the main result: how the comparison changes when each economy starts from its own stationary distribution, whether the investment effects reflect survivorship, and the role of inherited rollover exposure and of selection into exposure states. It closes with the sensitivity of debt prices to issuance, which underlies the endogenous cost of obtaining duration.

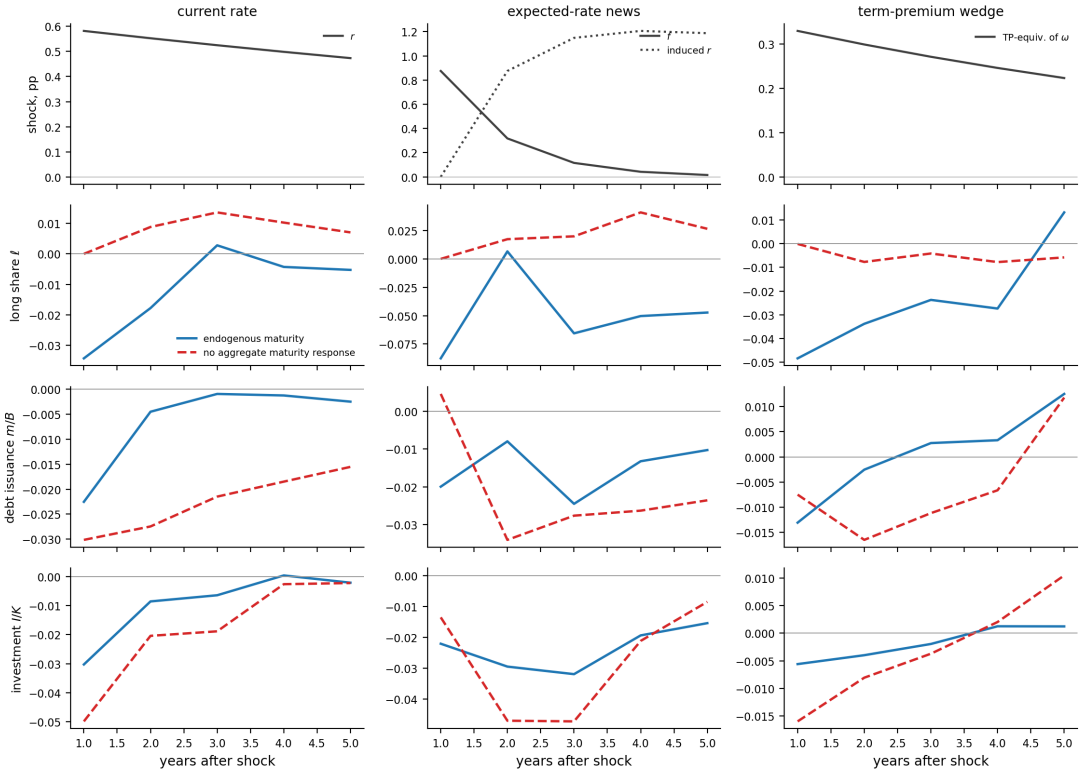


Figure 16: Aggregate Maturity Adjustment: Full Dynamics. The figure reports shock responses in the benchmark economy and the counterfactual that removes the direct dependence of maturity choice on aggregate financing conditions. The top row reports the aggregate shocks, including the short-rate path induced by expected-rate news; ω is expressed in term-premium-equivalent yields. The remaining rows report the long-term share of new issuance, gross debt issuance, and investment. Solid lines denote the benchmark; dashed lines denote the restricted economy. Results are averaged across 1,000 simulated economies.

E.1 Maturity Adjustment: Full Dynamics

Figure 16 reports the impulse responses underlying Table 13. Investment and debt issuance diverge immediately between the benchmark and restricted economies following current-rate and term-premium shocks; following the term-premium shock the investment gap closes by year four, as the wedge decays and the restricted economy's long-dated issuance stops being repriced. Following expected-rate news, the investment difference emerges primarily as the anticipated increase in short rates materializes.

Table 32: Value of Maturity Flexibility: Expected-Rate and Current-Rate Shocks with Lower Duration Costs

Horizon (years)	1	3	5
Investment (cumulative investment / capital, pp)			
News with lower duration costs	+0.2 (0.2)	+2.7 (0.2)	+3.1 (0.2)
Current rate with lower duration costs	+0.5 (0.3)	+3.3 (0.2)	+4.0 (0.2)
Debt issuance (cumulative gross issuance / outstanding debt, pp)			
News with lower duration costs	−0.9 (0.3)	−0.3 (0.5)	+1.2 (0.7)
Current rate with lower duration costs	+0.5 (0.4)	+2.7 (0.4)	+4.7 (0.3)
Long-term issuance share (level response, pp)			
News with lower duration costs	−4.0 (0.3)	−5.4 (0.2)	−4.3 (0.3)
Current rate with lower duration costs	+2.2 (0.7)	−2.2 (0.2)	−3.1 (0.2)
Credit spread (annual yield, pp; average response through horizon)			
News with lower duration costs	−1.0 (0.4)	−2.9 (0.3)	−1.8 (0.3)
Current rate with lower duration costs	−2.3 (0.2)	−0.3 (0.1)	+0.8 (0.1)
Default (cumulative incidence, pp)			
News with lower duration costs	—	−1.5 (0.2)	−3.7 (0.3)
Current rate with lower duration costs	—	−0.8 (0.1)	−0.6 (0.1)

Notes: Entries are the benchmark-minus-restricted difference in shock responses, as in Table 13, to a one-standard-deviation expected-rate news shock or current-rate shock combined with a one-standard-deviation increase or decrease in the term-premium wedge in the shock year; each factor then follows its estimated law of motion. The responses to each monetary shock alone are the corresponding rows of Table 13, and the rows with higher duration costs repeat Table 14. Restricted firm policies and debt-price schedules, units, horizons, and the simulation design are as in Table 13. Horizon-1 default entries are omitted because default does not respond on impact under the model's timing. Monte Carlo standard errors of the mean are in parentheses.

E.2 Joint Tightening: Lower Duration Costs and Current-Rate Shocks

Table 32 completes the joint tightening experiment of Section 8.3 with the mirror case of a one-standard-deviation *decrease* in the term-premium wedge and with the same three cases for a current-rate shock. Two facts stand out. First, when long-duration financing becomes cheaper as expected refinancing costs rise, flexible firms shorten considerably less than under news alone (4.0 rather than 8.8 percentage points on impact) and retain more maturity protection; the investment cushion after three years is 2.7 percentage points, close to the news-only value, and the five-year reduction in default rises from 2.7 to 3.7 points. Second, the interaction between the price of duration and the value of flexibility is specific to expected refinancing risk. Adding the same term-premium wedge increase to a current-rate shock leaves the three-year investment value of flexibility unchanged (4.5 against 4.4 percentage points), whereas it raises it from 2.4 to 3.9 points under expected-rate news; a lower wedge reduces it under the current-rate shock (3.3 points) but not under news. The incremental maturity response to the wedge is similar for the two monetary shocks (shortening of

9.6 against 3.4 points for the current-rate shock, and 16.4 against 8.8 for news), so the difference lies in the payoff of that adjustment: when expected future refinancing costs rise, maturity choice is an insurance decision and the price of duration changes the value of having it; under a current-rate shock the same repricing of duration adds a similar cost to both economies.

E.3 Monetary Transmission across Duration-Cost States

Figure 17 reports the responses to the same one-standard-deviation expected-rate news and current-rate shocks when they arrive in a low or a high duration-cost environment. The exercise uses the flexible economy throughout and therefore characterizes state-dependent transmission rather than the causal contribution of maturity flexibility. Under the current-rate shock the investment ordering is the same, with three-year cumulative investment falling by 2.7 percentage points when duration is cheap and by 5.4 points when it is expensive, but the maturity response is small and short-lived in both states: the long-term issuance share is 0.7 and 1.9 points below its no-shock path after three years, against 2.1 and 10.8 points under expected-rate news. The price of duration shapes transmission through the maturity margin only when the shock concerns future refinancing costs.

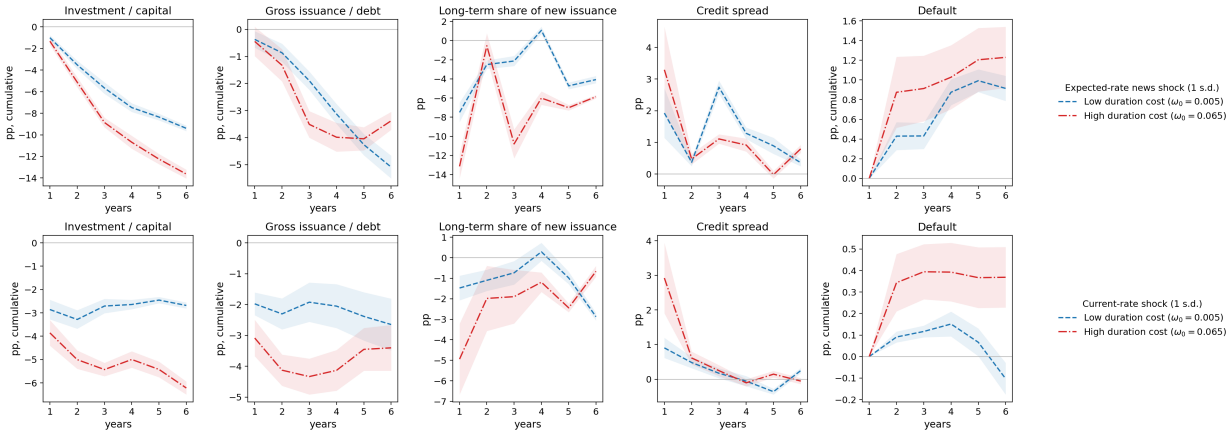


Figure 17: Monetary Transmission across Duration-Cost States

Notes: Responses of the benchmark economy to a one-standard-deviation expected-rate news shock (top row) and current-rate shock (bottom row) arriving when the term-premium wedge is low ($\omega_0 = 0.005$, dashed) or high ($\omega_0 = 0.065$, dash-dotted). The economy starts in both cases from the same stationary cross-section of firms as in Table 13; the wedge then reverts toward its mean along its estimated law of motion, and responses are measured relative to the same path without the innovation. Investment, gross issuance, and default are cumulative; the long-term share of new issuance and the credit spread are level responses at each horizon. The horizon stops at six years because in year seven the rate path induced by the news innovation crosses to a lower node of the aggregate grid and the responses thereafter reflect the grid rather than the shock. Shaded bands are 95 percent Monte Carlo confidence intervals of the mean across 1,000 simulated economies of 10,000 firms.

E.4 Maturity and Interest-Rate Fixation

This appendix reports the full results for the contract comparisons in Section 8.2. Table 33 repeats the maturity-flexibility counterfactual in the floating-coupon economy. In the floating-coupon economy, firms can postpone principal repayment but cannot lock in the interest rate on outstanding long-term debt. We conduct the maturity-flexibility counterfactual of Section 8.1 in this economy; the comparison identifies the contribution of maturity adjustment when coupon fixation is unavailable. It shows that maturity flexibility continues to protect firms against default when long-term

Table 33: Maturity Flexibility in the Floating-Coupon Economy

Horizon (years)	1	3	5
Investment (cumulative investment / capital, pp)			
Current rate	+0.5 (0.2)	+0.3 (0.2)	+0.1 (0.2)
Expected-rate news	+4.2 (0.3)	−1.1 (0.4)	−1.4 (0.3)
Term-premium wedge	+3.2 (0.3)	+0.5 (0.3)	+0.1 (0.1)
Debt issuance (cumulative gross issuance / outstanding debt, pp)			
Current rate	−0.1 (0.3)	+4.5 (0.5)	+10.1 (0.7)
Expected-rate news	+0.9 (0.4)	+2.9 (0.7)	+7.2 (0.9)
Term-premium wedge	+1.9 (0.3)	+8.2 (0.6)	+11.8 (0.7)
Long-term issuance share (level response, pp)			
Current rate	−16.8 (1.0)	−10.8 (0.5)	−7.1 (0.3)
Expected-rate news	−29.8 (1.4)	−10.9 (0.4)	−9.4 (0.3)
Term-premium wedge	−20.9 (1.1)	−8.8 (0.6)	+0.7 (0.5)
Default (cumulative incidence, pp)			
Current rate	—	−0.8 (0.2)	−1.2 (0.2)
Expected-rate news	—	−2.4 (0.3)	−4.2 (0.3)
Term-premium wedge	—	−0.7 (0.2)	−1.9 (0.1)

Notes: Entries report differences in shock responses between the economies named in each panel (first minus second). Floating-coupon debt retains the benchmark amortization schedule but reprices with the short rate. Investment, gross issuance, and default are cumulative; the long-term issuance share is measured at each horizon. Default is recorded beginning in the period after the shock. The horizon-1 default entries are therefore omitted. All economies start from the fixed-coupon benchmark's stationary cross-section and face identical idiosyncratic draws. Policies and debt prices are re-solved in each economy. Results average 1,000 economies of 10,000 firms following one-standard-deviation shocks with no subsequent aggregate innovations. Monte Carlo standard errors are in parentheses.

coupons float. Restricting the aggregate maturity response raises five-year cumulative default by 1.2, 4.2, and 1.9 percentage points following current-rate, expected-rate news, and term-premium shocks, respectively. The effect is particularly pronounced following expected-rate news: although existing floating-rate debt reprices as short rates rise, firms can still use maturity adjustment to change when principal must be refinanced.

The financing and investment effects are more nuanced. Maturity flexibility induces larger adjustments in new borrowing when coupons float: in the first year, the long-term issuance share falls by 16.8, 29.8, and 20.9 percentage points more under endogenous than restricted maturity following current-rate, expected-rate news, and term-premium shocks, respectively, and gross issuance remains higher under endogenous maturity across all three shocks at the five-year horizon. The investment effect, however, is concentrated early in the response: following expected-rate news, flexibility raises cumulative investment by 4.2 percentage points in the first year, but the difference reverses to −1.4 points by year five. The survival benefit therefore does not imply a persistent

increase in investment.

Holding the maturity regime fixed and comparing default under fixed and floating coupons gives the complementary result. When firms can adjust maturity, fixing coupons reduces five-year cumulative default by 1.1 percentage points following a current-rate shock and 2.0 points following expected-rate news. When maturity adjustment is restricted, these reductions increase to 2.5 and 3.4 points, respectively. Coupon fixation thus provides greater survival protection when firms have less scope to restructure principal repayments, while maturity flexibility is an important source of survival protection when coupons float. Maturity determines when principal must be refinanced; coupon fixation determines when interest payments on outstanding debt adjust. The two contract features provide distinct, partially substitutable forms of protection against shocks to current and expected short rates.

E.5 The 2022–24 Path: Components and the Restricted Economy

Figure 18 shows the realized paths fed to the model and the benchmark responses, and Table 34 reports the responses of the benchmark and restricted economies to the realized 2022–24 aggregate path and to its single and pairwise components. Firms enter the episode with balance sheets conditioned on the 2021 aggregate state, a short rate at the floor and Joslin et al. (2011) decomposition duration. The economy then experiences higher expected future rates and duration costs in 2022, the realized increase in short rates in 2023, and declining expected rates but persistently expensive duration in 2024.

In the benchmark economy, in 2022, before most of the increase in short rates materializes, the long-term share of new issuance falls by 22 percentage points, gross issuance by 4 points, and investment by 4.5 points of capital. The contraction deepens in 2023 as short rates rise: the long-term share falls by 36 points, issuance by 6 points, and investment by 12 points of capital in that year. In 2024, annual issuance and investment return to their no-shock paths as expected rates decline, but the long-term share remains 12 points lower because duration is still expensive. Cumulative investment through 2024 is 18 percentage points of capital below the no-shock path. The realized short-rate path accounts for most of this contraction; expected-rate news reduces subsequent defaults and credit spreads by inducing earlier financing adjustments, while higher duration costs amplify maturity shortening and the investment decline.

These responses are large, for two reasons. First, the episode delivers two large innovations in sequence: relative to the calibrated processes, the short-rate innovations of 2022 and 2023 are about three and two standard deviations and the 2022 innovation to the term-premium wedge about 2.4, so that by 2023 the short rate is 1.5 unconditional standard deviations above its mean, at the sixth of the seven grid nodes, whereas the impulse responses of Section 8.1 use single one-standard-deviation innovations; because external equity is unavailable and debt prices steepen as firms approach default, the responses are not linear in the size of the shock. Scaling all realized deviations from the calibrated means to three quarters of their size reduces the benchmark's cumulative investment loss from 18 to 12 percentage points and its cumulative default response from 1.4 to 0.9 points, close to proportionally, so the benchmark responses grow smoothly with the severity of the tightening. Second, the model is nominal and partial equilibrium: the real rate rises one for one with the policy rate, and neither inflation, aggregate demand, nor bank credit supply absorbs part of the tightening.

Along the full realized path, maturity flexibility preserves 6.7 percentage points of cumulative investment and reduces cumulative default by 7.3 points through 2024 relative to the restricted economy. The default difference is the one object that lives in the nonlinear region: the restricted

economy carries its 2021 long-term issuance share, priced at the 2021 cost of duration, into 2023 and cannot adjust to the tightening, so it experiences a sharp contraction in financing and a 30-percentage-point increase in credit spreads in that year; at three quarters of the realized shocks the default gap is 4.0 points while the investment gap is 6.4. These magnitudes reflect the model's nonlinear response to an unusually large tightening and the transition from inherited benchmark balance sheets, and should be read as upper bounds. The investment and default comparisons therefore quantify the role of maturity flexibility conditional on this initial distribution and shock path, rather than the effects of a permanently restricted maturity regime.

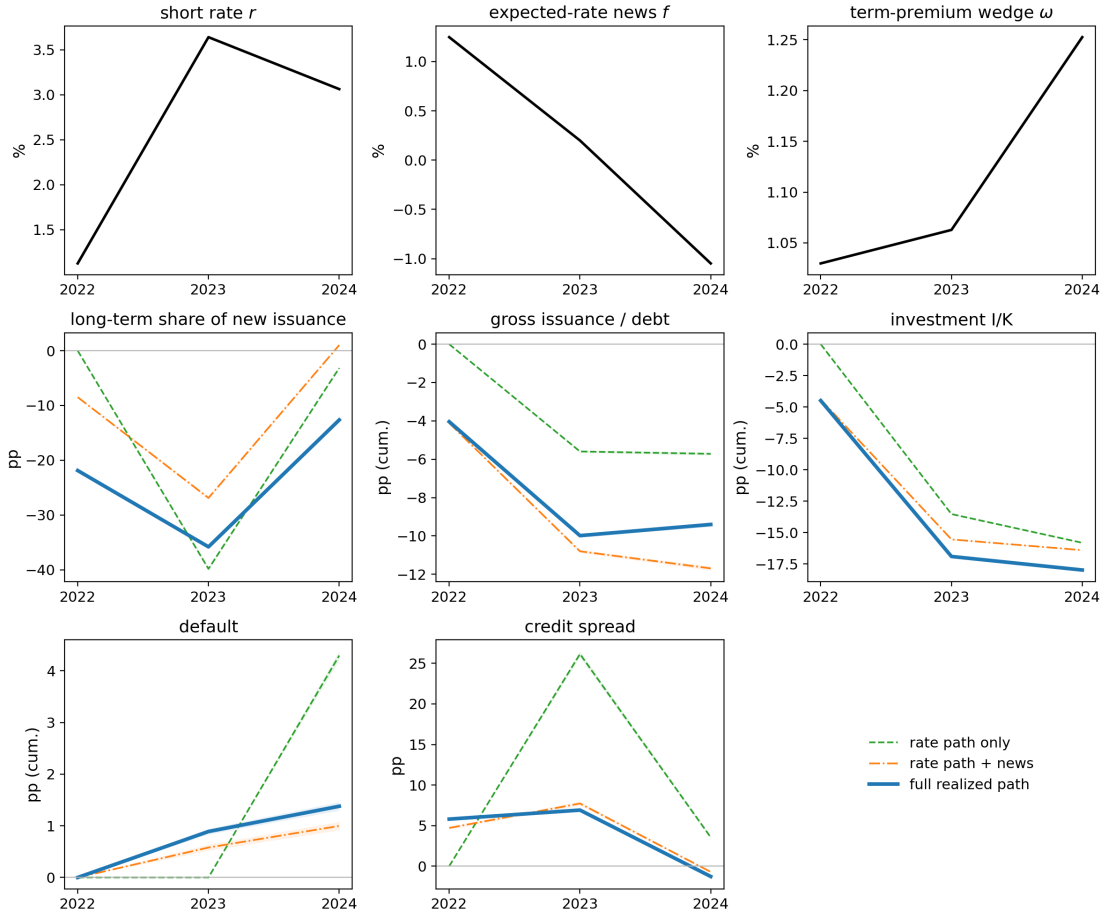


Figure 18: The 2022–24 Path: Shocks and Benchmark Responses. Top row: the realized annual path of the short rate, expected-rate news, and the term-premium wedge (in term-premium-equivalent yield) fed to the model. Remaining panels: responses of the benchmark economy, in percentage points, relative to the no-shock path from the same firms; gross issuance, investment, and default are cumulated from 2022. The solid line feeds the full realized path; the dashed line feeds the realized short-rate path alone, holding news and the wedge at their calibrated means; the dash-dotted line adds the realized expected-rate news to the rate path. The credit spread is the issuance-weighted mean, whose movements reflect the share of issuance priced close to default. Means across 50 simulated economies of 10,000 firms; shaded bands are 95% Monte Carlo intervals.

Table 34: The 2022–24 Path and Its Components: Benchmark and Restricted Economies

Cumulative through 2024, pp	LT share	Issuance	Investment	Default	Credit spread
Benchmark economy					
Realized rate path only	−14.3 (0.1)	−5.7 (0.0)	−15.8 (0.0)	+4.3 (0.0)	+9.9 (0.1)
Expected-rate news only	−0.7 (0.0)	+0.6 (0.0)	+0.0 (0.0)	+0.3 (0.0)	+0.8 (0.0)
Duration costs only	−8.1 (0.0)	−0.5 (0.0)	−0.9 (0.0)	−0.0 (0.0)	−0.2 (0.0)
Rate path with news	−11.5 (0.1)	−11.7 (0.1)	−16.4 (0.0)	+1.0 (0.0)	+3.9 (0.1)
Rate path with duration costs	−27.3 (0.1)	−6.5 (0.0)	−18.7 (0.0)	+3.6 (0.0)	+8.7 (0.1)
News with duration costs	−7.0 (0.1)	−1.3 (0.0)	−3.9 (0.0)	+0.6 (0.0)	+1.3 (0.0)
Full realized path	−23.4 (0.1)	−9.4 (0.1)	−18.0 (0.0)	+1.4 (0.0)	+3.8 (0.1)
Restricted economy					
Realized rate path only	+2.9 (0.0)	−7.2 (0.1)	−24.5 (0.1)	+9.2 (0.1)	+15.4 (0.1)
Expected-rate news only	+1.2 (0.0)	+3.5 (0.1)	+0.8 (0.1)	+2.6 (0.1)	+0.8 (0.0)
Duration costs only	+0.5 (0.0)	−2.6 (0.1)	−3.8 (0.0)	+1.7 (0.1)	+1.6 (0.0)
Rate path with news	+3.8 (0.0)	−5.6 (0.1)	−23.9 (0.1)	+8.6 (0.1)	+12.6 (0.1)
Rate path with duration costs	+3.1 (0.0)	−8.8 (0.1)	−26.2 (0.0)	+9.1 (0.1)	+16.5 (0.1)
News with duration costs	+1.4 (0.0)	−0.9 (0.1)	−2.6 (0.0)	+2.9 (0.1)	+2.1 (0.0)
Full realized path	+3.9 (0.0)	−8.7 (0.1)	−24.7 (0.1)	+8.7 (0.1)	+13.6 (0.1)

Notes: Responses to the realized 2022–24 aggregate path and to its single and pairwise component paths, relative to the no-shock path from the same 2021-conditioned cross-section, in the benchmark economy and in the restricted economy of Table 13. The long-term share of new issuance and the credit spread are averages of the annual responses; gross issuance, investment, and default are cumulated through 2024. The credit spread is the issuance-weighted mean spread on new claims; its level and its responses are dominated by the share of issuance priced close to default (7 percent of issuance carries a spread above 20 percentage points on the no-shock path), so the column measures the incidence of near-default pricing rather than the borrowing cost of the typical firm, whose median spread moves by less than half a point. Each component path holds the other aggregates at their calibrated means. Means across 50 simulated economies of 10,000 firms; Monte Carlo standard errors of the mean are in parentheses.

Table 35: Investment Effects and Survivorship

	Current Rate	Expected-Rate News	Term-Premium Wedge
Survivor mean	+4.2	+2.9	+3.5
Initial-cohort mean	+4.1	+3.6	+3.1
Cohort aggregate	+5.3	+3.9	+3.5

Notes: Entries report the benchmark-minus-counterfactual difference in cumulative investment responses over five years, in percentage points of capital, as in Table 13. Positive values indicate that endogenous maturity flexibility preserves investment. "Survivor mean" compares the surviving cross-section in each economy, including replacement entrants. "Initial-cohort mean" restricts the sample to surviving firms of the cohort alive at the shock date. "Cohort aggregate" retains the full initial cohort, counting the capital of firms that subsequently default as destroyed.
Results are averages across 1000 simulated economies.

E.6 Investment Effects and Survivorship

Because the maturity restriction changes default, differences in average investment could reflect changes in the composition of surviving firms rather than changes in firms' investment decisions. Table 35 recomputes the five-year investment gap of the main counterfactual on the surviving cross-section, on the cohort of firms alive at the shock date, and on the full initial cohort with the capital of defaulting firms counted as destroyed. The gap remains positive in every case and is larger, not smaller, for the cohort aggregate after current-rate and expected-rate shocks, so the main counterfactual does not arise mechanically from differential survivorship.

E.7 Inherited Rollover Exposure and Selection

The main counterfactual restricts firms' ability to adjust maturity after aggregate financing conditions change. Two exercises isolate the role of inherited rollover exposure itself; in both, subsequent maturity choices remain unrestricted and only firms' initial exposure is changed. The first imposes rollover exposure exogenously on otherwise identical firms. The share of debt in one-period claims is set to 0.1 for the low-exposure group and 0.9 for the high-exposure group, corresponding to $s_{i0} = (1 - \rho_L) + \rho_L \times 0.1 \approx 0.27$ and $s_{i0} \approx 0.92$, after which both populations face identical shocks and idiosyncratic draws under benchmark policies. On impact, high-exposure firms reduce investment by 2.0 percentage points more following a current-rate shock and by 1.7 percentage points more following expected-rate news. Conditional on the remaining firm state, inherited refinancing exposure itself amplifies the real response to financing shocks. Because the experiment places firms at two extreme initial exposure levels, their subsequent refinancing cycles become synchronized, so we interpret the impact comparison rather than the cumulative path. The second asks whether the cross-sectional distribution of the initial maturity wall matters for aggregate transmission. Starting from the benchmark stationary distribution, we compress each firm's initial rollover exposure toward the debt-weighted mean,

$$s_{i0}^{\lambda} = (1 - \lambda)s_{i0} + \lambda \bar{s}_0, \qquad \bar{s}_0 = \frac{\sum_i B_{i0} s_{i0}}{\sum_i B_{i0}}, \qquad \lambda \in [0, 1] \tag{77}$$

which preserves aggregate principal coming due while changing only its allocation across firms. Complete equalization has modest aggregate effects: across the three financing shocks, cumulative default changes by at most 0.2 percentage points, and the cross-sectional dispersion of cumulative

investment responses changes by less than 1%. The initial distribution of exposure is not persistent: once firms reoptimize, differences in rollover exposure re-emerge through subsequent maturity choices. Holding other states fixed, greater rollover exposure makes individual firms more sensitive to financing shocks, but redistributing a given aggregate maturity wall at a single date has limited aggregate effects. The quantitatively important margin is firms' ability to adjust maturity as financing conditions change. The same distinction explains why the raw cross-sectional relation between exposure and default is not the structural effect of exposure. Observed rollover exposure is an equilibrium outcome: firms with different characteristics and past financing choices select different maturity structures. Figure 19 shows the two objects side by side. Across the equilibrium distribution (Panel A), the relation between exposure and default risk is not monotone, because firms with different capital, leverage, productivity, and previous maturity choices populate different points of the distribution. Holding (z, K, B) fixed and varying exposure alone (Panel B), next-period default risk rises sharply as a larger fraction of principal comes due. Cross-sectional exposure reflects both treatment and selection, which is why the counterfactuals distinguish inherited exposure from firms' ability to adjust maturity.

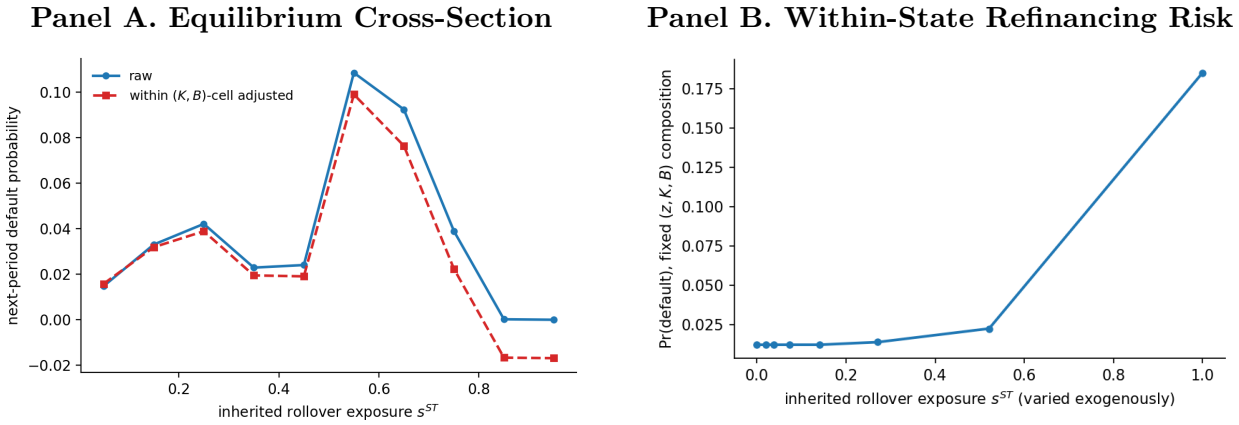


Figure 19: Rollover Exposure, Refinancing Risk, and Equilibrium Selection. Panel A reports default risk across the equilibrium distribution of rollover exposure. Because exposure reflects both refinancing needs and firms' past maturity choices, its unconditional relationship with default need not be monotone. The adjusted series residualizes composition within (K, B) cells. Panel B varies inherited rollover exposure while holding (z, K, B) fixed. Conditional on these firm states, next-period default risk rises as a larger share of principal comes due.

E.8 Debt-Price Sensitivity to Issuance

Because the long-term claim is priced recursively, current issuance affects not only next-period repayment risk but also the value of the surviving claim. Long-term debt prices may therefore be more sensitive to borrowing than the price of a one-period claim. We quantify this sensitivity at each stationary state by moving next-period debt one grid point above and below the policy, holding capital and the maturity share fixed, and re-pricing both claims using the same price schedules as in the firm's problem. We compute

$$\varepsilon^j = \frac{\Delta \log Q^j}{\Delta \log m}, \qquad j \in \{S, L\}$$

and report stationary-distribution-weighted statistics among issuing, non-defaulting firms at the mean aggregate state in Table 36. The difference across contracts is large. The mean price sensitivity is only 0.014 for short-term debt, compared with 0.092 for long-term debt, and the long-debt price is more sensitive to issuance for 98 percent of stationary issuing mass. The difference also increases with refinancing needs. At the amortization floor, the mean long-debt sensitivity is 0.063; among firms with high rollover exposure it rises to 0.182, while the short-debt price remains substantially less responsive. The same ranking obtains at high expected-rate and high term-premium wedge states. Thus, the firms for whom long maturity provides the greatest protection against future refinancing needs also face a steeper endogenous price schedule for obtaining that protection.

Table 36: Debt-Price Sensitivity to Gross Issuance

	Issuing mass	Mean $-\varepsilon^S$	Mean $-\varepsilon^L$	Share $\lvert\varepsilon^L\rvert > \lvert\varepsilon^S\rvert$
All issuers	1.00	0.014	0.092	0.98
At amortization floor	0.46	0.006	0.063	0.99
High rollover exposure	0.27	0.040	0.182	0.94

Notes: $-\varepsilon^j = -\Delta \log Q^j / \Delta \log m$ measures the sensitivity of the price of claim $j \in \{S, L\}$ to an increase in gross issuance. Statistics are weighted by the stationary distribution among issuing, non-defaulting firms at the mean aggregate state. "At amortization floor" denotes $s = 1-\rho_L$; "High rollover exposure" denotes $s > 0.22$, the upper half of the above-floor exposure distribution. Because adjacent debt-grid points imply sizable changes in issuance, the reported objects are secant rather than local elasticities.

E.9 Contract Structure: Short-Only and Long-Only Economies

The main counterfactual removes the response of maturity to aggregate conditions. A complementary experiment removes the contract menu itself. Holding the calibrated parameters fixed, we re-solve the firm problem in a short-only economy, in which all new borrowing is one-period debt ($\ell_{it} = 0$), and in a long-only economy, in which all new borrowing is the amortizing long-term claim ($\ell_{it} = 1$), with prices re-solved in each. Each economy is simulated from its own stationary distribution under common random numbers, so the comparison is between economies that have adapted to their contract structure. It is therefore a different experiment from the main counterfactual, which holds inherited balance sheets fixed at the shock date. The adaptation is large: the short-only economy holds about one-third of the benchmark's debt, all of it due within the year, and the long-only economy about two-thirds, with rollover exposure pinned at the amortization floor. Panel A of Table 37 reports the levels of the three economies along their no-shock paths, and Panel B the cumulative responses to one-standard-deviation shocks at three and five years, as benchmark-minus-alternative differences. The stationary comparisons differ from the levels reported by Deng and Fang (2022). In their model, the investment rate is lower when firms can issue only short-term debt (20.0% against 20.5% in their benchmark). In ours, the short-only economy has a higher investment-to-capital ratio than the benchmark (11.8% versus 10.9%), but a 21% smaller capital stock, substantially less debt relative to capital (0.35 versus 0.88), and a lower default rate (0.3% versus 1.7%). The absence of long-term borrowing is therefore associated with a smaller and less indebted economy, rather than a lower investment rate. The long-only economy is closer to the benchmark in capital and leverage, although its default rate is higher (2.1% versus 1.7%). These level comparisons characterize economies that have adapted to different contract menus; they do not isolate the contribution of maturity adjustment for firms entering a shock with identical balance sheets. The two restricted economies reveal why endogenous maturity

choice is valuable. Under current-rate tightening, both contract extremes generate substantially larger investment losses than the benchmark: short-only firms are fully exposed to refinancing, while long-only firms cannot economize on long-duration funding when financing conditions deteriorate. Expected-rate news exposes a different trade-off. The long-only economy suffers larger investment losses, higher default, and wider spreads because firms are forced to purchase duration even as its current price deteriorates. Short-only firms initially lose more investment than in the benchmark, but the difference largely disappears by five years because the economy has adapted to its contract structure by carrying substantially less debt. A term-premium shock reverses the comparison: the short-only economy is mechanically insulated from the direct increase in the price of duration, whereas the long-only economy experiences lower investment and issuance. Thus, no fixed contract structure dominates across financing shocks. The value of the benchmark contract menu is that firms can adjust the amount of refinancing insurance they purchase to the source of the change in financing conditions.

Table 37: Short-Only and Long-Only Economies

Panel A. Levels along the no-shock path (averages over years 3–10; MC s.e.)	Benchmark	Short debt only	Long debt only
Investment rate I/K (%)	10.9 (0.1)	11.8 (0.1)	11.0 (0.1)
Mean capital	0.511 (0.002)	0.406 (0.001)	0.497 (0.002)
Debt to capital	0.88 (0.00)	0.35 (0.00)	0.68 (0.00)
Default rate (%)	1.7 (0.0)	0.3 (0.0)	2.1 (0.0)
Rollover exposure s	0.23	1.00	0.19
Long-term issuance share	0.79	0.00	1.00

Panel B. Shock responses, benchmark minus alternative	Current rate		Expected-rate news		Term-premium wedge	
Horizon (years)	3	5	3	5	3	5
Investment (cumulative I/K, pp)						
Short debt only ($\ell = 0$)	+2.7 (0.2)	+4.1 (0.1)	+1.7 (0.2)	+0.1 (0.1)	−1.1 (0.1)	−0.9 (0.0)
Long debt only ($\ell = 1$)	+2.9 (0.2)	+4.5 (0.2)	+0.8 (0.3)	+1.4 (0.3)	+1.3 (0.2)	+0.9 (0.2)
Debt issuance (cumulative m/B, pp)						
Short debt only ($\ell = 0$)	+5.5 (0.2)	+4.3 (0.2)	+5.3 (0.3)	+0.6 (0.2)	−1.3 (0.2)	+0.3 (0.1)
Long debt only ($\ell = 1$)	+2.7 (0.2)	+2.8 (0.3)	−0.6 (0.2)	−1.3 (0.3)	+3.8 (0.2)	+4.0 (0.3)
Default (cumulative, pp)						
Short debt only ($\ell = 0$)	+0.1 (0.1)	+0.1 (0.1)	+0.4 (0.1)	+0.4 (0.1)	0.0	0.0
Long debt only ($\ell = 1$)	−0.5 (0.1)	−0.7 (0.1)	−1.6 (0.2)	−2.1 (0.2)	0.0	+0.3 (0.0)
Credit spread (average annual, pp)						
Short debt only ($\ell = 0$)	+0.6 (0.2)	+0.2 (0.1)	+1.5 (0.2)	+0.6 (0.1)	−0.4 (0.0)	−0.7 (0.0)
Long debt only ($\ell = 1$)	−0.4 (0.1)	−0.5 (0.1)	−0.3 (0.1)	−0.7 (0.1)	−0.4 (0.1)	−0.5 (0.1)

Notes: Panel A reports averages over years three to ten of the no-shock paths of the three economies, each initialized from its own stationary distribution with aggregate states held at their means. Because the stationary distributions reflect fluctuating aggregate states, all three economies adjust slowly along this path (capital rises by 2–4% and the investment rate falls by about one percentage point over the window); the differences between economies are stable across it. Panel B reports the benchmark-minus-alternative difference in cumulative responses to a one-standard-deviation shock (shock minus no-shock path in each economy); positive values indicate a higher response in the benchmark. Positive investment and issuance entries therefore mean that the benchmark loses less than the alternative economy, and negative default and spread entries mean smaller increases in default and financing costs in the benchmark. Both alternative economies use the calibrated parameters, with prices re-solved, and are simulated from their own stationary distribution, unlike the main counterfactual, which starts every economy from the benchmark cross-section; all economies face common random numbers. The short-only economy issues no long-term debt and is therefore mechanically unaffected by the term-premium wedge. Means across 1000 simulated economies of 10,000 firms; Monte Carlo standard errors of the mean in parentheses.

E.10 Policy-Function Diagnostics

To examine the heterogeneity underlying the simulated-regression results, we evaluate firms' maturity choices at the balance-sheet positions observed in the simulated histories. For each firm, we compare the long-term issuance share chosen when the aggregate financing states are at their means with the share chosen after a one-standard-deviation increase in expected-rate news or in the term-premium wedge, the other states at their means, and we average the difference within

three groups of rollover exposure, giving each exposure node within a group equal weight. Holding individual states fixed isolates the conditional maturity response to each aggregate factor within a group; across groups, the average response can still reflect differences in capital, productivity, leverage, and proximity to the threshold between short- and long-term issuance.

Figure 20 reports the conditional impact response of long-term issuance shares across the three groups. Following expected-rate news, firms at the amortization floor reduce their long-term issuance share by 15.4 percentage points, compared with 9.6 and 8.5 points among firms with intermediate and high exposure. Firms facing greater inherited refinancing needs therefore shorten relatively less when expected future rates rise. By contrast, a higher term-premium wedge induces maturity shortening that increases with exposure, 4.3, 5.4, and 9.6 points: more-exposed firms shorten more, with no relative protection. These conditional policy responses illustrate the heterogeneity underlying the model's maturity choices and the opposite signs of the two empirical interactions.

The figure describes impact responses at mean aggregate financing conditions. The relative news response is conditional on the initial financing state and on the horizon. The quantitative comparison with the empirical interactions therefore relies on the fixed-effects regressions estimated on simulated firm histories in Section 7.4, which incorporate the evolution of firm states and aggregate financing conditions and match the empirical specification. Figure 5 complements this exercise by decomposing the incentives underlying maturity choice into the insurance value of postponing refinancing and the valuation cost of long-duration funding.

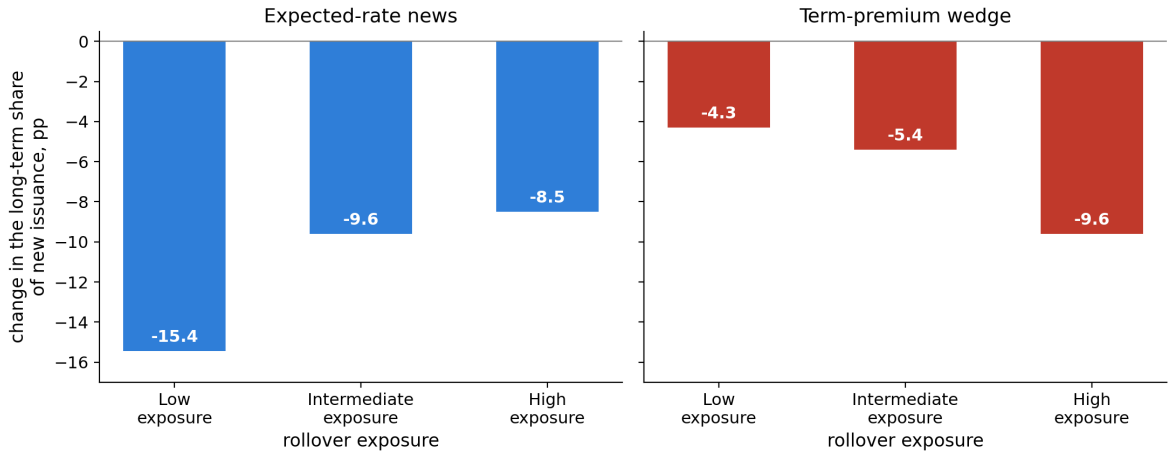


Figure 20: Conditional Maturity Responses by Rollover Exposure. Change in the long-term share of new issuance, in percentage points, when expected-rate news (left) or the term-premium wedge (right) rises by one grid node from its mean, about one innovation standard deviation, with the other aggregate states at their means. Each bar averages the responses of the same firms at the same balance sheets, those observed in the simulated histories, over the grid nodes in the group with equal weight per node. Low exposure is the amortization floor, $s = 1 - \rho_L = 0.19$, with 49 percent of issuing firms; intermediate exposure comprises the nodes between 0.21 and 0.25, with 25 percent; high exposure comprises the nodes between 0.31 and 1, with 26 percent. Means across 50 simulated histories.